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*** START OF THE PROJECT GUTENBERG EBOOK THE READER'S GUIDE TO THE ENCYCLOPAEDIA
BRITANNICA ***

THE READER'S GUIDE
TO THE
ENCYCLOPAEDIA BRITANNICA

A HANDBOOK CONTAINING SIXTY-SIX COURSES OF SYSTEMATIC STUDY OR
OCCASIONAL READING

THE ENCYCLOPAEDIA BRITANNICA COMPANY, Limited
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INTRODUCTION

In your ordinary use of the Encyclopaedia Britannica, you give your attention to the _one_ article that will answer the _one_ question you have in your mind. The aim of this Guide is to enable you to use the Britannica for an altogether different purpose, namely, for systematic study or occasional reading on any subject.

The volumes of the Encyclopaedia Britannica contain forty-four million words—as much matter as 440 books of the ordinary octavo size. And the subjects treated—in other words, the whole sum of human knowledge—may be divided into 289 separate classes, each one completely covering the field of some one art, science, industry or other department of knowledge. By the mere use of scissors and paste the alphabetical arrangement of the articles could be done away with, and the Britannica could be reshaped into 289 different books containing, on the average, about half as much again as an ordinary octavo volume. It would misrepresent the Britannica to say that you would then have 289 _text-books_, because there is an essential difference in tone and purpose. A text-book is really a book intended to be used under the direction and with the assistance of a teacher, who explains it and comments upon it. The Britannica, on the other hand, owes the position it has enjoyed since the first edition appeared in 1768 to the fact that it has succeeded, as no other book has succeeded, in teaching without the interposition of a teacher.

It is not, of course, claimed that the idea of reading certain groups of Britannica articles in the order in which they will combine themselves into complete books is a novel invention. Thousands of men owe the greater part of their educational equipment to a previous edition of the Britannica. And not only did they lay out their own courses of reading without the aid of such a Guide as this, but the material at their

disposal was by no means so complete as is the 11th Edition. Every edition of the Britannica before this one, and every other book of comparable size previously published, appeared volume by volume. In the case of the last complete edition before the present, no less than 14 years elapsed between the publication of the first volume and the last. It is obvious that when editors have to deal with one volume at a time, and are unable to deal with the work as a whole, there cannot be that exact fitting of the edges of one article to the edges of another which is so conspicuously a merit of the 11th Edition. All the articles in this edition were completed before a single volume was printed, and the work stood, at one stage of its preparation, in precisely the form which, as has already been said, might be given to it by merely rearranging the articles according to their subjects.

In this Guide, the principal articles dealing with the subject of each chapter are named in the order in which you may most profitably study them, and the summaries of the larger articles afford such a preliminary survey as may assist you in making your choice among the courses. Besides, where it seems necessary, there is added to the chapter a fairly complete list of all articles in the Britannica on the subject, so that the reader may make his study exhaustive.

A brief review of the six parts into which the Guide is divided will show the general features of its plan, of which a more detailed analysis is given in the Table of Contents.

Part 1 contains 30 chapters, each designed for readers engaged in, or preparing for, some specific occupation. To the beginner, who still has everything to learn, the advantages derived from such a course of study may well be so great as to make the difference between success and

failure in life, and to those who have already overcome the first difficulties, to whom the only question is how marked a success awaits them, the Britannica can render invaluable service of another kind. No amount of technical training and of actual experience will lead a man of sound judgment to believe that he alone knows everything that all his competitors put together know; or that his knowledge and theirs is all that ever will be known. The 1500 contributors in 21 different countries who wrote the articles in the Britannica include the men who have made the latest advances in every department of knowledge, and who can forecast most authoritatively the results to be expected from the new methods which are now being experimentally applied in every field of activity. The experienced merchant, manufacturer, or engineer, or the man who is already firmly established in any other profession or business, will naturally find in some of the articles facts and figures which are not new to him, but he can profit by the opportunity to review, confirm, reconsider and "brush up" his previous knowledge.

Part 2 contains 30 chapters, each devoted to a course of systematic study designed to supplement, or to take the place of, some part of the usual school and college curriculum. The educational articles in the Britannica are the work of 704 professors in 146 universities and colleges in 21 different countries. No institution of learning in the world has a faculty so numerous, so authoritative, or so highly specialized. Nor has any system of home study ever been devised by which the student is brought into contact with teachers so trustworthy and so stimulating. The fascination of first-hand knowledge and the pleasure of studying pages intended not for reluctant drudges submitting themselves to a routine, but for students eager to make rapid progress, are factors in the educational value of the Britannica that cannot be overestimated, and the elasticity with which any selected course of study can be enlarged and varied is in full accordance with the modern theories of

higher education.

Part 3 is devoted to the interests of children. The first of its chapters describes Britannica articles of the utmost practical value to parents, dealing with the care of children's health, with their mental and bodily training, and with the intelligent direction of their pastimes. The second chapter indicates varied readings in the Britannica for children themselves, showing how their work at school can be made more interesting and profitable to them by entertaining reading on subjects allied to those included in their studies. The third chapter in this Part gives a number of specific questions such as children are prone to ask, as well as questions which may be put to them in order to guide their natural inquisitiveness to good purpose. The references to pages in the Britannica show where these questions are clearly and instructively answered.

Part 4 suggests readings on questions of the day which relate to American citizenship and to current politics. A study of the articles indicated in this section of the Guide will aid the reader not only to form sound opinions for himself, but also to exercise in private or public life the influence for good which arises from a clear view of the arguments on both sides of controverted questions. It is no exaggeration to say that the Britannica is the only existing work in which such subjects as tariffs, trusts, immigration, labour and the relation between legislative and judiciary powers are treated without partisan bias and with adequate fulness.

Part 5, especially for women, deals with their legal and political status in various parts of the world, their achievements in scholarship, art and science, as well as with home-making, domestic science and kindred subjects. The important part which women, both among the

contributors and on the editorial staff of the Britannica, took in the preparation of the work sufficiently indicates that the editor-in-chief made ample provision for the subjects peculiarly within their sphere.

Part 6 is an analysis of the many departments of the Britannica which relate to recreation and vacations, travel at home and abroad, photography, motoring, out-door and indoor games and other forms of relaxation and of exercise. The extent to which the work can be used in planning motoring tours, and the superiority, in such a connection, of its articles to the scant information found in ordinary guide books, are shown in the extracts, included in this Part 6, relating to a trip from New York through the Berkshire Hills to the White Mountains.

It will be seen from this brief survey of the field covered by the Guide that provision has been made for every purpose which can dictate the choice of a course of reading. But as you proceed to examine its contents for yourself, you should remember that the lists it gives name only a fraction of the articles in the Britannica, and that for a fuller summary of the work as a whole you should turn to the Table on pp. 881–947 of Vol. 29.

Finally, the form in which this Guide is printed may call for a word of justification. It is inevitable that chapters, of an analytical character, bespattered with references to the numbers of volumes and of pages, and terminating with lists of the titles of articles, should bear a certain air of formality. There is no danger that the possessor of the Britannica, familiar with the fascination of its pages and the beauty of the illustrations which enhance their charm would permit his impression of the work itself to be affected by the bleak appearance of the Guide. But he may feel that because a list has a forbidding aspect the pleasure

he has derived from browsing at will in the Britannica would give place to a sense of constraint if he rigidly pursued a course of reading. It may easily be shown that such a fear would be groundless, for the Britannica articles are all the better reading when one carries forward the interest which one of them has excited to others of related attraction. But to anyone who is firmly determined that he shall not be persuaded to read systematically, the Guide will none the less be useful, for he may flit from one chapter to another, selecting here and there an article merely because the account which is given of it pleases him. Or, better yet, he may find, in one portion only of a selected course, a series of only three or four articles which will, in combination, make the best of occasional reading.

THE EDITORS.

TABLE OF CONTENTS

PART I

COURSES OF READING ESPECIALLY USEFUL TO THOSE ENGAGED IN CERTAIN OCCUPATIONS, OR PREPARING FOR THEM

	Page
Chapter 1. For Farmers	3
„ 2. For Stock-Raisers	10
„ 3. For Dairy Farmers	14
„ 4. For Merchants and Manufacturers, General and	
Introductory	19

„ 5.	Textiles	21
„ 6.	Machinery	28
„ 7.	Metals, Hardware, Glass and China	33
„ 8.	Furniture	39
„ 9.	Leather and Leather Goods	44
„ 10.	Jewelry, Clocks and Watches	48
„ 11.	Electrical Machinery and Supplies	55
„ 12.	Chemicals and Drugs	58
„ 13.	Food Products	63
„ 14.	For Insurance Men	69
„ 15.	For Architects	71
„ 16.	For Builders and Contractors	79
„ 17.	For Decorators and Designers	83
„ 18.	For Railroad Men	90
„ 19.	For Marine Transportation Men	94
„ 20.	For Engineers	100
„ 21.	For Printers, Binders, Paper-makers and All who Love Books	109
„ 22.	For Journalists and Authors	117
„ 23.	For Teachers	122
„ 24.	For Ministers	127
„ 25.	For Physicians, Surgeons and Dentists	135
„ 26.	For Lawyers	143
„ 27.	For Bankers and Financiers	151
„ 28.	For Civil Service Men	156
„ 29.	For Army Officers	158
„ 30.	For Naval Officers	168

PART II

COURSES OF EDUCATIONAL READING TO SUPPLEMENT OR TAKE THE PLACE OF
SCHOOL OR UNIVERSITY STUDIES

Chapter 31. Music	175
„ 32. The Fine Arts, Introductory and General	187
„ 33. Painting, Drawing, Etc.	189
„ 34. Sculpture	198
„ 35. Language and Writing	207
„ 36. Literature, Introductory and General	214
„ 37. American	218
„ 38. English	224
„ 39. German	230
„ 40. Greek	234
„ 41. Bible Study	237
„ 42. History, Introductory and General	246
„ 43. American	248
„ 44. Canadian	270
„ 45. English, Scotch and Irish	272
„ 46. French	278
„ 47. The Far East: India, China, Japan	281
„ 48. Economics and Social Science	288
„ 49. Health and Disease	294
„ 50. Geography and Exploration	300
„ 51. Anthropology and Ethnology	308
„ 52. Mathematics	316
„ 53. Astronomy	322
„ 54. Physics	329
„ 55. Chemistry	334
„ 56. Geology	338
„ 57. Biology, General and Introductory	344
„ 58. Botany	347

„ 59. Zoology	353
„ 60. Philosophy and Psychology	361

PART III

DEVOTED TO THE INTERESTS OF CHILDREN

Chapter 61. Readings for Parents	371
„ 62. Readings for School Children	379
„ 63. Questions Children sometimes ask, and Some Questions to ask Children	387

PART IV

READINGS ON QUESTIONS OF THE DAY

Chapter 64.	393
Education, Training of Defectives, Psychology	
Crime, Juvenile Courts, Alcoholism	
Heredity and Eugenics	
Wages and Labour, Labour Organization	
Immigration, The Negro Problem	
Trusts, Finance, Tariff, Banking, Insurance	
Socialism and its Tendencies	
Initiative, Referendum and Recall, Government by Commission	
Suffrage and the Suffrage Question	
International Relations, Peace Arbitration	
The Greater United States	

PART V

FOR WOMEN

Chapter 65. 411

The many subjects on which Women contributed to the
Britannica

Accomplishments of Women in Scholarship, Art and
Science

Women's Legal Position in the United States and
elsewhere

Their Disabilities in Great Britain

Home-making, Domestic Science, the Table

Food Preservation and Food Values

Costume and Ornament

Women famous in History and Literature, and on the
Stage

PART VI

READINGS FOR RECREATION AND VACATION

Chapter 66. 425

Motoring, a Specimen Trip: New York to the White
Mountains

Photography

Out-door Games and Athletic Sports

Hunting, Fishing and Taxidermy

Sailing, Canoeing and Boating
Mountaineering and Winter Sports
Driving, Riding, Polo and Horse-racing
Gardening and Plants
In-door Games and Pastimes, Bridge, Needlework
Dancing, the Stage
Travel at Home and Abroad

PART I

COURSES OF READING ESPECIALLY USEFUL TO THOSE ENGAGED IN CERTAIN
OCCUPATIONS OR PREPARING FOR THEM

CHAPTER I FOR FARMERS

SEE ALSO CHAPTER II, FOR STOCK-RAISERS, CHAPTER III, FOR DAIRY FARMERS

Every farmer in the United States knows that farming is to-day an industry which calls for study of the world's agricultural products, processes, and markets as well as for scientific knowledge of soils, crops, and animals. Fifty years ago the farmer sold for consumption in his immediate neighborhood the small surplus of his crops that was not needed for his own household and live stock. To-day he competes, in all the world's great markets, with all the world's farmers, and is the

chief among American exporters. The Russian wheat fields and the Argentine cattle ranches are really nearer to him than a farm in the next township was to his grandfather. He lives better, does more for his children and pays higher wages than do farmers in other parts of the world, and yet he can successfully compete with them, because, as the article on Agriculture in the Encyclopædia Britannica says, in speaking of the United States, "there is no other considerable country where as much mental activity and alertness has been applied to the cultivation of the soil as to trade and manufactures." American farmers "have been the same kind of men, out of precisely the same houses, generally with the same training, as those who filled the learned professions or who were engaged in manufacturing or commercial pursuits"; and their competitors abroad have been, for the most part, ignorant peasants. The course of reading indicated here is designed for wide-awake farmers who intend to be large farmers—by whom the latest information and the broadest outlook are recognized as essential to their calling. _If you think the articles named here cover a great deal of ground, remember that the Massachusetts Agricultural College provides no less than sixty-four distinct courses of instruction, and that the subjects included in all the sixty-four are treated in the Britannica._

GETTING "GROUNDWORK" KNOWLEDGE

You may think, as you look at the titles of articles mentioned in these pages, that there are some which you need not read because you have already read bulletins of the United States Department of Agriculture or of your State Experiment Station. These official publications are most valuable, but naturally, they do not attempt to cover the whole range of agricultural subjects as the Britannica does—they are not intended for that purpose. Their arrangement and the way in which they are issued

shows that they are designed to meet only certain special needs, not to give a general view of all the branches of farming. One subject may for example be discussed in three different bulletins, published in three different years, and the first may be out of print before the third appears. In the Britannica you get information that forms the very foundation of a thorough knowledge of farming and that also extends over the widest field. Of course it would be absurd to say that merely reading these articles will make any man a successful farmer as to say that a medical student who works hard at his books will always develop the tact and the sound judgment that a doctor needs. But unless the medical student has studied those text books he will never make a successful doctor; and similarly the information in the Britannica will give the farmer new advantages, no matter how much practical experience and special training he has had.

[Sidenote: Scope of the Articles]

There are in the Encyclopædia Britannica 1,186 articles dealing with animal and vegetable life; and among the 11,341 geographical articles a great many give important information about the production, distribution and consumption of farm products. Those upon continents, countries, states and provinces describe the local crops and any local methods of farming that are of special interest. _There are some 600 articles on individual plants_, of which a list will be found on pp. 889 and 890 of Vol. 29 (the index volume). If any one of these thousands of articles were not in the Britannica, it would not be quite so valuable as it is to you, for you may, any day, want to find out about any plant that grows, or about farming in any part of the world. A professor in an agricultural college would of course be glad to study the whole series. But in this Course of Reading only the articles which are of most immediate use to all practical farmers are mentioned, and the contents

of each of these is described, so that you can omit any article that goes into details which you think you do not want. If you do skip any of them, it will, however, be a good plan to mark their titles in this list, for you may like to come back to them later when you realize how practical and understandable all the Britannica articles are—even those with dullsounding names.

Of course you will begin by reading the article AGRICULTURE (Vol. 1, p. 338), by Dr. Fream and Roland Truslove, which is the key to the whole subject. And remember that this chapter of the Readers' Guide _mentions only those subjects that are treated more fully in other parts of the Britannica than in that article_, so that the chapter does not attempt to tell the whole story.

[Sidenote: Soil and Subsoil]

The first thing a farmer has to deal with is the ground from which his crops are to come. The whole surface of the earth was originally hard rock. The article on PETROLOGY, the science of rocks (Vol. 21, p. 323), by J. S. Flett, and the second part (Vol. 11, p. 659) of the article GEOLOGY, by Sir Archibald Geikie, deal with the "weathering" of rock, which has in great part broken it down into the small particles of stone that, mixed with decayed roots and plants, form the soil or subsoil. It may seem that it is going very far back into the origin of things for a farmer to read about the sources from which soil comes, but the nature of the mineral substances in it has a great deal to do with its power to nourish plants, and you cannot know too much about the material on which your principal work is done. The article which should next be read, SOIL (Vol. 25, p. 345), continues the story of these particles of rock and shows how sand and clay must be combined with decaying vegetable or animal matter in order to make the best soil. This mixture is in turn

“weathered” by air, heat, frost, and moisture; and not only the size of the grains in which it lies, but also their shape—which makes them pack more or less tightly—affect the pores, or spaces between the grains, through which the roots of the plants must push their way, and through which air and water must reach these roots. The article EARTHWORM (Vol. 8, p. 825) describes the useful part that worms play in stirring the mixture, while the natural and artificial fertilizers, which supply whatever ingredients the soil lacks, are discussed in the article MANURES AND MANURING (Vol. 17, p. 610). An important part of this article deals with the best methods of keeping farm yard manure in such a way that it does not lose its value before it is spread over the fields, and with the use, in this connection, of the liquid-manure tank. The microbes in the soil render the farmer an enormous service by changing crude nitrogen, which plants cannot digest, into the forms in which it is indispensable to them, and this process is described in the article BACTERIOLOGY (Vol. 3, p. 164), by Professor Marshall Ward, Professor Blackman, and Professor Muir.

[Sidenote: Sunlight and Shade, Heat and Cold, Water Enough—and Not too Much]

The action of light, the supply of which is just as necessary in causing growth as the warmth the sun gives, and the action of water and of heat and cold, are explained in the section “Physiology” (Vol. 21, p. 745) of the article on PLANTS. The proper method of working each farm, with a view to using these four in the right proportions, is influenced by the latitude in which it lies, its height above sea level, the protection that mountains give it, the slope at which the fields face the sun or turn away from it, the rain-fall, the relative dampness or dryness of the air when it is not raining, and the moisture of the soil. Every one of these subjects is vital to the farmer, and the Britannica brings to

its readers the latest information regarding them in articles written by the leaders of progress. You will find the latest scientific guidance, in the most practical shape, in the articles CLIMATE (Vol. 6, p. 509), by Professor R. de C. Ward, of Harvard, METEOROLOGY (Vol. 18, p. 264), by Professor Cleveland Abbe, of the United States Weather Bureau, and ACCLIMATIZATION (Vol. 1, p. 114). The distribution of heat in the soil is described in the article CONDUCTION OF HEAT (Vol. 6, p. 893), where the diagram showing variations of temperature at different depths in the soil should be carefully studied.

[Sidenote: Drainage and Irrigation]

The brackish water that troubles farmers near tidal creeks, the alkali water that often occurs West of the Mississippi, and the stagnant water that never does the farm any good, are all as bad in their way as the river-floods or the merely sodden soil in which nothing will grow but coarse grass that is always unsafe pasturage. Drains and embankments need very careful planning, and sound information will be found in the articles DRAINAGE OF LAND (Vol. 8, p. 471), RECLAMATION OF LAND (Vol. 22, p. 954), and RIVER ENGINEERING (Vol. 23, p. 374), the latter by Professor L. F. Vernon H. Harcourt, the leading authority on such subjects the world over.

The saving of water and the method of bringing it to the farm and distributing it over the fields are authoritatively discussed in the articles IRRIGATION (Vol. 14, p. 841), WATER SUPPLY (Vol. 28, p. 387), by G. F. Deacon, WINDMILL (Vol. 28, p. 710), PUMP (Vol. 22, p. 645), and in the section headed "Utility of Forests" (Vol. 10, p. 646) of the article FORESTS AND FORESTRY, by Gifford Pinchot, formerly U. S. Chief Forester. The other parts of this article, dealing with the timber industry, are of course important to farmers whose land includes any

lumber. WATER RIGHTS (Vol. 28, p. 385) explains the laws which regulate the taking of water from streams and lakes, and the article LAKE (Vol. 16, p. 86) is also of interest in connection with irrigation.

[Sidenote: Farm Buildings and Fences]

When the farmer, who has to be everything by turns, has been an engineer long enough to get the water off his farm or on his farm—and perhaps he has to do both in different parts of the same farm—he must next take on the builder's job. He will be reminded of a good many precautions and economies that are often overlooked, and may find, too, some hints that are quite new to him, in the excellent series of articles, all by experts in the building trade: FARM BUILDINGS (Vol. 10, p. 180), BUILDING (Vol. 4, p. 762), FOUNDATIONS (Vol. 10, p. 738), BRICKWORK (Vol. 4, p. 521), STONE (Vol. 25, p. 958), MASONRY (Vol. 17, p. 841), TIMBER (Vol. 26, p. 978), CARPENTRY (Vol. 5, p. 386), and ROOFS (Vol. 23, p. 697). The use of concrete for buildings, tanks, irrigation works, etc., has proved so successful, and is so rapidly increasing, that you will be especially interested by the article CONCRETE (Vol. 6, p. 835). BARBED WIRE (Vol. 3, p. 384), in which the meshed field fencing, of late increasing in favor, is also dealt with, is another practical article.

[Sidenote: Agricultural Machines]

Advertisers no doubt supply you with more literature about farm machinery than you find time to read, but that makes it all the more essential to get sound information that has no trade bias. The Britannica goes into the principles of construction and helps you to see the good and bad points in the new models you are constantly offered. You can learn a great deal from the articles PLOUGH (Vol. 21, p. 850), HARROW (Vol. 13, p. 27), CULTIVATOR (Vol. 7, p. 618), HOE (Vol. 13, p.

559), and the sections on machines in the articles HAY (Vol. 13, p. 106), REAPING (Vol. 22, p. 944), SOWING (Vol. 25, p. 523) and THRASHING (Vol. 26, p. 887). OIL ENGINE (Vol. 20, p. 35), WATER MOTORS (Vol. 28, p. 382) and TRACTION (Vol. 27, p. 118) are also of importance.

Farm horses and the other live-stock required in general farming fall under Chapter II of this Guide.

[Sidenote: Farm Finance]

You cannot read the articles already mentioned, and consider all that has to be done in merely getting a farm ready to be worked, without realizing how grossly unfair it is that the American farmer should be hampered, as he is, by the want of proper banking facilities when he is making a start. And after he has bought and prepared his land and equipped and stocked his farm he needs, each year, money to finance his crops. For any loan used in the purchase of land and in permanent improvements such as buildings, drainage, irrigation, a mortgage is the natural security; but the short-term farm mortgages—five years at most—customary in the United States, do not give the farmer as much time as he needs for repayment, no matter how successful he may be. The average farm offers quite as good a certainty of continued earning power as does the average railroad, and farm mortgages should be—in fairness—regarded not as opportunities for short loans, but as sound standing investments, just as suitable as railroad bonds for conservative investors. The farmer's position is even worse when he needs a short loan that he will be able to repay as soon as his crops have been sold, for he is then expected either to give a mortgage as security or to pay exorbitant interest.

Notwithstanding the prosperous conditions of farming in the United

States, the country as a whole produces only half as much grain for every acre of farm land as is produced in Europe, and the only reason is that most of our farmers lack the capital needed in order to get the fullest yield from their land. In the chief European countries, the system of banking facilities for farmers, described in the article CO-OPERATION (Vol. 7, p. 86), by Aneurin Williams, shows what can be done, and sooner or later will be done, in the United States. This article fully describes the admirable Raiffeisen banks in Germany, which are based upon the idea that a society of farmers (restricted to the neighborhood, so that each member's honesty and capability are known to the other members) make themselves jointly responsible for loans to the members. A promissory note is the only security required. The French, Italian, Austrian, and other systems are also discussed in the Britannica, but the German plan is that which offers the best example to America.

[Sidenote: Plants and Crops]

This course of reading has now covered the conditions and the material required for farming, and it is time to get down to something that grows. In the old books everything about the life of a plant was treated as a part of the science of botany, and if you remember the botany you were taught at school, you remember a string of long names and very little else. There is of course an article on botany in the Britannica, but it deals chiefly with the history of botanical science, and the life of the plant is treated under another heading, and in a novel, interesting, and practical way. The article PLANTS (Vol. 21, p. 728) is indeed one of the most important and unusual in the Encyclopædia, giving the results of recent investigation which you could not find in any other book. It is written by eight contributors, all men who have done a great deal of original work. The section on classes of

plants is by Dr. Rendle, that on the anatomy of plants by A. G. Tansley, that on the healthy life of plants by Professor J. Reynolds Green, that on their diseases by Professor H. Marshall Ward, that on the relation between plants and their surroundings by Dr. C. E. Moss, that on plant cells by Harold Wager, that on the forms and organs of plants by Professor S. H. Vines, and that on the distribution of plants in various parts of the world by Sir. W. Thiselton-Dyer. Special accounts of the chief parts of the plant are given in the articles LEAF (Vol. 16, p. 322), STEM (Vol. 25, p. 875), and ROOT (Vol. 23, p. 712). The success of artificial fertilization or impregnation is explained (Vol. 13, p. 744) in the article HORTICULTURE.

Apart from the diseases described in the section, already mentioned, of the article PLANTS, the greatest danger to which crops are exposed is that of insect pests, and the special article ECONOMIC ENTOMOLOGY, dealing with them (Vol. 8, p. 896), gives a full account of each of the remedies that have proved useful. The cotton boll weevil is the subject of a most interesting section of the article COTTON (Vol. 7, p. 261). Separate articles are devoted to individual pests, such as LOCUST (Vol. 16, p. 857), and—turning to a larger enemy—RABBIT (Vol. 22, p. 767). There is no bird that troubles the farmer, or helps him by killing insects, upon which there is not an article, for more than 200 distinct bird articles are listed under the heading “Birds” on p. 891 of Vol. 29 (the index volume), in addition to the information in the article BIRD (Vol. 3, p. 959), and the article on families of birds (Vol. 20, p. 299).

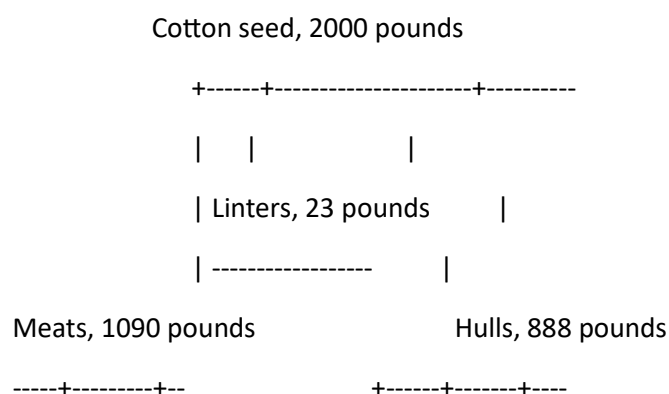
The crops of all climates are treated in general in the article AGRICULTURE, and in particular under their individual names, all of which are so familiar, and indeed so fully listed on p. 889 of Vol. 29 (the index volume), that they need not be repeated here. Naturally you

will include in this course of reading the crops with which you are personally concerned, and in any case you ought to read GRASS AND GRASSLAND (Vol. 12, p. 367), and GRASSES (Vol. 12, p. 369).

[Sidenote: Wheat]

The article WHEAT (Vol. 28, p. 576) deals with one of the chief products of "the greatest cereal producing region of the world." It begins the story of a wheat crop with the burning of the old straw of the previous year, then takes up ploughing, harrowing, seeding, thrashing, labor in connection with all these operations, and transportation and marketing. At this point, the article FLOUR AND FLOUR MANUFACTURE (Vol. 10, p. 548), by G. F. Zimmer, takes up the later history of wheat. It may surprise you to learn from the Britannica that wheat first found its way to America through a few grains being accidentally mixed with some rice. BARLEY (Vol. 3, p. 405) is an interesting article on the grain that is the oldest cereal food of the human race, and that is also remarkable for its power to grow over a greater range of latitude than any other grain. COTTON (Vol. 7, p. 256), by Professor Chapman, is an article of which the vast importance may be judged by the following table taken from page 261:

[Illustration: PRODUCTS FROM A TON OF COTTON SEED]



Cake, 800 pounds					
---+-----+					
Meal					
-----+					
(Feeding stuff. Fertilizer)			Fibre		Bran
-----+			---+---+		-+--
		-----+-----+-----+-----			
Crude Oil, 290 pounds			(High-grade paper) (Cattle food)		
---+-----+---			-----+-----		
Summer Yellow		Soap stock	(Fuel)		
+-----+-----		+---+---	--+-----+-----+		
(Winter Cotton seed					
yellow stearin)		Soaps	Ashes	-----+-----	
+-----+-----		----	--+--	(Cattle food)	
			with the meal		
	Salad oil				
+-----					
	Summer white	Fertilizer			
+---+-----		These together,			
Lard		a very valuable			
+---+		manure			
Cottolene (with beef stearin, cooking oil)					
+-----					
Miners' oil					
+-----					
Soap					

+----

Every one of the other cereal and general crops produced in any part of the world is treated in the Britannica with the same fullness of information and with the same practical detail which characterizes these articles on wheat, barley and cotton.

Some of the principal articles on the routine of farming such as sowing, reaping, and the like, have already been mentioned in connection with agricultural machinery. The articles on individual countries contain sections on the crops of each of them, and you will find CANADA (Vol. 5, p. 152), and GERMANY (Vol. 11, p. 810), of special interest. The special features of tropical farming are described in the articles on tropical crops.

[Sidenote: Fruit and Flower Growing]

The article FRUIT AND FLOWER FARMING (Vol. 11, p. 260) covers fruit culture in general, and, in the section of it which deals with the United States (Vol. 11, p. 268), the American fruit crops. This section describes the wonderful development of the fruit industry since cold transportation and cold storage enabled consumers in every part of the country, and in Europe as well, to purchase fruit grown in whatever state most advantageously produces any one variety. You should select, from the twenty separate articles on individual fruits, not only those on the varieties which you are already growing, but those on any others that are possible in the part of the country where your land lies. The section on fruit in the article on HORTICULTURE (Vol. 13, p. 775) is devoted to growing on a smaller scale, in gardens. It contains (Vol. 13, p. 780) a practical calendar to show each month's work.

Flower culture is the subject of special sections in both the articles above named and there is a descriptive list (Vol. 13, p. 766) of more than three hundred hardy annuals, biennials, and perennials, full of practical information. The calendar already mentioned indicates the dates for indoor and out-door operations. From the many articles on individual flower plants listed at the end of Part 3 of this chapter you can make your own choice.

[Sidenote: Poultry and Bees]

Poultry and their rearing are dealt with in the articles POULTRY AND POULTRY FARMING (Vol. 22, p. 213), FOWL (Vol. 10, p. 760), TURKEY (Vol. 27, p. 467), GUINEA FOWL (Vol. 12, p. 697), DUCK (Vol. 8, p. 630), GOOSE (Vol. 12, p. 241), and INCUBATION and INCUBATORS (Vol. 14, p. 359). Bee-keeping and the honey industry are treated in the articles BEE (Vol. 3, p. 625) and HONEY (Vol. 13, p. 653). Truck farming is treated in the section dealing with vegetables (Vol. 13, p. 776), of the article HORTICULTURE. Apart from the law as to water rights already mentioned the legal doctrine most particularly affecting farmers is that of EMBLEMENTS (Vol. 9, p. 308). GRAIN TRADE (Vol. 12, p. 322), and GRANARIES (Vol. 12, p. 336), the latter describing the latest type of grain elevators, are articles of great interest to farmers who specialize in cereal crops.

The new system of purchase of grain by the government, which is working admirably in Western Canada, protects the farmer against the speculators who buy standing crops for less than a fair price, and it is to be hoped that some similar plan may be adopted in the United States.

ECONOMICS (Vol. 8, p. 899), by Professor Hewins, CO-OPERATION (Vol. 7, p. 82), and TARIFF (Vol. 26, p. 422), deal with topics related to the

marketing of all agricultural products. The articles on learned societies have an extensive section (Vol. 25, p. 317) on the agricultural societies of all countries.

[Sidenote: The History of Farming]

Agricultural history is, naturally, based upon the history of vegetable life, and the fossil plants described in the article PALÆOBOTANY (Vol. 20, p. 524), long as their appearance preceded that of man, greatly affected the nature of the earth's crust which he was to occupy.

The earliest of all known writings, the Code of Khammurabi, described in the article on Babylonian Law, shows (Vol. 3, p. 117) that agriculture was the subject of careful legislation under the oldest government of which a contemporary record has survived; and the provisions as to the working of land on the "metayer" system, under which the landowner received from the landholder a share of the crops, and as to irrigation, are most explicit and practical. Ancient Egyptian implements of agriculture are fully described (Vol. 9, p. 69) in the article EGYPT, and pictures of them appear on page 72 of the same volume. If the ancient history of farming interests you, it is only necessary for you to turn to the heading "Agriculture," in the Index (Vol. 29), where you will find references to a number of other articles on the early civilizations.

From these articles, as from the historical section of the guiding article AGRICULTURE, and the passages relating to agriculture in many of the 6,292 articles on the histories of races and countries, the reader may learn that agriculture has been the key to all history. The earliest migrations of the human race, as definitely as the comparatively recent development of America, Australasia and the interior of Africa, were

based upon an agricultural impetus. And his reading upon other subjects in the Encyclopædia Britannica will often remind him that the wool and cotton and linen and leather that we wear, the carpets and blankets and sheets in our houses, all originated in farming of one kind or another; while every food that nourishes us, save fish and game, is directly an agricultural product. All the bustle of the great cities, all the wheels that turn in the mills, all the intricate mechanism of industry and commerce, all the world's work and thought and happiness, depend upon the mysterious and inimitable processes by which the brown soil yields green growth. For all the progress science has made, we are no nearer to replacing these processes by any short cut of chemistry than were the first farmers whose husbandry is recorded in history. If all the little roots ceased for one year to do their work in the dark, the human race would hopelessly starve to death.

The alphabetical list of articles at the end of Chapter III of this Guide will make it easy for you to add to this course of reading, choosing for yourself the line that will be most attractive to you. In making your choice, do not forget that plant-life is a subject you cannot study too closely. No matter what crop you make your specialty, you have to _educate_ the plants that produce it to do their work, just as carefully as a teacher trains children. Another fact to keep in mind is that just as a doctor is dealing with organs in the human body which he cannot see, so you are particularly concerned with the roots down in the soil, and the more you know about the way they eat and drink, the better for your farm.

The names of many of the writers of these articles are given in the table of the 1,500 Contributors to the Britannica, beginning at page 949 of Vol. 29 (the index volume); a glance will show you what authoritative positions they occupy and how thoroughly they command your confidence.

[_See list of articles on subjects connected with farming, at the end of Chapter III of this Guide._]

CHAPTER II

FOR STOCK-RAISERS

Stock-raising in the United States was, until quite recent years, under the evil influence of the careless methods which had been handed down from the old days of the range-cattle industry. Chicago men still tell the story of the Chicago banker, afterwards Secretary of the Treasury, who declared, in reply to a request for a loan on the security of range-cattle, that he “would as soon lend money on a shoal of mackerel in the Atlantic Ocean.” The vague possession and the vague methods of breeding and marketing which suggested this comparison did not form the habits of close observation and incessant care which became necessary when land and food began to cost money. The lesson has been learned, and the present conditions of the industry are infinitely better for the country at large. It has been proved that fattening as well as breeding can be successfully undertaken in almost every part of the United States. Even in the North West, the tendency to-day is to turn from exclusive grain growing to a combination of cropping and feeding. Cattle, and also work horses of the right type, for which the demand is always greater than the supply, are yielding fair profits on many of the New England farms which had been neglected for years.

[Sidenote: Staying on the Land]

One of the most encouraging features of the present situation is that the broader distribution of the live-stock industry encourages farm-bred boys to remain at home. It has long been a popular belief that the attraction of the cities lies largely in the facilities for amusement which they offer; but the best class of young men who have left the farms have done so because they did not believe that plowing and sowing and reaping gave enough scope for their intelligence and their initiative. When stock-raising is combined with tillage, there is not only a greater interest in farm life and a greater chance to make general knowledge effective, but there are also better opportunities for a young man to make a small venture of his own while he is still a farm hand. It is certainly true that stock-raising needs the young man who is determined to know something about everything and all there is to know about one thing. To him the articles in the Britannica which are indicated in this chapter should be of the greatest value, for they cover a broad range, and they are written by specialists of the highest authority. They do not profess to teach what can only be learnt in the course of practical experience, but they will make each day's work more interesting and more effective.

[Sidenote: Cattle]

You cannot do better than to begin your reading with the article (Vol. 4, p. 337) on the family of animals to which cattle belong, a family so varied that it includes so small a creature as the hare, and so large a one as the rhinoceros. The article CATTLE (Vol. 5, p. 359), by Professor Wallace and Dr. Fream, begins by reminding you that the idea of cattle owning has always been so closely associated with the idea of wealth that the two words "capital" and "cattle" have the same root, and that our word "pecuniary" is taken from the Latin term for cattle. This

article, illustrated with photographs of the best specimens of bulls and cows of different breeds, deals with Shorthorns, Herefords, Devons, Holsteins, Dutch Belteds, Sussexes, Longhorns, Aberdeen-Angus, Red Polleds, Galloways, Highlands, Kerry's, Dexters, Jerseys and Guernseys, and has a section on the rearing of calves. OX (Vol. 20, p. 398) is chiefly about the origin of domestic cattle. AGRICULTURE (Vol. 1, p. 388) contains information of a more general kind as to practical stock-raising. The best methods of mating are described fully in BREEDS and BREEDING (Vol. 4, p. 487), VARIATION and SELECTION (Vol. 27, p. 906), and HEREDITY (Vol. 13, p. 350), by Dr. Chalmers Mitchell. MENDELISM (Vol. 18, p. 115) will tell you all about the theory which is nowadays the great subject of discussion among experts in breeding. EMBRYOLOGY (Vol. 9, p. 314), by Dr. Hans Driesch, and REPRODUCTION (Vol. 23, p. 116), by Professor Vines, contain the results of the latest investigations, and the article SEX (Vol. 24, p. 747) describes the recent experiments undertaken with the hope that breeders may at some future time be enabled to vary at will the proportion of males and females. TELEGONY (Vol. 26, p. 509) gives you the evidence for and against the belief that offspring are influenced by a previous mate of the dam. FOOD PRESERVATION (Vol. 10, p. 612) and REFRIGERATING (Vol. 23, p. 30) cover the cold shipping and cold storage of beef. LEATHER (Vol. 16, p. 330), by Dr. J. G. Parker, one of the foremost technical experts on this subject, follows hides through the market to their final distribution and industrial uses.

[Sidenote: Horses and Mules]

Notwithstanding the harm that trolley cars and automobiles and mechanically propelled agricultural machines have done to important branches of the horse business, and notwithstanding the competition which American exporters find in Europe from the Argentine ranches,

there is still an active market for farm horses and for stock suited to trucking and light delivery work in cities. You no doubt find, in whatever part of the United States your interests lie, that you need to watch the market very closely, and that you must always be ready to change your plans at short notice. But it is to the quick-witted man who is always prepared to vary his methods that the Britannica offers the greatest practical services. The article on the horse family in general (Vol. 9, p. 720) is very interesting, but you will give more time to the elaborate article HORSE (Vol. 13, p. 712), by Richard Lyddeker, E. D. Brickwood, Sir William Flower, and Professor Wallace. The illustrations are unusually valuable, for instead of following the usual custom of making all the photographs the same size, the Editors of the Britannica showed good sense and originality by making each one to scale. The breeds are separately described, and the sections on feeding and breaking are full of useful hints. The history of the thoroughbred strain is carefully traced, the pedigree of one famous type being shown in a table naming more than one hundred ancestors. The article HORSE-RACING (Vol. 13, p. 726), by Alfred Watson, shows how the sport has influenced breeding, and the description of American trotting goes back to the day when "Boston Blue," in 1818, trotted a mile in three minutes, "a feat deemed impossible" at that period! The English race meetings, in which American owners and jockeys now play so conspicuous a part, are described in special sections, as well as the training at Newmarket. RIDING (Vol. 23, p. 317), and DRIVING (Vol. 8, p. 585), are by practical experts, and TRACTION (Vol. 27, p. 118) contains an interesting table analyzing the draft power of the horse. The section on Arab horses in the article ARABIA (Vol. 2, p. 261) should be read, for it adds to the information, in the articles already named, on the breed that has influenced every variety of horse. MULE (Vol. 18, p. 959) will tell you about the varieties not only in the United States and Mexico, but also in France, Spain, Portugal, Italy, Asia Minor, Syria, Egypt,

Algeria and North China. The section on Hybrids (Vol. 13, p. 713) of the article HORSE deals with all the attempts that have been made to get a perfect type of mule by introducing various strains of blood.

[Sidenote: Sheep and the Wool Market]

SHEEP (Vol. 24, p. 817) contains separate descriptions of the 28 best breeds, discussing their values both for wool and for the meat trade. Breeding, feeding, dipping and lambing are fully treated. Sheepdogs and other breeds useful to the stock-raiser fall under the article DOG (Vol. 8, p. 374). WOOL (Vol. 28, p. 805), by Professor Aldred Barker, is an article in which you will at once be impressed by the splendid thoroughness that is characteristic of the Britannica. It goes to the very foundation of the subject by giving you microscopic photographs, on a scale of 320 to 1, of each of the six great varieties of wool, and explaining the structure of the fibres. The article FIBRES (Vol. 10, p. 309) will enable you to compare another microscopic photograph of wool fibre with similar pictures of silk, flax, cotton, jute, and other textile materials. The article wool deals next with wool-yolk and wool-fat, and then goes on to show why greasy wool is better than wool washed before shearing. Wool classing and sorting are next described, and then scouring. From this point the treatment of wool hardly comes within the jurisdiction of the sheep-man, although he cannot know too much about the qualities of the yarns obtained from different kinds of wool. It is interesting to note in this article that the first fulling mill in America was built at Rowley, Mass., in 1643, only thirty-four years after the first sheep was brought to America, and only twenty-three years after the Pilgrims landed on Plymouth Rock.

[Sidenote: Pigs and Pork]

The article SWINE (Vol. 26, p. 236) deals with the swine family in general, and the article PIG (Vol. 21, p. 594), containing a fine full-page plate, gives a detailed account of the breeds most profitable on the farm, including the Poland-China, the Berkshire, the Duroc, and the Chester White. Eleven breeds in all are particularized. The breeding and fattening of hogs, although it is now successfully followed as a distinct branch of the live-stock industry, must always remain in great part a mere branch of general farming; for the pig's power of thriving on many kinds of food, enables the farmer to utilize produce that cannot advantageously be shipped, and to keep his pigs following his cattle over the fields. Much information will be found all through the article AGRICULTURE (Vol. 1, p. 388). TRICHINOSIS (Vol. 27, p. 266) deals with a disease that has sometimes seriously affected the pork market, and been made the excuse, too, for some very harsh restrictions on American exportation.

[Sidenote: Diseases and Parasites of Live-stock]

You will find in the Britannica (Vol. 28, p. 6) a very full and clear account of the diseases of all domestic animals, by Dr. Fleming and Professor McQueen, with special sections on the maladies of the horse, of cattle, of sheep, and of pigs, and on the parasites that infest them. TUBERCULOSIS (Vol. 27, p. 354) calls for special study, for it is a "disease of civilization" almost unknown among wild animals in their natural state and among the uncivilized races of mankind. The connection between the disease in cattle and its spread among human beings is fully explained in this article. PLEURO PNEUMONIA (Vol. 21, p. 838) deals with the lung disease from which cattle are the only sufferers, RINDERPEST (Vol. 23, p. 348), with the infectious fever which affects both cattle and sheep, and ANTHRAX (Vol. 2, p. 106), with the terribly infectious carbuncles communicated from cattle and sheep to man by the microbes

carried in wool and hides. GLANDERS (Vol. 12, p. 76) describes the form in which this disease of horses and mules afflicts human beings, the symptoms and course of which, in the animals themselves, fall under the subject of horse diseases (Vol. 28, p. 8). The microbe by which this disease is carried is shown in the plate facing one of the pages (Vol. 20, p. 770) of the article PARASITIC DISEASES. FOOT AND MOUTH DISEASE (Vol. 10, p. 617) afflicts cattle, sheep, and pigs, and occasionally human beings.

Among the articles on continents and countries which contain special information on stock-raising, you should not miss the interesting general review of the European live-stock industry in the article EUROPE (Vol. 9, p. 914), the section on live-stock in CANADA (Vol. 5, p. 153), that in ARGENTINA (Vol. 2, p. 465), in AUSTRALIA (Vol. 2, p. 950), and in NEW ZEALAND (Vol. 19, p. 627) The history of stock-raising is fully treated at the beginning of the article AGRICULTURE (Vol. 1, p. 388).

[Sidenote: How to "Even Up"]

When you have read the articles mentioned in the three parts of this chapter on Farming, do not turn away with the idea that you have got from the Britannica all that it can give you to help you in your business. Remember that you have to judge men, as well as live-stock, in order to succeed, and that general knowledge is of the greatest use in doing that. The one sure sign of the kind of man you cannot rely upon is that he talks confidently about subjects of which he really knows little, and the more you yourself know, the more readily you can detect the pretentious people who might make you think too well of them.

If you turn over the pages of this guide, and ask yourself, as you glance at the chapters, in what departments of general knowledge you are

weakest, you will see what courses of reading will do most to make you an “evened up” man, without any weak threads in your intellectual texture. And, whatever you read, do not forget that the Britannica is a book of reference as well as for reading: that you are debasing your mind every time you leave unanswered any question that comes up in the course of the day’s work or talk, or while you are reading your newspaper. A vigorous mind wants an answer whenever it becomes conscious of a question or of a doubt, and if you fail to feed it with the information it asks for, it loses health. Now that you have the Britannica, the food is in the store-room, do not leave it there!

[_See list of articles on subjects connected with stock-raising and other branches of farming, at the end of Chapter III of this Guide._]

CHAPTER III

FOR DAIRY FARMERS

SEE ALSO CHAPTER I, FOR FARMERS, AND CHAPTER II, FOR STOCK-RAISERS

The admirable set of rules for dairy farmers issued by the United States Department of Agriculture begins by telling you to “read current literature and keep posted on new ideas.” And you can easily see that the information on dairy-farming and the many subjects connected with it, supplied by the Britannica, must cover a much broader field of new ideas than can be included in any periodical or dairying manual. The branches of science in which the greatest advance has been made since the beginning of the present century happen to be those that have most

to do with dairying; and the industry itself has been completely revolutionized since the days when cities got their milk from ramshackle cow-sheds in their suburbs, and when butter-making was regarded as one of the “chores” to be done at odd times.

The key article in the Britannica, DAIRY AND DAIRY FARMING (Vol. 7, p. 737), deals with the best milking breeds, the installation, equipment, and management of a dairy farm, the values of various kinds of pasturage and fodder; with the milk trade, with butter-making and cheese-making, with condensed milk, skim milk, and milk powder and with the organization and operation of creameries, cheeseries, and dairy factories in general. Such subjects as soil, grass, hay and other fodder crops fall under Part I of this chapter, and the articles dealing with the breeding and rearing of dairy cattle are mentioned in Part II, “For Stock-Raisers.”

[Sidenote: Dairy-Herd Diseases]

Cattle diseases in general are also covered by the course of reading suggested in Part II; but the dairy farmer has a special interest in contagious mammitis, milk fever, contagious abortion, and cowpox, all of which are described (Vol. 28, p. 10) in the article on VETERINARY SCIENCE. You cannot study too carefully the article on TUBERCULOSIS (Vol. 27, p. 354), for this terrible infection is not only a standing danger to your herd, but also affects the transportation and marketing of milk. Dr. Hennessey, who wrote the article, is an expert of the first rank and, like most other great authorities, is not inclined to encourage the popular exaggeration of the dangers for which newspaper “sensations” are responsible.

[Sidenote: Milk and the Milk Market]

You get to the very foundation of the supply of milk in Professor Parson's and Dr. Edmund Owen's article MAMMARY GLAND (Vol. 17, p. 528), in which the comparative anatomy of the milk yielding organ is fully treated. The article MILK (Vol. 18, p. 451) discusses the chemistry of many kinds of milk and the diseases carried by milk, and deals with the gravest problems of the industry: the difficulty of sterilizing milk, so that tuberculosis and typhoid cannot be carried by it, and the difficulty of sterilizing cream, so that butter may be quite safe, without making the milk less nutritious and the butter less delicate in flavor. The article BACTERIOLOGY (Vol. 3, p. 156), by Professor H. Marshall Ward and Professor Blackman, goes to the root of this whole question of infection. Milk is, on the other hand, used to convey into the human system the "friendly microbes," and the use of soured milk and cheese for this purpose is explained in the articles THERAPEUTICS (Vol. 26, p. 800) and LONGEVITY (Vol. 16, p. 977), which deal with Metchnikoff's system of treatment. PEPSIN (Vol. 21, p. 130) describes the process by which milk is rendered more digestible, and INFANCY (Vol. 14, p. 513) deals with the preparation of milk to be sold for the use of young children. There is so general a demand for prepared milk which is from every point of view wholesome that you will find it worth while to read, in this connection, FOOD (Vol. 10, p. 611), NUTRITION (Vol. 19, p. 920) and DIETETICS (Vol. 8, p. 214).

[Sidenote: Products and Marketing]

BUTTER (Vol. 4, p. 889,) and CHEESE (Vol. 6, p. 22) are brief articles which you should not overlook, although they refer you to the key and article on dairying for details; and OILS contains (Vol. 20, p. 47) an interesting analytical table in which butter is compared with other animal fats. FOOD PRESERVATION (Vol. 10, p. 612) deals with the cold

storage of butter, cheese, condensed milk and milk powder; and REFRIGERATING (Vol. 23, p. 30) with the processes and machinery employed. KOUMISS (Vol. 15, p. 920) describes the milk-wine or milk-brandy prepared by fermenting mare's milk, and the similar product "kerif" made from cow's milk. Although the special developments of dairying in various parts of the world are discussed in the article DAIRY AND DAIRY-FARMING, the articles on individual countries also contain information of value. The section on dairying (Vol. 5, p. 154) in the article CANADA, and the account of co-operative dairying (Vol. 7, p. 87) in DENMARK should not be overlooked.

In reading these articles in Britannica, and thinking of the present conditions of this great business, you will be reminded that dairying is an industry of peculiar importance to the whole people of the United States, not only because of the money made out of it, and not only because it gives hundreds of thousands of men employment on the land instead of in crowded cities, but also because it promises to develop the co-operative action which harmonizes with the best ideals of democracy. The co-operative plants which are beginning to be established by dairy farmers are the only institutions our modern civilization has created in which you find the neighborly spirit that the first American settlers showed in the days when they joined to defend themselves against the Indians. At political meetings, in machine shops and cotton mills and shoe factories, you hear unhappy talk about the relations of capital and labor, about strikes and trusts, about the man on top and the man underneath. But where the farmer's wagons clatter up to the separator platform, there is combination in the best sense of the word. The Britannica article on co-operation says that the word "in its widest usage, _means the creed that life may best be ordered not by the competition of individuals, where each seeks the interest of himself and his family, but by mutual help, by each individual consciously striving

for the good of the social body of which he forms part, and the social body in return caring for each individual; 'each for all, and all for each' is its accepted motto. Thus it proposes to replace among rational and moral things the struggle for existence by voluntary combination for life._”

ALPHABETICAL LIST OF ARTICLES IN THE BRITANNICA ON SUBJECTS CONNECTED
WITH FARMING, STOCK-RAISING AND DAIRYING

(The more important articles have already been mentioned in the preceding pages, but the following list includes many others in which valuable information will be found.)

Aal
Aaron's Rod
Abaca
Abutilon
Acacia
Acanthus
Acaulescent
Acerose
Achimenes
Acinus
Acorn
Acorus Calamus
Acotyledones
Acrogenæ
Adonis
African Lily
Agave

Agrimony
Ailanthus
Alburnum
Alder
Aleurites
Alexanders
Algæ
Algum or Almug
Alismaceæa
Allamanda
Alliaria Officinalis
Allium
Almond
Aloe
Amadou
Amanita
Amaranth
Amaryllis
Amentiferæ
Ammoniacum
Ampelopsis
Anatto
Anemone
Angelica
Angiosperms
Angulate
Anime
Anise
Antirrhinum
Apiculture
Apple

Apricot
Araucaria
Arbor Day
Arbor Vitæ
Arboretum
Arboriculture
Archil
Aristolochia
Aroideæ
Arrowroot
Artichoke
Ascus
Ash
Asparagus
Aspen
Ashpodel
Aspidistra
Aster
Aubergine
Aucuba
Auricula
Autogamy
Auxanometer
Averruncator
Avocado Pear
Axile or Axial
Azalea
Bael Fruit
Balm
Bamboo
Banana

Baneberry
Banksia
Baobab
Barberry
Barley
Bdellium
Bean
Bee
Beech
Beet
Begonia
Benzoin
Betel-nut
Bilberry
Birch
Bird's Eye
Blackberry
Bladder-wort
Boletus
Boll
Borage
Boraginaceæ
Botryis
Bottle-brush plants
Bouvardia
Boxwood
Bracket-fungi
Bramble
Bran
Brazil Nuts
Brazil Wood

Bread-fruit
Breed and Breeding
Bromeliaceæ
Brooklime
Broom
Broom-rape
Bryophyta
Buchu
Buck-bean
Buckthorn
Buckwheat
Bulrush
Bur, or Burr
Burnet
Buttercup
Butter-nut
Butterwort
Cabbage
Cactus
Caducous
Cæspitose
Calabash
Calabash Tree
Calceolaria
Calf
Camellia
Campanula
Candytuft
Cane
Cannon-ball Tree
Capers

Caprifoliaceæ

Capsule

Caraway

Cardamon

Cardoon

Carnation

Carrageen

Carrot

Caryophyllaceæ

Cashew Nut

Cassava

Cassia

Casuarina

Catalpa

Cataphyll

Catha

Cattle

Cayenne Pepper

Ceanothus

Cecropia

Cedar

Celandine

Celery

Centaurea

Centaury

Chantarelle

Chenopodium

Cherry

Chestnut

Chicory

Chive

Chlorosis
Chrysanthemum
Churn
Cicely
Cimicifuga
Cinchona
Cineraria
Cinnamon
Citron
Cleavers
Clematis
Climbing Fern
Cloudberry
Clover
Cloves
Cocoa, or Cuca
Cocculus Indicus
Cock's-comb
Cocoa
Coco de Mer
Coco-nut Palm
Codiæum
Coffee
Colchicum
Coleus
Colleter
Colocynth
Colt's-foot
Columbine
Compass plant
Compositæ

Convolvulaceæ
Copaiba
Copal
Coppice
Coriander
Cork
Corn
Corn-salad or Lamb's Lettuce
Correa
Cotoneaster
Cotton
Cow-tree
Cranberry
Crassulaceæ
Crazy Weed
Cress
Crinum
Crocus
Crowberry
Cruciferæ
Cryptomeria
Cucumber
Cucurbitaceæ
Cumin or Cummin
Cupulliferæ
Cultivator
Currant
Custard Apple
Cyclamen
Cyperaceæ
Cypress

Cystolith
Daffodil
Dairy & Dairy Farming
Dahlia
Daisy
Dame's Violet
Dammar
Dandelion
Daphne
Darlingtonia
Date Palm
Deciduous
Dewberry
Diatomaceæ
Dicotyledons
Dictyogens
Dividivi
Dock
Dodder
Dogwood
Dracæna
Dragons Blood
Drainage
Dropwort
Duck
Duckweed
Dulse
Duramen
Durian
Durra
Earth-nut

Earth-star
Ebony
Economic Entomology
Edelweiss
Eglantine
Elder
Elecampine
Elephant's foot
Elm
Endive
Ensilage
Entada
Ericaceæ
Espalier
Esparto
Eucharis
Eunonymus
Euphorbia
Euphorbiaceæ
Evergreen
Everlasting
Fairy Ring
Fallow
Farm
Farm Buildings
Fennel
Fenugreek
Fern
Fig
Filmy Ferns
Finger-and-toe

Fir

Flail

Flax

Flower

Fool's Parsley

Forage

Forests & Forestry

Forget-me-not

Fork

Foxglove

Freesia

Fritillary

Frog-bit

Fruit

Fruit & Flower Farming

Fuchsia

Fumitory

Fungi

Funkia

Furze

Fustic

Gale

Galls

Gardenia

Garlic

Genista

Gentian

Gentianaceæ

Geoponici

Geraniaceæ

Geranium

Geum
Gillyflower
Ginger
Gladiolus
Glasswort
Glaucous
Gloriosa
Gloxinia
Goat
Golden Rod
Goose
Gooseberry
Goose Grass
Gorse
Gourd
Graft
Grains of Paradise
Gram or Chick-pea
Granadilla
Grass and Grassland
Grass of Parnassus
Grasses
Greenheart
Ground Nut
Groundsel
Guano
Guava
Guelder Rose
Gulfweed
Gum
Gumbo

Gutta Percha
Gymnosperms
Hacienda
Hackberry
Harebell
Harrow
Hawthorn
Hay
Hazel
Heath
Hedges and Fences
Heifer
Heliotrope
Hellebore
Hemlock
Hemp
Hen
Henbane
Henna
Herb
Herbarium
Hickory
Hippeastrum
Hoe
Holly
Hollyhock
Honey
Honey Locust
Honeysuckle
Hop
Horehound

Hornbeam
Horse
Horseradish
Horsetail
Horticulture
Houseleek
Huckleberry
Humus
Huon Pine
Hyacinth
Hydrangea
Hydrocharideæ
Hyssop
Ice-plant
Iceland Moss
Idioblast
Immortelle
Impatiens
India Hemp
Indian Corn
Insectivorous Plants
Iridaceæ
Iris
Irish Moss
Iron-wood
Ivy
Jarrah Wood
Jasmine
Jew's Ears
Job's Tears
Judas Tree

Jujube

Juncaceæ

Juniper

Jute

Kaffir Bread

Kauri Pine

Kerguelen's Land Cabbage

Kumquat

Labiataæ

Labrador Tea

Laburnum

Lac

Lace-bark Tree

Lancewood

Larch

Larkspur

Lattice Leaf Plant

Laurel

Laurustinæ

Lavender

Leaf

Leek

Leguminosæ

Lemon

Lentil

Lettuce

Lichens

Lilac or Pipe Tree

Liliacæ

Lily

Lime or Linden

Liquidambar

Litchi

Lobelia

Loco-weeds

Locust

Loosestrife

Loquat

Lotus

Lucerne

Lupine

Lycopodium

Madder

Magnolia

Mahogany

Maidenhair

Maize

Mallow

Malvaceæ

Mammee Apple

Mandrake

Mangel-wurzel

Mango

Mangosteen

Mangrove

Manila Hemp

Manna

Manures

Maple

Marcescent

Mare's-tail

Marguerite

Marigold
Marjoram
Mastic
Mate
Mattock
Medlar
Melon
Meristem
Mesquite
Merino
Mignonette
Mildew
Milkwort
Millet
Mimosa
Mimulus
Mint
Mistletoe
Moly
Momordica
Moonseed
Moonwort
Moraceæ
Moreton Bay Chestnut
Mucuna
Mulberry
Mushroom
Mustard
Myrobalans
Myrrh
Myrtle

Narcissus
Nard
Nasturtium
Nettle
Nettle Tree
New England Flax
Nightshade
Nut
Nutmeg
Oak
Oat
Okra
Oleander
Oleaster
Olive
Onagraceæ
Onion
Orach or Mountain Spinach
Orange
Orchard
Orchids
Orris-Root
Osier
Ox
Oxalis
Pæony
Palm
Palmetto
Pansy or Heartsease
Papyrus
Paraguay Tea

Parsley
Parsnip
Passionflower
Pea
Peach
Pear
Pellitory
Pennyroyal
Pentstemon
Pepper
Peppermint
Pepper Tree
Persimmon
Petunia
Phlox
Phormium
Pig
Pimento
Pine
Pine-apple
Pin-eyed
Pink
Pistachio Nut
Pistil
Pitcher Plants
Plane
Plantain
Plough and Ploughing
Plum
Poinsettia
Pokeberry

Pollination

Polyanthus

Polygonaceæ

Polypodium

Pomegranate

Pondweed

Poplar

Poppy

Potato

Potentilla

Poultry & Poultry Farming

Primrose

Primulaceæ

Privet

Pteridophyta

Puff-ball

Pumpkin

Purslane

Pyrethrum

Quince

Radish

Ram

Ramie

Ramsons

Ranch

Ranunculas

Ranunculaceæ

Rape

Raspberry

Reaping

Reed

Rhododendron
Rice
Richardia
Robinia
Rocambole
Roller
Root
Rosaceæ
Rose
Rosemary
Rosewood
Rosin or Colophony
Royal Fern
Rubraceæ
Rubber
Ruderal
Rue
Rush
Rye
Sabicu Wood
Safflower
Saffron
Sago
Sainfoin
St. John's Wort
Salsafy or Salsify
Salvia
Sap
Sapan Wood
Sarcocarp
Sarmentose

Sarracenia
Satin Wood
Saxifrage
Saxifragaceæ
Scammony
Scion
Scorzonera
Screw-pine
Scrophulariaceæ
Scythe
Sea-kale
Seawrack
Sedum
Secund
Seed
Sequoia
Service Tree
Sesame
Shaddock
Shallot
Sheep
Sisal Hemp
Skirret
Snake-root
Snapdragon
Snowdrop
Soap-bark
Soil
Solanaceæ
Sorghum
Sorrel

Sowing
Spade
Spanish Broom
Spanish Grass
Spikenard
Spinach
Spruce
Stem
Stink-wood
Strawberry
Strophanthus
Sudd
Sumach
Sundew
Sunflower
Sunn
Sweet Gum
Sweet Potato
Sweet-sop
Swine
Switch-plants
Synanthry
Tallow Tree
Tamarind
Tamarisk
Tea
Teak
Teasel
Terebinth
Thistle
Thorn

Thrashing
Thrum-eyed
Thyme
Tiger-flower
Toadstool
Tobacco
Tomato
Tonqua Bean
Toothwart
Topiary
Traveller's Tree
Tree
Tree-fern
Trowel
Truffle
Tuberoze
Tulip
Tulip Tree
Tumble-weed
Turkey
Turmeric
Turnip
Turnsole
Umbelliferæ
Urticaceæ
Vanilla
Vegetable
Vegetable Marrow
Venus's Fly Trap
Venus's Looking Glass
Veratrum

Verbena
Vetch
Vine
Violet
Walnut
Water-lily
Water-thyme
Wax-tree
Wheat
Whin
Whortleberry
Willow
Willow-herb
Wintergreen
Winter's-bark
Witch Brooms
Witch Hazel
Woad
Wormwood
Yam
Yew
Yucca
Zinnia

CHAPTER IV

FOR MERCHANTS AND MANUFACTURERS: GENERAL AND INTRODUCTORY

[Sidenote: Technical Education for Manufacturer and Merchant]

The article on TECHNICAL EDUCATION in the new (Eleventh) Edition of the Encyclopædia Britannica (Vol. 26, p. 487), written by Philip Magnus, one of the greatest educational authorities in the world, says that:

“The widespread appreciation of the advantages of the higher education among all classes of the American people, and the general recognition among manufacturers, engineers and employers of labour, of the value to them in their own work, of the services of college-trained men, has largely helped to increase the number of students in attendance at the universities and technical institutions.”

A still broader truth is that the men who have learned to think clearly, by whatever study or _reading_ they may have developed that power, possess the greatest of all advantages. As the Britannica article on EDUCATION indicates, the true value of education (not simply school education, but all education) lies as much in the influence which intelligently directed study exerts upon the mind as in the immediate usefulness of the information acquired, and the articles in the Britannica not only supply the most recent and authoritative information, but are so logically arranged, one dove-tailing into another, that they give the reader precisely that _orderly_ view of knowledge which is the foundation of all mental training.

Since all of the series of chapters which immediately follow and which are intended for merchants and manufacturers, deal with commerce and manufactures, it will be for the reader's convenience to begin by dealing with those two subjects in general. But certain branches of industrial and manufacturing knowledge are dealt with in special chapters. The articles on banking and finance are described fully in

this Guide in the chapter _For Bankers and Financiers_, those on insurance in the chapter _For Insurance Men_, and those on law in the chapter _For Lawyers_. Three of the legal articles should, however, be mentioned here, as they are on especially important subjects: SALE OF GOODS (Vol. 24, p. 63), COMPANY (Vol. 6, p. 795), which deals with the laws in various countries regulating corporations, and EMPLOYERS' LIABILITY (Vol. 9, p. 356), on this topic so important in modern industrial law and in the relations between capital and labour.

[Sidenote: Practical Economics for Practical Men]

The broad questions of commercial and industrial policy are discussed in ECONOMICS (Vol. 8, p. 899), by Prof. Hewins; COMMERCE (Vol. 6, p. 766); TRUSTS (Vol. 27, p. 334); MONOPOLY (Vol. 18, p. 733), and TRADE ORGANIZATION (Vol. 27, p. 335), which describes commercial associations in the United States, the work of the consular service, and the organizations in Germany, France, Great Britain and other countries. BOOK-KEEPING (Vol. 4, p. 225), with its up-to-date account of modern accounting methods, card ledgers and loose leaf systems; ADVERTISEMENT (Vol. 1, p. 235), and MERCANTILE AGENCIES (Vol. 18, p. 148) may be named as specimens of the many practical articles on business methods which need not all be enumerated here.

[Sidenote: Imports and Exports]

Much of what you read and hear about the tariff systems of the United States and various other countries and about their influence upon trade is so vague and confusing that you will be delighted with the group of clear, common-sense articles in the Britannica. TARIFF (Vol. 26, p. 422) is by one of the most famous American economists, Prof. Taussig of Harvard, and is a very full and fair discussion of the points in

controversy. PROTECTION (Vol. 22, p. 464) is by Prof. James of the University of Illinois, and FREE TRADE (Vol. 11, p. 89) by William Cunningham. You should read with care CUSTOMS DUTIES (Vol. 7, p. 669); FREE PORTS (Vol. 11, p. 88), and BOUNTY (Vol. 4, p. 324). BALANCE OF TRADE (Vol. 3, p. 235) and TAXATION (Vol. 26, p. 458) are both by Sir Robert Giffen. EXCHANGE (Vol. 10, p. 50), by E. M. Harvey, a partner in one of the largest firms of bullion brokers in the world, deals with the movement of gold. COMMERCIAL TREATIES (Vol. 6, p. 771) is by Sir C. M. Kennedy. Freights are discussed in AFFREIGHTMENT (Vol. 1, p. 302) by Sir Joseph Walton. LIEN (Vol. 16, p. 594), with its section on "Stoppage in transitu," is by F. W. Raikes; SALVAGE (Vol. 24, p. 97), by T. G. Carver, and BLOCKADE (Vol. 4, p. 72), by Sir Thomas Barclay, the great international lawyer in Paris. Marine insurance, indemnity, Lloyds, and other insurance subjects fall under the chapter of this Guide _For Insurance Men_ to which you should refer. Cargo-carrying and merchant shipping are further covered by SHIPPING (Vol. 24, p. 983). This article is by Douglas Owen, honorary secretary and treasurer of the Society of National Research, and author of _Ports and Docks_; it contains information about the great freight carrying lines of the world that can be found in no other book. Railroad freighting is covered by the article RAILWAYS (Vol. 22, p. 819), in which there is a special section (p. 854b) on the new models of American freight cars.

[Sidenote: Manufacturing and Consuming Nations]

In the article UNITED STATES, which contains more matter than a whole book of ordinary size and more information than a dozen ordinary books, the sections (Vol. 27, p. 639) on manufactures and on foreign and domestic commerce, are by F. S. Philbrick, Ph.D. _The internal commerce of the United States, as this article states, is in itself greater than the total international commerce of the world_, and is so far from

exhausting the country's power of production and consumption, that even when coastwise traffic is disregarded, New York is the most active port in the world. A section (Vol. 9, p. 916) of the article EUROPE deals with European commerce in general. The articles on the great manufacturing towns of Europe contain much information as to industries. Great Britain's industries are dealt with in the article UNITED KINGDOM (Vol. 27, p. 691). The industries of England alone are separately treated in a section (Vol. 9, p. 426) of the article ENGLAND. Germany's industries are the subject of sections (Vol. 11, p. 811) of the article GERMANY; and it is interesting to note that although Germany has outranked France in cotton manufactures since Mülhausen, Colmar and other important milling centres of Alsace became German, France has retorted by overtaking and passing Germany in the production of linen. The sections (Vol. 10, p. 785) on foreign commerce in the article FRANCE show her position as in the main a self-supporting country, though only a fourth of the cargoes loaded and discharged in French ports are carried under the French flag. It would be a waste of space to enumerate here the articles on Belgium, Switzerland, Italy and other countries, which you will consult in relation to those of their exports in which you are especially interested; but you should not overlook the article on Japan. The Britannica has done commerce a great service in giving to the world at last a good account of this extraordinary country.

The body of the article JAPAN (Vol. 15, p. 156) is by Capt. Brinkley, long editor of the Japan _Mail_, whose opportunities of seeing Japanese life from the inside have been greater than those of any other foreign observer. Baron Dairoku Kikuchi, President of the Imperial University of Kyoto, a statesman of great experience and authority, contributes to the article a section (Vol. 15, p. 273) dealing with Japan's international position. His remarks upon the commercial morality of the Japanese are so ingenuous and so candid that an extract from them cannot be omitted:

Now when foreign trade was first opened, it was naturally not firms with long-established credit and methods that first ventured upon the new field of business—some few that did failed owing to their want of experience—it was rather enterprising and adventurous spirits with little capital or credit who eagerly flocked to the newly opened ports to try their fortune. It was not to be expected that all or most of those should be very scrupulous in their dealings with the foreigners; the majority of those adventurers failed, while a few of the abler men, generally those who believed in and practised honesty as the best policy, succeeded and came to occupy an honourable position as business men.... Commerce and trade are now regarded as highly honourable professions, merchants and business men occupy the highest social positions, several of them having been lately raised to the peerage, and are as honourable a set of men as can be met anywhere. It is, however, to be regretted that in introducing Western business methods, it has not been quite possible to exclude some of their evils, such as promotion of swindling companies, tampering with members of legislature, and so forth.

The account (Vol. 15, p. 201) by Capt. Brinkley of the curious system of creating branches of Japanese business houses is another part of this article which should not be overlooked.

[Sidenote: Mill Labour]

The proportion of labour cost to the total cost of production is in most industries so great that you cannot study too carefully every aspect of the labour question. The chief articles are LABOUR LEGISLATION (Vol. 16, p. 7), jointly written by the late Dr. Carroll D. Wright, the great American authority on the subject, and Miss A. M. Anderson, Principal

Lady Inspector of Factories to the British government; TRADES UNION (Vol. 27, p. 140); STRIKES AND LOCKOUTS (Vol. 25, p. 1024); WAGES (Vol. 28, p. 229), by Prof. J. S. Nicholson; PROFIT SHARING (Vol. 22, p. 423), by Aneurin Williams and APPRENTICESHIP (Vol. 2, p. 228), by J. S. Ballin. The article EMPLOYERS' LIABILITY (Vol. 9, p. 356), has already been mentioned.

CHAPTER V

FOR MERCHANTS AND MANUFACTURERS OF TEXTILES

[Sidenote: Practical Men Among the Contributors]

The Course of Reading outlined in this chapter will help anyone who has to do with the making or with the buying and selling of textiles, in three ways, at least, each of the greatest importance to him—and possibly in many more. Taking up these three:—In the first place, it will teach him many facts about manufacturing and merchandizing in general, and about dry goods in particular, that he could learn nowhere else, because the scope of the Britannica is broader than that of any other book—or, for that matter, than the scope of any collegiate course can well be. In the second place, the number of distinguished men who have devoted their exclusive attention to the subjects upon which they write, and have given to the Britannica the results of their research and of their experience as practical experts—in many cases, indeed, as successful business men—is far greater than the number of men who form the faculty of any university in the world. The fifteen hundred contributors in fact include no less than 704 connected with the staffs

of 151 different universities, technological and commercial institutes and colleges in twenty countries. The reader thus gets the benefit of contact with the thought of many, of varied, and always of authoritative, personalities. In the third place, the _textile trade is peculiarly an international trade_, the raw materials often traveling from one end of the world to the other before manufacture, and making as long a journey in the finished form, before they reach the consumer, and the international character of the Britannica gives equal weight to the articles which deal with the textiles and with the markets of all countries—a statement which it would certainly not be safe to make about any other book.

[Sidenote: Textile Fibres and their Treatment]

The article FIBERS (Vol. 10, p. 309), by C. F. Cross, whose name has been much before the public in connection with the recent scientific investigation of the subject, compares the fibres yielded by all the vegetable and animal substances used in textiles. The 18 microscopic photographs on the full page plates (facing pp. 310 and 311) and the table of vegetable fibres (p. 311) should be carefully studied.

CELLULOSE (Vol. 5, p. 606) deals with the “body” of cotton, flax, hemp and jute fibres. CARDING (Vol. 5, p. 324) deals with the brushing and combing of fibres. SPINNING (Vol. 25, p. 685) covers both cotton and linen, and it is curious to note from this article that in preparing yarns for the exquisite Dacca muslins one pound of cotton has been spun into a thread 252 miles long; while the article DACCA says that a piece 15 feet by 3 was once woven that weighed only 900 grains. YARN (Vol. 28, p. 906) deals with cotton, woollen and silk yarns. WEAVING (Vol. 28, p. 440), by Prof. T. W. Fox, author of _Mechanics of Weaving_, and Alan Cole, is the first article you should read in a group dealing with processes applied to more than one material. The first section is on the

various combinations of warp and weft, and contains 23 illustrations showing the chief weaving “schemes.” A section on weaving machinery follows, and then one on weaving as an art, illustrated with a number of reproductions of famous specimens of hand-loom work. The whole article is full of practical every-day information of the kind the merchant and manufacturer wants to know. BLEACHING (Vol. 4, p. 49) describes the chemical processes which have expedited the bleaching of cotton, wool, linen and silk, which it used to take all summer to complete. DYEING (Vol. 8, p. 744), by Prof. Hummel, author of *The Dyeing of Textile Fabrics*, and Prof. Knecht, author of *A Manual of Dyeing*, is another of the thorough articles which entitle the Britannica to rank as a great original work on textiles. Every dye is separately treated, and the latest models of dyeing machinery are carefully described. FINISHING (Vol. 10, p. 378) deals with the processes used for cotton, woollens, worsteds, pile fabrics, silks and yarns. TEXTILE-PRINTING (Vol. 26, p. 694) is by Prof. Knecht and Alan Cole, author of *Ornament in European Silks*, and not only describes all the styles of printing, but gives sixty recipes for various shades of colour. The full page plates reproduce fine specimens of early printing. The art of textile-printing “is very ancient, probably originating in the East. It has been practised in China and India from time immemorial, and the Chinese, at least, are known to have made use of engraved wood-blocks many centuries before any kind of printing was known in Europe.”

[Sidenote: Cotton and Cotton Fabrics]

The elaborate article COTTON (Vol. 7, p. 256) begins by discussing the peculiar twist of the hairs on the cotton seed which by facilitating spinning gives cotton its predominant position as a textile material. The section on cultivation, by W. G. Freeman, deals with the soils, bedding, planting, hoeing and picking, then with ginning and baling. A

section on diseases and pests of the cotton plant follows, then a discussion of the improvement of yield by seed selection. The section on marketing and supply is by Prof. Chapman, and his practical study of “futures,” “options,” and “straddles” shows how greatly the movement of prices is affected by speculation and often by artificial manipulation.

COTTON MANUFACTURING (Vol. 7, p. 281) describes the industry in England, that of the United States, with a special section on the recent developments in the two Carolinas, Georgia and Alabama, and also the mills in Germany, France, Russia, Switzerland, Italy and in other countries, including India, China and Japan. It is interesting to note (p. 293) that “Americans were making vast strides in industrial efficiency even before the period when American theories and American enterprise were monopolizing in a wonderful degree the attention of the business world” abroad. As far back as 1875 progress in the United States was so rapid that the production for each operative had increased during the ten years 1865–75, by 100% in Massachusetts as against only 23% in England. One explanation of American success is that the American employer “tries to save in labour but not in wages, if a generalization may be ventured. The good workman gets high pay, but he is kept at tasks requiring his powers and is not suffered to waste his time doing the work of unskilled or boy labour.”

COTTON SPINNING MACHINERY (Vol. 7, p. 301) describes all the machines in great detail and contains a number of full-page plates and other illustrations. MERCERIZING (Vol. 18, p. 150) is another important article.

[Sidenote: Wool, Linen and Silk]

WOOL, WORSTED AND WOOLLEN MANUFACTURES (Vol. 28, p. 805) is by Prof.

Aldred F. Barker. The development in wool production of various countries is first described and then the wool fibre is studied and microscopic photographs reproduced to show the structure of different varieties. A diagram of a fleece shows the qualities obtained from various parts of the animal, ranging from the shoulders, where the finest is found, to the hind quarters. Lamb, hogg and wether wools are compared and the article discusses shearing, classing, sorting, scouring, drying, teasing, burring, mule spinning, combing, drawing and spinning. The centres of the industry are then compared, with details as to the special products of each. The article contains illustrations of a number of machines. Articles dealing with certain sources of wool or of the wool-like hair used in textiles, and with the finished products, are: ALPACA (Vol. 1, p. 721), the history of its manufacture being "one of the romances of commerce;" MOHAIR (Vol. 18, p. 647), which deals with the hair of the Angora goat, familiar from discussions of the Underwood Tariff bill, and dealing with its weaving and the imitations of the cloth; LLAMA (Vol. 16, p. 827); and the articles GUANACO (Vol. 12, p. 649) and VICUGNA (Vol. 28, p. 47), on the two wild animals from whose hair high priced materials, extraordinarily warm and light, are woven.

FLAX (Vol. 10, p. 484) describes the cultivation of the crops which are harvested by being "pulled," roots and all, instead of being cut, the process of separating the capsules from the branches, and the subsequent stages of preparation. LINEN AND LINEN MANUFACTURES (Vol. 16, p. 724), by Thomas Woodhouse, takes up the story where the flax fibre is ready for market and carries it to the point where the yarn is delivered for weaving. The winding, warping, dressing and beaming, and the looms employed, are virtually the same processes and machines that are used for cotton. The article states that the finest linen threads used for lace are produced by Belgian hand spinners who can only get the desired results by working in damp cellars, the spinner being guided by touch

alone, as the filament is too fine for him to see. This thread is said to have been sold for as much as \$72 an ounce.

JUTE (Vol. 15, p. 603) deals with the vegetable fibre which ranks, in its industrial importance, next after cotton and flax and with the processes employed in its manufacture.

SILK (Vol. 25, p. 96) contains illustrations of cocoons and worms, microscopic photographs of fibre, and pictures of the moths which produce wild silk. The section on the fibre and its production and preparation is by Frank Warner, president of the Silk Association of Great Britain and Ireland; and that on the silk trade by Arthur Mellor, a well known manufacturer of Macclesfield, the great British center. The degree of fineness to which silk thread can be spun is stated (Vol. 28, p. 906) to be such that 450,000 yards of thread have been produced from one pound of silk, and this is slightly in excess of the fineness of the Dacca cotton thread already mentioned as producing 252 miles for a pound. But at Cambrai the lace maker's linen thread already described has been made as fine as 272 miles to the pound, and the drawing of platinum wire to the fifty-thousandth part of an inch in thickness (Vol. 28, p. 738) seems hardly more wonderful than this. Spider silk is as valuable as the best qualities of the silkworm product, but spiders are such fierce cannibals that it is necessary to keep each one in a separate cage, and the cost of doing this has prevented the fibre from being generally used (Vol. 25, p. 664). Artificial or "viscose" silk is described in the article CELLULOSE (Vol. 5, p. 609), and is a textile of which the importance is rapidly increasing.

Felting is an even older textile process than weaving, just as weaving, which no doubt originated in basket making (Vol. 3, p. 481) is older than spinning. The article FELT (Vol. 10, p. 245) deals with asphalted

felts used for roofing as well as with the hat felts; and the article HAT (Vol. 13, p. 60) gives further details as to both woollen and fur felts and describes the machinery for hatmaking, which originated in the United States.

Save that gold, silver and other metals are occasionally used in cloth or gauze, ASBESTOS (Vol. 2, p. 714) is the only mineral employed in textiles, and its value for jacketing steam pipes and boilers and for insulating fabrics and fireproofing gives it great importance. RAMIE (Vol. 22, p. 875) is not so largely used in textiles, but experiments in the production of better fibre are being made.

SHODDY (Vol. 24, p. 992) is an article which shows how unfair it is to treat the re-manufacture of "devilled" fabric as an illegitimate if not absolutely fraudulent branch of the textile industry, for really serviceable cloths are woven from it, and masses of poor people who would otherwise be in rags are thus comfortably clad. "Mungo," another re-manufactured cloth, is described (Vol. 28, p. 906) in the article YARN. Pine-apple fibre is described (Vol. 10, p. 311) as of exceptional fineness and is used in yarn cloths of the best quality. The article PINE-APPLE (Vol. 21, p. 625) describes its culture. SISAL HEMP (Vol. 25, p. 158) is used in bagging as well as cordage, and the same is true of PHORMIUM (Vol. 21, p. 471), sometimes called New Zealand flax. Paper pulp yields a yarn which is used in some cheap fabrics as described (Vol. 5, p. 609) in the article CELLULOSE already mentioned.

[Sidenote: Textile Merchandise]

The many varieties of woven cloths are described in the articles already mentioned in the manufacture of cotton, linen, wool, and silk, and in articles on special fabrics. HOSIERY (Vol. 13, p. 788) covers the

textiles that are produced by knitting or looping, and gives an account, with illustrations, of the machinery employed. NET (Vol. 19, p. 412) covers the textiles of which the mesh is knotted.

LACE (Vol. 16, p. 37), by Alan Cole, contains some of the most beautiful full-page plates and other illustrations to be found in the Britannica, and is a very full treatise on the history and the present state of the lacemaking art.

FLANNEL (Vol. 10, p. 480) describes the true flannels made from wool, and FLANNELETTE (Vol. 10, p. 481) the cotton imitations and the new fire-resisting fabrics of this class. DRILL (Vol. 8, p. 580) covers both the cotton and linen tissues sold under this name. CREPE (Vol. 7, p. 379) mentions the curious fact that the Chinese and Japanese makers of soft crepe guard their secret processes, which are still unknown to western manufacturers, so carefully that the different stages of their production are carried on in towns far distant from one another.

CARPET (Vol. 5, p. 392) contains full-page plates of rare specimens and describes pile carpets, flat-surfaced carpets and the printed carpetings.

TAPESTRY (Vol. 26, p. 403) deals with another luxurious branch of the textile industry, and is illustrated with photographs of the finest specimens and with pictures showing the methods of weaving. BROCADE (Vol. 4, p. 620) describes and illustrates this stately class of fabrics. EMBROIDERY (Vol. 9, p. 309) with six full-page plates and SHAWL (Vol. 24, p. 814) deal with other art textiles.

TARTAN (Vol. 26, p. 431) describes the colours and patterns of all Scottish clan tartans. DAMASK (Vol. 7, p. 785) discusses this fine class

of fabrics, the weaving of which is the subject of a special section (Vol. 28, p. 454) of the article WEAVING. The enormous consumption of coarse bags for the packing of raw cotton and of sugar gives importance to the articles BAGGING (Vol. 3, p. 200) and SACKING AND SACK MANUFACTURE (Vol. 23, p. 975). CANVAS (Vol. 5, p. 223) discusses sail cloth and artists' canvas, and TARPAULIN (Vol. 26, p. 430) deals with waterproof covers.

[Sidenote: The Seventy Articles on Special Fabrics]

It is unnecessary to describe one by one the seventy articles on other fabrics and tissues, ranging through the alphabet from Alpaca to Velveteen; but they are all included in the list at the end of this chapter, and all are fully described in the Britannica. COSTUME (Vol. 7, p. 224) is a long and important article, with a full page plate and many other illustrations. The section on dress in general is by T. A. Joyce, of the British Museum staff, that on ancient costumes by H. S. Jones, director of the British School at Rome, and that on modern costume by Oswald Barron, editor of *The Ancestor*. The account of underclothing is of especial interest, as most books on costume altogether neglect this branch of the subject. Another section of this article is on national and official costumes by W. Alison Phillips, principal assistant editor of the Britannica. The study of ceremonial robes is carried into further detail by the article ROBE (Vol. 23, p. 408), with its five richly colored plates, in one of which the judicial robes of the U. S. Supreme Court Justices are shown. Liturgical vestments are dealt with in VESTMENTS (Vol. 28, p. 27) and in a series of articles such as DALMATIC (Vol. 7, p. 776) and ALB (Vol. 1, p. 497).

[Sidenote: Inventors of Textile Machinery and Great Textile Merchants]

Among the biographies which are of interest in connection with textiles are those of ARKWRIGHT, RICHARD (Vol. 2, p. 556), the barber who invented the spinning frame; CARTWRIGHT, EDMUND, (Vol. 5, p. 425), inventor of the power loom; CROMPTON, SAMUEL (Vol. 7, p. 486), inventor of the spinning mule; SALT, TITUS (Vol. 23, p. 87), who created the alpaca industry; STRUTT, JEDEDIAH (Vol. 25, p. 1044), who did much to perfect the manufacture of cotton; and of WHITNEY, ELI (Vol. 28, p. 611), who went from Yale to Savannah to secure a position as school teacher and then, being disappointed, turned his attention to a device for separating the cotton fibre from the seeds and refuse, and invented the gin which has "profoundly influenced American industrial economic and social history." Another name of a great American inventor who individually rendered great services to the textile industry is that of HOWE, ELIAS (Vol. 13, p. 835), who invented the sewing machine. You will also be interested in the lives of successful merchants such as CANYNGES, WILLIAM (Vol. 5, p. 223), the great 15th Century cloth manufacturer who became a clergyman after making a large fortune; MACKINTOSH, CHARLES (Vol. 17, p. 250), who introduced lightweight waterproof garments; WANAMAKER, JOHN (Vol. 28, p. 302), who began life as an errand boy in a book store; FIELD, MARSHALL (Vol. 10, p. 322), who when Chicago was a comparatively unimportant city founded there what has become the finest dry goods store in the world; STEWART, A. T. (Vol. 25, p. 912), who after studying for the ministry in Dublin, immigrated to New York and gradually built up the largest retail store in the city; PEASE, EDWARD (Vol. 21, p. 31), founder of a famous Quaker family of textile manufacturers in England; and CLAFLIN, H. B. (Vol. 6, p. 418), who came from Worcester, Mass., to New York where he for years controlled "the greatest mercantile business in the world." If you turn to the Article WORCESTER (Vol. 28, p. 823) you will note the associations of the locality with Elias Howe, Eli Whitney, Samuel Crompton, already mentioned, L. J. Knowles, another inventor who helped

to perfect the power loom, and Erastus Bigelow, who invented the carpet-weaving machine (Vol. 6, p. 530) and was one of the incorporators of the Massachusetts Institute of Technology. Other lives of successful textile makers and dealers are those of RYLANDS, JOHN (Vol. 23, p. 950), founder of the largest cotton mills in Lancashire; DEXTER, TIMOTHY (Vol. 8, p. 141), the eccentric New England merchant of the 18th Century who beat his wife for not weeping heartily enough at the rehearsal of his funeral; HORROCKS, JOHN (Vol. 13, p. 712), the great English cotton manufacturer who was far ahead of his time and died of brain fever produced by overwork in 1804; WORTH, C. F. (Vol. 28, p. 834), the famous Paris dressmaker who began life as a London draper's apprentice; WHITELEY, WILLIAM (Vol. 28, p. 605), "the Universal Provider," of London; and TATA, J. N. (Vol. 26, p. 448), the great Parsee textile manufacturer.

A LIST OF THE PRINCIPAL ARTICLES IN THE BRITANNICA OF SPECIAL INTEREST
TO MERCHANTS AND MANUFACTURERS OF TEXTILE GOODS

Alb

Alpaca

Apprenticeship

Arkwright, Richard

Artel

Asbestos

Bagging

Baize

Bleaching

Bombazine or Bombasine

Book-keeping

Bounty

Brocade
Buckram
Bunting
Calender
Calico
Cambric
Camel
Canvas
Canynges, William
Carding
Carpet
Cartwright, Edmund
Cellulose
Chasuble
Cheese Cloth
Chenille
Chintz
Claflin, H. B.
Cloth
Clouting
Codilla
Coir
Commerce
Corduroy
Costume
Cotton
Cotton Manufacture
Cotton Spinning Machinery
Crash
Cravat
Crepe

Cretonne
Crompton, Samuel
Dalmatic
Damask
Denim
Dexter, Timothy
Demurrage
Diaper
Die
Dimity
Dowlas
Drill
Duck
Dyeing
Embroidery
Felt
Fibres
Field, Marshall
Finishing
Flannel
Flannelette
Flax
Flock
Floorcloth
Frock
Fustian
Gante
Gauze
Gimp
Gingham
Girdle

Glass Cloth
Guanaco
Gunny
Haberdasher
Hat
Hessian
Holland
Honeycomb
Horrocks, John
Hosiery
Hose-pipe
Howe, Elias,
Huckaback
Jute
Knitting
Lace
Lawn
Linen
Llama
Longcloth
Manila Hemp
Macintosh, Charles
Maniple
Mantle
Matting
Mercantile System
Mercerizing
Merchant
Mohair
Moleskin
Mull

Muslin

Nankeen

Net

Osnaburg

Padding

Pease, Edward

Petticoat

Phormium

Pine-apple

Plaid

Plush

Poplin or Tabinet

Print

Protection

Ramie

Rep

Ribbons

Ring

Robes

Rylands, John

Sacking

Salt, Titus

Salvage

Scarf

Scrim

Shawl

Sheet

Shoddy

Silk

Sisal Hemp

Sleeve

Spinning

Stewart, A. T.

Stocking

Stole

Strutt, Jedediah

Tare and Tret

Tariff

Tarpaulin

Tartan

Tata, J. N.

Tapestry

Technical Education

Textile-printing

Ticking

Tow

Towel

Trousers

Tulle

Twill

Veil

Velvet

Velveteen

Vestments

Vicugna

Wanamaker, John

Weaving

Whiteley, William

Whitney, Eli

Wool, Worsted and Woollen Manufactures

Worth, C. F.

Yarn

CHAPTER VI

FOR MERCHANTS AND MANUFACTURERS OF MACHINERY

[Sidenote: A Change in Public Opinion]

An appreciation of the science of mechanical engineering is so indispensable to the manufacture and sale of machinery that the reader of this Guide might simply have been referred to the chapter _For Engineers_ as covering the industry, if it were not that the Britannica contains (as the list at the end of this chapter shows) a great number of articles dealing with individual machines. The amount of space which the new Britannica devotes to mechanical subjects, and the great number of expert contributors whose collaboration was enlisted in this connection, are significant from more than one point of view. All other general encyclopaedias, including earlier editions of the Britannica itself, seem to have been influenced by the old-fashioned fetish of "pure" scholarship and "pure" science, treating theory as a subject of study much more dignified than the application of knowledge to the practical affairs of life. Until recent days the great universities of such important manufacturing countries as England, Germany and France were almost exclusively devoted to the teaching of philosophy, history, Greek and Latin, mathematics and pure or natural science. The older universities of the United States, too, were for a long time reluctant to recognize the growing importance of technical education, and the necessity, apart from technical education, of giving the general student some knowledge of mechanics. And it is a significant fact that the

Britannica, the first encyclopaedia that has ever been published by a university, should be, although it comes from one of the oldest of all universities, the first to give full recognition to the importance of this department of knowledge.

Men in the machinery trade will welcome this change of attitude in the Britannica, not because they crave a public acknowledgment of the great share of the world's work that they are doing, but because public ignorance of mechanical subjects results in the abuse of machines and in unreasonable complaints against manufacturers when improperly used machinery fails to do its work. A curious illustration of the general disregard of the subject is supplied by the fact—as true of the United States as of England, Germany or France—that representative government is, in practice, chiefly government by lawyers, and that in this age of machinery it is the exception to find in the cabinet which directs the affairs of any country, a single member who has any knowledge of mechanics. The same ignorance is conspicuous in newspaper offices. Even the most dignified dailies seem unable to deal with any news that has to do with machinery without making ridiculous blunders.

[Sidenote: Influence of Automobiles]

Fortunately, the automobile is beginning to stimulate interest in practical mechanics, for no one can attempt to drive his own car, or even to obtain proper service from his chauffeur and from garage workmen, without realizing that he failed, at school, to learn some of the most useful of lessons. Before long the authorities responsible for our public schools may realize that it is absolute barbarism to neglect mechanical teaching as they do; and the new Britannica is already doing good service in stimulating public interest in the subject.

An examination of the articles mentioned in detail in the following summary, and a glance at the long list of articles at the end of the chapter, will show the comprehensiveness with which the Britannica treats all types of machinery. The materials employed are, logically, the first subjects upon which information will be desired.

IRON AND STEEL (Vol. 14, p. 801), by Professor H. M. Howe of Columbia University, is a mine of information about the properties and uses of the different varieties of the indispensable metal of which 50,000,000 tons per annum are employed. In the manufacture of electrical apparatus COPPER (Vol. 7, p. 102) is largely employed, and for this reason alone the article has great value for the manufacturer. Almost as important is ALLOYS (Vol. 1, p. 704). Its chief author, Sir William Chandler Roberts-Austen, is the greatest living authority on alloys, and it is full of interesting facts about new admixtures.

The processes of ANNEALING, HARDENING AND TEMPERING are described in J. G. Horner's article under that title (Vol. 2, p. 70). This authority explains clearly the difference between hardening and tempering and gives valuable advice as to the most efficient methods of hardening. FOUNDRY (Vol. 10, p. 743), also by J. G. Horner, is fully illustrated, and the question of the highest economies of machine moulding are among the practical matters considered. FORGING (Vol. 10, p. 663), with 19 illustrations, discusses fullering, swaging, upsetting, bending, welding, pinching, cutting-off, and die-forging. There is also a separate article, WELDING (Vol. 28), in which the section on _Electric Welding_ is written by Elihu Thomson, who invented the process. A table of energy used in electric welding is added. See also BRAZING AND SOLDERING (Vol. 4, p. 463).

[Sidenote: Manufacturing Methods]

The designer of machinery will find much practical information in DRAWING, *_Drawing Office Work_* (Vol. 8, p. 556), and SUN-COPYING (Vol. 26, p. 93). It is a remarkable fact that prints identical in scale with the originals are now made up to a length of 22 feet.

BEARINGS (Vol. 3, p. 578), illustrated, is written by Professor Dalby of the South Kensington Central Technical College. The article TOOL (Vol. 27, p. 14), by J. G. Horner, is 33 pages in length and has 79 illustrations. The whole subject is completely covered. In the section on *_Machine Tools_* are discussed turning lathes, reciprocating machines, machines with drill and bore holes, milling machines, machines for cutting the teeth of gear wheels, grinding machinery, sawing machines, shearing and punching machines, hammers and presses, portable tools, appliances, wood-working machines, and measurement. In regard to the last subject great advances have lately been made. A thousandth of an inch is now considered a coarse dimension in the machine shop, where gauges within one five-thousandth of an inch are often used. This article is an invaluable manual for the machine-shop, and supplies many hints which should be given to workmen, for, to use the author's words, "a clumsy workman is as much out of place in a modern machine-shop as he would be in a watch-factory." Another article useful to the mechanic is SCREW (Vol. 24, p. 477), with 10 illustrations, by J. G. Horner, with a section on the *_Errors of Screws_*, by the late Henry A. Rowland, the American physicist, whose skill, shown in the construction of dividing engines of extraordinary precision and delicacy, made him famous the world over. See also GRADUATION (Vol. 12, p. 312).

[Sidenote: Engines and Motors]

The articles on the prime-movers are an important and noteworthy part of

the new Britannica. Professor Ewing, of Cambridge University, contributes AIR ENGINE (Vol. 1, p. 443) and STEAM ENGINE (Vol. 25, p. 818), both fully illustrated. The latter has a most interesting preliminary historical account of engines from the aeolipile of Hero of Alexandria (about 130 B.C.) to the steam-turbine, the most modern type of all. The newest forms of internal combustion motors, OIL ENGINE (Vol. 20, p. 35) and GAS ENGINE (Vol. 11, p. 495), are described by Dugald Clerk, inventor of the Clerk cycle gas engine, and the articles are fully illustrated. Under HYDRAULICS (Vol. 14, p. 91) will be found complete information as to the construction of water-pressure engines, water-wheels, turbines, and also pumps. The article is written by Professor W. C. Unwin, and has been universally declared to be the best treatise on the subject that has yet appeared. There is a separate illustrated article WATER-MOTORS (Vol. 28, p. 382), by Professor Beare of Edinburgh University. See also WINDMILL (Vol. 28, p. 710).

Designers and constructors of electrical machinery will be greatly interested in C. C. Hawkins' illustrated article DYNAMO (Vol. 8, p. 764), which explains fully how the dynamo is constructed and gives its history from Faraday's discovery of the principle in 1831. Dr. Louis Bell, of the General Electric Co., writes on MOTORS, ELECTRIC (Vol. 18, p. 910).

[Sidenote: Machinery for Special Purposes]

In hundreds of articles on manufacturing and manufactured products there are excellent descriptions of the machinery employed. COTTON-SPINNING MACHINERY (Vol. 7, p. 301), by Professor Fox, of Manchester University, gives details, with illustrations, of the modern systems of spinning, all founded on the inventions of Paul, Arkwright, Hargreaves and Crompton, while an historical account of primitive machines as well as

much practical information, will be found under SPINNING (Vol. 25, p. 685). WEAVING has a section _Weaving Machinery_ (Vol. 28, p. 443). An account of the special machinery and appliances used in the manufacture of woollens is included in Professor Barker's illustrated article WOOL, WORSTED AND WOOLLEN MANUFACTURES (Vol. 28, p. 805). In HOSIERY (Vol. 13, p. 788) we learn about framework knitting and warp-knitting machines. It is recorded that up to the middle of the 19th century only a flat web could be knitted, and that a circular knitting machine of American origin is the type of machine on which is produced the seamless hosiery of to-day. This was introduced by J. W. Lamb in 1863. ROPE AND ROPE MAKING (Vol. 23, p. 713), by Thomas Woodhouse, of the Dundee Technical College, is richly illustrated with pictures of the most modern type of machinery for the manufacture of fibre and wire ropes. The various machines and apparatus for sugar making are carefully described in SUGAR, _Sugar Manufacture_ (Vol. 26, p. 35). For milling machinery see FLOUR AND FLOUR MANUFACTURE (Vol. 10, p. 548), by George F. Zimmer, author of _Mechanical Handling of Material_. The latest designs in agricultural machines, with illustrations, as well as a history of their development, will be found under PLOUGH AND PLOUGHING (Vol. 21, p. 850), SOWING (Vol. 25, p. 523), HARROW (Vol. 13, p. 27), REAPING (Vol. 22, p. 944), THRASHING (Vol. 26, p. 887), etc. It is a matter of interest that the first successful reaping-machine was invented by a Scotch clergyman in 1826. For machinery used in the modern dairy see DAIRY AND DAIRY PRODUCTS (Vol. 7, p. 750). The germ of the sewing machine dates back to 1755, and the whole story of its development is told in SEWING MACHINES (Vol. 24, p. 744). [Sidenote: A Vast Encyclopaedia of Machinery] The descriptions of machinery of various kinds are continued under such headings as BREWING, _Brewing Operations_ (Vol. 4, p. 506), illustrated; BELLOWS AND BLOWING MACHINES (Vol. 3, p. 705), illustrated; PIN (Vol. 21, p. 615); NEEDLE (Vol. 19, p. 338); TYPOGRAPHY, _Modern Practical Typography_ (Vol. 27, p. 542), illustrated; PRINTING (Vol. 22, p. 350),

illustrated; BOOKBINDING, _Modern Methods_ (Vol. 4, p. 218),
illustrated; TEXTILE PRINTING (Vol. 26, p. 694); ALKALI MANUFACTURE
(Vol. 1, p. 674), illustrated; REFRIGERATING AND ICE MAKING (Vol. 23, p.
30); SILK, _Silk Manufacture_ (Vol. 25, p. 102); LACE, _Machine-made
Lace_ (Vol. 16, p. 44), illustrated; CARPET, _Modern Machinery_ (Vol. 5,
p. 396); LEATHER (Vol. 16, p. 330), illustrated; BICYCLE (Vol. 3, p.
913), illustrated; TYPEWRITER (Vol. 27, p. 501), illustrated; DREDGE AND
DREDGING (Vol. 8, p. 562), illustrated; and PAPER, _Paper Manufacture_
(Vol. 20, p. 727), illustrated.

Biographies of many inventors, designers and builders of machines are
included in the list of articles at the end of the chapter _For
Engineers_ in this Guide, and are therefore omitted in the following
alphabetical summary.

ALPHABETICAL LIST OF THE PRINCIPAL MACHINES AND APPLIANCES DESCRIBED IN THE BRITANNICA AND GENERAL SUBJECTS AND ARTICLES ON MACHINERY

Accumulator

Acetylene Generator

Aerating Apparatus

Aeroplane

Air Brake

Alternators

Alloys

Ammunition Hoist

Anemometer

Annealing, Hardening and Tempering

Archimedes, Screw of Babbitt's Metal

Back-starching Mangle

Bale-breakers
Band-knife Cutting Machine
Barbed Wire Machinery
Barker's Mill
Barrel Organ
Bearings
Beating Machine
Beetling Machine
Bellows and Blowing Machines
Bessemer Convertor
Bevel
Bicycle
Black-ash Revolving Furnace
Blast Furnace
Blocking Machine
Boiler
Bolt-screwing Machines
Book-sewing Machine
Boring Tools
Brake, Hydraulic
Brass
Brazing and Soldering
Breaker Card
Brewing Machinery
Bronze
Bundling Press
Burner
Butter Worker
Butyrometer
Calculating Machines
Calender Machine

Calipers
Calorimeter
Carburetter
Carding Engine
Carpet-making Machinery
Case-making Machine
Casing-in Machine
Centrifugal Machines
Chisel
Chronograph
Chucks
Churn, Mechanical
Clepsydra, or Waterclock
Clock
Coal-cutting Machines
Coal-wedging Machines
Coal-weighing Machine
Coining Press
Comber
Compressed-air Machines
Continuous Press
Conveyors
Copper
Copying Machines
Core-making
Cotton-gin
Cotton-spinning Machinery
Cranes
Crushing Machine
Cultivator
Current Meter

Curvometer
Cutting Machines
Cutting Tools
Damping Machines
Dash Wheel
Depth Recorder
Die
Differential Machines
Dividing Engines
Diving Bell
Doubblers
Dough Kneaders
Dough Dividers and Moulders
Dough Mixers
Drawing-box
Drawing-frame
Drawing-office
Dredgers
Dressing Machine
Drill
Drop Hammer
Drying Machine, Horizontal
Dye-jigger
Dynamo
Dynamometer
Eccentric
Elevators, Lifts and Hoists
Error of Screws
Fans, Rotary
Fire-engines
Flour-sifters

Flying Machines

Fly-shuttle

Forging

Forging Press, Hydraulic

Founding

Friction

Furnace

Gas Engine

Gas Plants

Gas Producers

Gill Frame

Glass-blowing Machine

Glass Press

Graduation

Gravity Stamp

Grinding Machinery

Gyroscope and Gyrostat

Hackling and Spreading Machine

Half-stuff Machine

Hammer

Hand Drill, Electric Harrow

Hat-making Machines

Hay Elevator

Hide Mill, or Double-Acting Stock

Hoe, Horse

Holden Burner

Hydraulic Machines

Hydraulics

Hydro-extractors

Ice-making Machines

Indicator

Injector

Integrators

Iron and Steel

Ironing Machines

Jigger, Hydraulic

Jigs

Jute-Crusher

Jute-Opener

Jute-softening Machine

Kier

Knitting Machines

Labour Legislation

Lace Machines

Lappet Looms

Lathe, Automatic

Laundry Machines

Lever

Lifts, Hydraulic

Linotype Machine

Liquid-air Machine

Lithographing Machines

Loaders

Lock

Locomotives

Loom

Lubricants

Luggage-weighing machine, Automatic

Machine

Machine Gun

Machine Moulding

Mandrel Lathe

Mangling Machines

Manometer

Measuring Machine

Mercerizing Machines

Metal-turning Tools

Meter, Electric

Micrometer

Microtome

Milling Cutters

Milling Machines

Milling Stock

Monotype Machine

Mortising Machine

Motors, Electric

Motor Vehicles

Mowers

Mule, Crompton's

Nail Machines

Needle Machines

Netting Machine

Oil Engine

Oil Muffle Furnace

Opening Machine

Ore-Breaker

Pantograph

Paper-making Machines

Patent logs

Patents

Perpetual Motion

Phonograph

Phosphor Bronze

Pin Machine

Pig-casting Machine

Planimeters

Planing Tools

Plug and Ring Gauge

Pneumatic Hammer

Potter's Wheel

Power-looms

Power Transmission

Price-computing Weighing Machine

Printing Presses

Pulley

Pumps

Purifiers

Rag Boiler, Revolving

Rag-breaking Engine

Rake, Horse

Reaping Machines

Reciprocating Machines

Rectifiers

Reel Paper-Cutter

Reels

Refrigerating Machines

Remontoire

Reverbatory Furnace

Rifling Machine

Ring-frame

Rock Drill

Rod Gauge

Roller Milling Machine

Roller Washing Machine

Rolling, Mill

Rope-making Machines

Rotary Washing Machines

Rounding and Backing Machines

Rusden and Eeles Burner

Salt-cake Furnace, Mechanical

Sawing Machines

Scalpers

Screw cutting

Screw-Gill Drawing Frames

Screw-thread gauge

Screw

Screwing Machine

Scutcher

Separators

Sewing Machines

Shaping Machines

Shearing and Punching Machines

Shuttles

Signal Lever

Silk-reeling Machine

Slide-rule

Slime-tables

Slotter Tools

Sowing Machines

Spinning-jinny

Splitting Machine

Steam Engine

Steam Hammers

Steam Plough

Steam Turbines

Stentering Frame

Still

Stocking Frame

Strength of Materials

Sugar-making Machinery

Sugar Weighing Machine, Automatic

Sulphuric-Acid Plant

Sun Copying

Swathe Turners

Sweep Rake

Table, Mathematical

Tea-weighing Machine

Teasel

Technical Education

Testing Machines

Thermodynamics

Thrashing Machines

Throstle

Tool

Tractors, Steam and Oil

Trepans

Turbine

Turning Lathes

Turret Lathe

Type-setting Machines

Typewriter

Units, Physical

Vacuum brake

Valve

Vanners

Vernier

Voting Machines
Vulcanizer
Washing Machines
Wash Mill
Watch
Water Motors
Water-pressure Engines
Water Wheels
Weaving Machinery
Weighing Machines
Welder, Automatic
Welding
Welding, Electric
Winding Machines
Windmill
Wire-winding Machine
Wiring Machine
Wood-working Machines
Woolen Mule

CHAPTER VII

FOR MERCHANTS AND MANUFACTURERS OF METALS, HARDWARE, GLASS AND CHINA

Elisée Reclus, the great French student of the origins of civilization, says, in the Britannica article FIRE (Vol. 10, p. 399), that “human culture may be said to have begun with fire, of which the uses increased in the same ratio as culture itself.” The industries grouped in the

present chapter all depend upon the curiously diverse effects of heat; the softening and tempering of metals, the hardening of clay and the changes by which sand becomes glass. It is for the reader himself to decide whether he wishes to begin his course of reading by a study of the article HEAT (Vol. 13, p. 135), and the allied articles to which it refers, and thus to understand how temperature plays its dominant part in the most useful of manufacturing processes.

[Sidenote: Knowledge in "Layers"]

It is, indeed, one of the most attractive features of the Britannica that it presents knowledge in _layers_. In text-books, the theoretical and practical aspects of an industry are so interwoven that you cannot separate them. But in the Britannica, if you desire only to examine the finished products of any branch of industry, as you might see them and hear them described at an exhibition or in a manufacturer's sample room, you can turn to articles and sections of articles in which critical comment and elaborate illustrations put clearly before you the varieties of, for example, plated ware, china or glass. Proceeding to the next "layer," you find technical information about the manufacture of these and all other goods; you have been permitted to pass from the sample room into the factory, which is not usually so easy of access. And in the scientific articles you arrive at the very substratum and foundation of knowledge; you have what the experts in the factory could not give you if they would: the clear teaching that only the great masters of science can supply.

The manufacturer, of course, absolutely _needs_ to know all that can be learned about the origin of his materials and the principles upon which his processes are based. But the dealer, in his turn, will be a shrewder buyer, a more convincing salesman and a better manager of the salesmen

under him, if he knows the whole history of his wares, of the ingredients that enter into their composition and of their manufacture. Factory experience is hardly more universal among wholesale men, most of whom begin as clerks, than among retailers, and it is impossible for a business man who has got his foot fairly on the ladder to drop his work and go through an apprenticeship or take a thorough course at a technical college. If, however, he will for a few months devote his spare time to the studies he can pursue, unaided, in the Britannica, the insight he obtains will give a new value to all the knowledge he picks up in the course of his business.

[Sidenote: Physics and Chemistry]

[Sidenote: Some of the Authorities]

The departments of physics and physical chemistry are of course those in which the Britannica's scientific contents especially interest those to whom this chapter is addressed, and the authority of the Britannica in those departments of knowledge is shown by a very striking fact. You may remember that Alfred Nobel, the great Swedish chemist, who made a fortune by the invention and manufacture of dynamite, devoted \$9,000,000 to the establishment of the annual Nobel prizes, to be awarded, irrespective of nationality, for eminence in scientific research and in the cause of peace. In physics and chemistry, _Britannica contributors have won, in eleven years, seven of these prizes_, these winners being: in 1901, Prof. J. H. van't Hoff, of the University of Berlin; in 1902, Prof. Lorentz, of the University of Leiden; in 1904, Lord Rayleigh, Chancellor of the University of Cambridge; in 1906, Sir J. J. Thomson, of the University of Cambridge; in 1909, Prof. Ostwald, of the University of Leipzig; in 1911, Prof. Van der Waals, of the University of Amsterdam. In other words, you find that the scientific committee who

award the Nobel prizes select for these unique distinctions the same men whom the editor of the Britannica selected as contributors. Now apply another test, in connection with the subject matter of this chapter. What is, by general consent, the most exquisitely finished product of any of the industries under discussion in the present section? To this question there can be but one answer: Optical glass. Where is the best glass made? At the Zeiss Works in Jena, Germany. Very well, Dr. Otto Henker and Dr. Eppenstein, both of the scientific staff of the Zeiss Works, wrote the optical articles in the Britannica which deal with the lens and with aberration in lenses. You should therefore remember, in reading the Britannica, that whether you are only going as far as the uppermost layer of knowledge, or reaching down to the very foundations of science, the men whose articles you are reading command the respect that you can pay to them by giving your very closest attention. Do not imagine that because the book contains forty-four million words, it is made to be skimmed; every article in it is condensed; and you cannot derive the fullest benefit from your reading unless you feel, as you would feel if you were fortunate enough to be brought into personal contact with any of these great men, that you have a privilege of which you must make the most.

[Sidenote: Metals]

Other chapters of this Guide also deal in detail with the scientific side of the industries mentioned here; and in examining the groups of industrial articles, those dealing with metals claim first consideration. The article METAL (Vol. 18, p. 198) is devoted to classification only, and would not occupy more than ten pages of this Guide. It contains information as to the physical properties of the metals, including a table in which the specific gravity of each of 42 metals is stated, a table of comparative ductility under the hammer, for

rolling and for wire drawing, a table of elasticities, and other tables showing the ratio of expansion under heat, the melting and boiling points, and the relative thermic and electric conductivity. A section is devoted to the action of chemical agents upon the simple metals.

METALLURGY (Vol. 18, p. 203), and ELECTROMETALLURGY (Vol. 9, p. 232), by W. G. McMillan, lecturer on metallurgy at Mason College, Birmingham, deal with all the methods of smelting ores. Your next reading should be the great article IRON AND STEEL (Vol. 14, p. 801), by Prof. H. M. Howe, of Columbia University, containing as much matter as would fill 110 pages of this Guide. At the beginning of this article Prof. Howe disposes of the much discussed question as to the true distinction between iron and steel, as to which there has been great confusion. Before 1860, the word "steel" was never applied to a metal that could not be hardened by tempering. But when the invention of the Bessemer and open-hearth processes introduced a new class of iron, "which lacked the essential property of steel, the hardening power, yet differed from the existing forms of wrought iron in freedom from slag," the men interested in the new product did not like to call it "wrought iron," which is what it really is, because that name would confuse it with a lower-priced grade of metal. They ought to have coined a new word for it, but they appropriated the name of steel—so that to-day "steel" means either true steel or the low-carbon, slagless variety of malleable iron. The article is divided into 133 sections, so that to analyze its contents would swamp this chapter of the Guide, but the reader will find in it the clearest and most authoritative account of the industry which has yet been published.

Among articles on the commercial metals are COPPER (Vol. 7, p. 102), LEAD (Vol. 16, p. 314), TIN (Vol. 26, p. 995), ZINC (Vol. 28, p. 981), ALUMINIUM (Vol. 1, p. 767), NICKEL (Vol. 19, p. 658), ANTIMONY (Vol. 2,

p. 127), and, on the precious metals, GOLD (Vol. 12, p. 192), SILVER (Vol. 25, p. 112), and PLATINUM (Vol. 21, p. 805).

The article ALLOYS, of which Sir W. C. Roberts-Austen, long chemist of the London Mint, is the chief contributor, with its photomicrographic illustrations, contains not only an account of the alloys already generally used in the metal industries, but also practical information as to the experiments which have been made recently with some of the newly discovered rare earths. In the article METALLOGRAPHY (Vol. 18, p. 202), by the same specialist, the microscopic examination and photography of metals and alloys is described.

Among articles on the metallic compounds are BRASS (Vol. 4, p. 433), in which "Dutch metal," "Mannheim gold," "similor" and "pinchbeck" are described; BRONZE (Vol. 4, p. 639), which deals with steel bronze, phosphor bronze, and other combinations; FUSIBLE METAL (Vol. 11, p. 369) is an important compound. PEWTER (Vol. 21, p. 338), by Malcolm Bell, author of *Pewter Plate*, etc., is of historical interest, and of value to the dealer or collector, while he who wishes to distinguish between the older and the more modern electroplated ware is referred to the article SHEFFIELD PLATE (Vol. 24, p. 824), also by Malcolm Bell. ELECTROPLATING (Vol. 9, p. 237) describes the art that put an end to the Sheffield plate industry. Other methods of coating metals are given under GALVANIZED IRON (Vol. 11, p. 428), TIN PLATE AND TERNE PLATE (Vol. 26, p. 1000), and GILDING (Vol. 12, p. 13). The art of making gold-leaf is described in GOLDBEATING (Vol. 12, p. 202).

In regard to manufacturing processes there are the separate articles: FORGING (Vol. 10, p. 663), with 19 illustrations; FOUNDRY (Vol. 10, p. 743), with 11 illustrations; ANNEALING, HARDENING AND TEMPERING (Vol. 2, p. 70), and BRAZING AND SOLDERING (Vol. 4, p. 463). These four articles

are by J. G. Horner. And see WELDING (Vol. 28, p. 500), also by Mr. Horner, with a section on _Electro-Welding_, by Elihu Thomson, inventor of the process of electric welding and expert for the General Electric Co. The article TOOL (Vol. 27, p. 14), another of Mr. Horner's valuable contributions, has 79 illustrations and possesses special interest for the manufacturer of metal-ware as well as the dealer in hardware.

[Sidenote: Metal-Ware]

Coming now to the production of metal wares, the article METAL-WORK (Vol. 18, p. 205), beautifully illustrated, is the work of three noted experts. The late J. H. Middleton, Slade Professor of Fine Art, Cambridge University, writes on _Methods of Manipulation in Metal Work_ and tells of the metal work of Greece, Italy, Spain, Germany, France, England, Persia and Damascus. J. S. Gardner, an expert metal worker, deals with _Modern Art Metal Work_, and J. G. Horner contributes the section on _Industrial Metal Working_, in which he deals with _Plater's Work_, _Coppersmith's Work_, _Raised Work_, _Cast Work_, _Methods of Union and Protection of Surfaces_. In connection with the last mentioned subject, see also JAPANNING (Vol. 15, p. 275), LACQUER (Vol. 16, p. 53), and PAINTER-WORK (Vol. 20, p. 457). Further information about lacquering, with valuable formulas, will be found in the article JAPAN (Vol. 15, p. 188). Some of the ornamental forms of metal work are described in REPOUSSÉ (Vol. 23, p. 108), by M. H. Spielmann, formerly editor of _The Magazine of Art_; INLAYING (Vol. 14, p. 574), and DAMASCENING (Vol. 7, p. 783). See also GRILLE (Vol. 12, p. 596).

PLATE (Vol. 21, p. 789), an article by H. R. Hall, of the British Museum, H. Stuart Jones, director of the British School at Rome, and E. A. Jones, author of _Old English Gold Plate_, etc., is a concise, complete hand-book on work in silver and gold of any class other than

those of personal ornaments and coins. It is profusely illustrated with plates and text-cuts, showing many exquisite models; and the reader can master the details of style in different periods and countries. The subjects of the assay of gold and silver plate and hall-marks are discussed, the former being treated more fully in ASSAYING (Vol. 2, p. 776), by A. A. Blair, chief chemist of the U. S. Geological Survey. The article ROMAN ART, by H. Stuart Jones, has a section devoted to _Work in Precious Metals_ (Vol. 23, p. 483).

CUTLERY (Vol. 7, p. 671) is one of the articles pertaining specifically to hardware manufacture and trade, in which general processes of manufacture are described; and of allied interest are KNIFE (Vol. 15, p. 850), FORK (Vol. 10, p. 666), SPOON (Vol. 25, p. 733), SCISSORS (Vol. 24, p. 407), SHEARS (Vol. 24, p. 815), RAZOR (Vol. 22, p. 937), CHAFING-DISH (Vol. 5, p. 800), NAIL (Vol. 19, p. 153), AXE (Vol. 3, p. 67), HAMMER (Vol. 12, p. 897), CHISEL (Vol. 6, p. 247), WIRE (Vol. 28, p. 738), and BARBED WIRE (Vol. 3, p. 384). Articles describing all forms of agricultural implements will be found under their respective headings.

[Sidenote: Glassware]

GLASS (Vol. 12, p. 86) is most complete in its consideration of the entire subject. The introductory section by H. J. Powell, of the Whitefriars Glass Works, London, author of _Glass Making_, and W. Rosenhain, of the National Physical Laboratory, London, deals with the manufacture of optical glass, blown glass and mechanically-pressed glass. The necessary qualities of each kind are stated and the newest processes of manufacture described, with full information about materials. The second part of the article is devoted to the _History of Glass Manufacture_, by Mr. Powell and Alexander Nesbitt, who wrote the

well-known Introduction to the South Kensington Museum Catalogue of Glass Vessels. Egyptian, Assyrian, Roman, Venetian, Bohemian and Oriental glass, as well as the modern types, are exhaustively described. The article is splendidly illustrated. DRINKING VESSELS (Vol. 8, p. 580), by Dr. Charles H. Read, of the British Museum, describes old forms of glass cups and goblets. It is most valuable for its information in regard to styles of different countries and periods, and the illustrations show many types.

Stained glass is the subject of the separate article GLASS, STAINED (Vol. 12, p. 105), illustrated, by the late Lewis F. Day, author of Windows, a Book about Stained Glass. It is both historical and descriptive in its nature, deals with painted and stained glass, contains a table of examples of important historical stained glass, and treats of the latest progress in the art, including the productions of La Farge and L. C. Tiffany in this country. The art of fitting and setting of glass is described in GLAZING (Vol. 12, p. 116), illustrated, by James Bartlett. Here we learn about the setting of window glass, the use of glass in decoration, systems of roof glazing and the use of wire glass.

Full information about glass for optical purposes will be found under LENS (Vol. 16, p. 421), illustrated, by Dr. Otto Henker, of the Carl Zeiss Factory, Jena, Germany; LIGHTHOUSE, Optical Apparatus (Vol. 16, p. 633), illustrated, by W. T. Douglass, who erected the Eddystone and Bishop Rock lighthouses, and Nicholas G. Gedy, chief engineer to the Tyne Improvement Commission; TELESCOPE, Instruments (Vol. 26, p. 561), illustrated, by H. Dennis Taylor and Sir David Gill; PHOTOGRAPHY, Photographic Objectives or Lenses (Vol. 21, p. 507), illustrated, by James Waterhouse; SPECTACLES (Vol. 25, p. 617).

[Sidenote: Chinaware, Pottery and Porcelain]

To those engaged in the china ware, pottery or porcelain manufacture and trade, the great article CERAMICS (Vol. 5, p. 703) will prove a revelation. It is the joint product of a number of experts, both practical and artistic, including William Burton, chairman, Joint Committee of Pottery Manufacturers of Great Britain, Henry R. H. Hall and Robert Lockhart Hobson, both of the British Museum, and A. Van de Put and Bernard Rackham, both of the Victoria and Albert Museum. It is 85,000 words in length and contains over a hundred beautiful illustrations, including six plates in colour. It deals fully with the artistic and economic phases of the subject, the methods of manufacture, the different varieties of ceramics, their history, decoration, etc. Japanese ceramics are treated separately in JAPAN, _Ceramics_ (Vol. 15, p. 183), illustrated, by the late Capt. Frank Brinkley.

CLAY (Vol. 6, p. 472), by Dr. J. S. Flett, describes the occurrence, composition and properties of the various clays used in ceramics.

TERRACOTTA (Vol. 26, p. 653), illustrated, by William Burton and H. B. Walters, of the British Museum, deals with the artistic use to which baked clay is put, while TILE (Vol. 26, p. 971), illustrated, also by William Burton, has great practical value for the present-day manufacturer.

KAOLIN (Vol. 15, p. 672), by F. W. Rudler, of the Museum of Practical Geology, London, deals specifically with china clay and its preparation for the market. GILDING (Vol. 12, p. 13) contains material on the subject of the gilding of pottery and porcelain, and PAINTING has a section _Painting with Coloured Vitreous Pastes_ (Vol. 20, p. 484), by Prof. G. B. Brown, of Edinburgh University, which describes the use of

these pastes in ceramics. ENAMEL (Vol. 9, p. 362), illustrated, by Alexander Fisher, yields equally valuable information for those concerned with the decoration of china.

In MURAL DECORATION, by Walter Crane and William Morris, there is a section devoted to _Wall-Linings of Glazed Brick or Tiles_ (Vol. 19, p. 17). Material of great archaeological interest relating to earthenware, etc., will be found in such articles as AEGEAN CIVILIZATION (Vol. 1, p. 245), illustrated, by D. G. Hogarth, of the Ashmolean Museum, Oxford; CRETE, _Archaeology_ (Vol. 7, p. 421), illustrated, by Arthur J. Evans, the famous Cretan explorer, and GREEK ART (Vol. 12, p. 470), illustrated, by Percy Gardner, the classical archaeologist.

The following is a partial list in alphabetical order of articles and subjects in this field treated in the Britannica.

ALPHABETICAL LIST OF ARTICLES AND SUBJECTS IN THE ENCYCLOPAEDIA
BRITANNICA OF SPECIAL INTEREST TO THOSE IN METAL, HARDWARE, GLASS AND
CHINA MANUFACTURE AND TRADE

Adze

Aegean Civilization

Ainmuller, M. E.

Alloy Steels

Alloys

Aluminium

Amphora

Andiron

Annealing, Hardening and Tempering

Antimony

Anvil

Armour Plate

Arms and Armour

Arretine Ware

Assaying

Auger

Awl

Axe

Barbed Wire

Banko Ware

Basin

Beaker

Belleek Ware

Bidri Work

Binocular Instrument

Biscuit

Bismuth

Bizen Ware

Bohemian Glass

Bottle

Bow Ware

Bradawl

Brass

Brasses, Monumental

Brazier

Brazing and Soldering

Bronze

Byzantine Glass

Caffieri, Jacques

Candlestick

Capo di Monte Ware

Capronnier, J. B.

Cast Work

Cellini, Benvenuto

Ceramics

Chafing Dish

Chalice

Chelsea Ware

China

China, _Art_

Chinese Porcelain

Chisel

Churn

Clay

Cookworthy, William

Coperta

Copper

Coppersmith's Work

Crete

Crown Glass

Cup

Cutlery

Cultivator

Damascening

Damask Steel, or Damascus Steel

Damascus Ware

Delft Ware

Della Robbia

Derby Ware

Doulton, Sir Henry

Dresden, or Meissen, Ware

Drinking Vessels

Dwight, John
Electrolier
Electroplating
Electrum
Enamel Painting
Etruscan Ware
Faience
Fender
File
Finiguerra, Maso
Fireback
Firing
Fire-irons
Flint Glass
Fork
Forging
Founding
Fusible Metal
Galvanized Iron
German (or Nickel) Silver
Gilding
Gimlet
Girandole
Glass
Glass, Ancient
Glass-blowing Machine
Glass Cutting and Engraving
Glass, Painted
Glass-press
Glass, Stained
Glazes

Glazing

Goblet

Gold

Gold and Silver Thread

Goldbeating

Gouge

Gombroon Ware

Gouthière, Pierre

Graffito Ware

Grate

Greek Art

Grille

Hall-marks

Hammer

Harrow

Hatchet

Henri-Deux, Oiron, or St. Porchaire Ware

Hispano-Moresque Ware

Hizen Ware

Hoe

Horseshoes

Ingot

Inlaying

Invar

Iron and Steel

Iron Work

Izumo Ware

Japan, _Ceramics_

Japanning

Jug

Kaolin

Kashi

Kiln

Kioto Ware

Knife

Kuang-Yao

Kuft Work

Kutani Ware

Lacquer

La Farge, John

Lang-Yao

Latten

Lead

Lens

Lighthouse Apparatus, Optical

Lock

Lubricants

Lustred Ware

Majolica

Meissonier, J. A.

Medal

Metal

Metallography

Metallurgy

Metal Work

Mezza Majolica

Minoan, or Kamares, Ware

Mirror

Monstrance

Morel-Ladeuil, L.

Mural Decoration

Nail

Needle

Nickel

Niello

Ormolu

Owari Ware

Painter-work

Palissy, Bernard

Palissy Ware

Painting

Pen

Persian Pottery

Pewter

Photographic Objectives or Lenses

Pin

Pitcher

Plaque

Plate

Plated Ware

Plate-glass

Plater's Work

Platinum

Plough

Porcelain

Pot-hook

Potteries, The

Potter's Marks

Potter's Wheel

Pottery

Protection of Surfaces

Raised Work

Rake

Razor

Reaper

Repoussé

Roman Art

Rookwood Ware

Royal Copenhagen Ware

Royal Worcester Ware

Salt Glaze

Salver

Samovar

Saracenic Glass

Satsuma Ware

Saw

Scissors

Sconce

Screen

Screw

Scythe

Sèvres Porcelain

Shears

Sheet Glass

Sheffield Plate

Shovel

Shuttle

Sieve

Silver

Smith

Solder

Spade

Spectacles

Spit

Spoon

Spade

Stone Ware

Table-ware

Takatori Ware

Tanagra Figures

Tankard

Tazza

Telescopic Instruments

Terracotta

Thrasher

Tiffany, C. L.

Tiffany Glass

Tiles

Tin

Tinker

Tin and Terne Plate

Tongs

Tool

Torchère

Tray

Tripod

Trivet

Tube-making, Glass

Turkish Pottery

Tweezers

Trowel

Vacuum Cleaner

Vase

Venetian Glass

Wedgewood, Josiah

Wedgewood Ware

Whitefriars Glass

Wire

Wired Glass

Yatsushiro Ware

Yi-Hsing-Yao

Zinc

CHAPTER VIII

FOR MERCHANTS AND MANUFACTURERS OF FURNITURE

[Sidenote: Art and Industry]

When you think of your home, making a picture in your mind of the familiar surroundings associated in your memory with your greatest pleasures, you are really thinking of furniture. Tradition makes the dwelling itself the tangible symbol of home, because when a primitive tribe ceased to be wanderers, the walls that excluded wild beasts and inclement weather and gave privacy were conspicuous evidences of a change for the better. But in our higher civilization our way of thinking has changed. Nothing seems to us more desolate than the bleak surfaces and harsh angles of an unfurnished house. Colour and softness and the curved lines which we instinctively love because they suggest softness come into the dwelling with furniture, and culture has progressed so far that the chair or bed must be a delight to the eye as well as to the weary limbs, that the dinner table and the bookcase must be so designed as to enhance the satisfaction we find in refreshing body

and mind. You would not get so much pleasure as you do from your Encyclopaedia Britannica if its paper and print and pictures and the colour and texture of the bindings did not make it one of the chief adornments of your home; the volumes might be just as useful in a less pleasing guise, but you would not feel the same affection for the book.

[Sidenote: Form and Embellishment]

To satisfy the spirit of home-love and house-pride in the making of furniture is an art, and the idea that furniture can only be artistic when it is made by hand, from a design that is to be used but once, is as nonsensical as it would be to say that a beautiful etching is not true art because a press produces it and others like it. "Fine art is everything which man does or makes _in one way rather than another_ ... in order to express and arouse emotion ... with results independent of direct utility." These words from Sir Sidney Colvin's delightful Britannica article FINE ARTS (Vol. 10, p. 361), and another passage (p. 370), in which he speaks of "the artificers who produce wares primarily for use, in a form, or with embellishments, that have the _secondary virtue of giving pleasure_" might well be quoted to the supercilious and superficial critic who condemns every product which machinery has brought within the reach of the less fortunately situated. Furniture, made in one form rather than another, because that one form gives greater pleasure, is artistic furniture whether it is made of machined pine chemically stained or of handworked and hand-polished rosewood. The manufacturer and dealer who ingeniously minimize the cost of production and distribution are benefiting the public just as truly as did Thomas Chippendale, "at once an artist and a prosperous man of business," or Thomas Sheraton, "the great artistic genius who lived in chronic poverty." The adaptation and variation of their ideas, under modern conditions of manufacture, have given pleasure to tens of thousands for

every one whose home was enriched by the original products.

[Sidenote: Related Subjects]

We have, then, in the furniture business, the combination of an art with an industry of the most practical and useful kind, and this art is one which does more than any other to “express and arouse” the home-cherishing emotions which solidify family life. The principles which underlie architecture, sculpture, painting, metal work, embroidery and the weaving of patterns all affect the design of furniture, since its contours and surfaces are obtained by the application of the structural and decorative laws of all of them, and it might therefore be said that the only course of reading in the Britannica which could fully justify the title of this chapter would be one which covered all these diverse fields. The reader can, however, with the assistance of other chapters of this Guide, easily find his way to the Britannica’s articles on each of these allied subjects, and an indication of the articles dealing specifically with furniture will at any rate serve his primary purpose.

[Sidenote: “Art Nouveau” School]

The keystone article FURNITURE (Vol. 11, p. 363) is by James Penderel-Brodhurst, one of the greatest of living authorities, to whom many of the subsidiary articles are also due. The 37 illustrations on plate paper include two large views of the most famous and resplendent piece of furniture ever constructed, the cylinder desk, now in the Louvre Museum in Paris, made for Louis XV by a number of “artist-artificers,” the chief among them Oeben and Riesener, with bronze mounts by Duplessis, Winant and Hervieux. The article explains the scanty attention paid to furniture in ancient Egypt, Rome and

Greece, and throughout the Middle Ages in Western Europe, as due to the routine of life in centuries during which people spent their days in the open air, and went to bed as soon as it was dark, therefore needing but few household appliances. The Renaissance was the first era of sumptuous and elaborately varied furniture; and it was not until the 18th century that the art of the cabinet-maker was fully developed. The English periods of Queen Anne and early Georgian craftsmanship and the reigns of Louis XV and Louis XVI brought the development to its high-water-mark. Since then, there has been no really new departure except the “art nouveau” school, which professed to be free from all traditions and to seek inspiration from nature alone. The revolution which was thus attempted was not successful, and the permanent influence of the movement will, in all probability, be less notable for its effect upon style than for the very great service it rendered in reviving the use of oak. Lightly polished, fumed or waxed, this wood, which was so long neglected, is the most effective that can be employed at moderate cost.

[Sidenote: Beds]

The oldest and most indispensable of all furnishings is treated in the article BED (Vol. 3, p. 612). The Egyptians had high bedsteads to which they ascended by steps, and the Assyrians, Medes and Persians followed the same custom. The Greek bed had a wooden frame, with a board at the head, and bands of hide laced across, upon which skins were laid. At a later period, as vase-paintings show, the Greeks used folding beds. Another ancient application of an idea commonly supposed to be of modern origin is found in the Roman use of bronze beds, and metal is so much more sanitary than wood for this purpose that it seems strange it was afterwards discarded for many centuries. The bed of the Emperor Eliogabalus was of solid silver, with counterpane and hangings of purple embroidered in gold. In Pompeii wall-niches for beds, like those still

used in Holland, are found, and were apparently closed by sliding partitions as well as by curtains. To our modern ideas, this arrangement seems to have been disgustingly devoid of ventilation, but the four-poster, with its “tester” roof and its curtains, which was widely used until the middle of the 19th century, was not much better. Mattresses developed very slowly, for in the 18th century pea-shucks and straw were the stuffing materials employed in houses of prosperous people, and hair had not come into use. The article gives a full and interesting account of the quaint custom, instituted by Louis XI of France, and followed by many of his royal successors, of a sovereign remaining in bed while he received the visits of his ministers and courtiers.

[Sidenote: Chests and Chairs]

The chair, to us the commonest of objects, did not come into general use until, as the articles BENCH (Vol. 3, p. 715) and STOOL (Vol. 25, p. 967) indicate, these two had long been the usual seats. The CHEST (Vol. 6, p. 106) was also used as a seat, and was the original form of wardrobe before hanging space and drawers were provided. The ecclesiastical chests, of great length in order that they might contain, without folding, church vestments stiff with embroidery, are the most ornate of all the models of furniture which have been preserved from the 13th and 14th centuries. The article CHAIR (Vol. 5, p. 801) shows that chairs were everywhere uncommon until the middle of the 16th century; and it was not until the 17th was well advanced that upholstery began to be employed for them. The typical Louis XVI chair, with its oval back and ample seat, descending arms, round-reeded legs and gay tapestry was the most beautiful and elaborate model that has ever been devised. But it was the original Chippendale design and the still lighter patterns of Hepplewhite, Sheraton and Adam that gave us the slender, compact and

easily moved chairs which will always be the more numerous. It is interesting to observe that the revolving chair, commonly regarded as an office convenience of modern origin, has a pedigree of no less than four centuries.

[Sidenote: Bookcases and Desks]

It would seem that the old English makers of furniture went somewhat astray when they gave themselves the general designation, still surviving, of “cabinet-makers”; for we learn from the article CABINET (Vol. 4, p. 918) that the elaborate cabinets which have come down to us from the 16th, 17th and 18th centuries are almost invariably of Italian, Dutch and French origin, and it was in other branches of work that the English were most successful. THE CUPBOARD (Vol. 7, p. 634) was used to contain books long before the BOOKCASE (Vol. 4, p. 221) had assumed a distinct form, and in the earlier bookcases the volumes were either placed on their sides, or, if upright, were ranged with their backs to the wall and their edges outwards. Until printing had cheapened books, it was not the custom to mark the title on the back, and the band of leather which closed the volume, like the strap on an old-fashioned wallet, bore the inscription. Sheraton’s satinwood bookcases were among the most elegant of all his pieces. The DESK (Vol. 8, p. 95) about the year 1750 had assumed the form which is now described as a library table—a flat top with a set of drawers on each side of the writer’s knees, when its vogue was interrupted by the invention of the cylinder-top desk. At first the cover was a solid piece of curved wood, but the “tambour,” or series of slats mounted on canvas proved more serviceable; and the American roll-top desk is now exported to all parts of the world. Other articles dealing with individual pieces of furniture are WARDROBE (Vol. 28, p. 323), SIDEBBOARD (Vol. 25, p. 38), DRESSER (Vol. 8, p. 577), CHEFFONIER (Vol. 6, p. 22), CRADLE (Vol. 7, p. 360),

BUFFET (Vol. 4, p. 757), and MIRROR (Vol. 18, p. 575).

[Sidenote: Technical Articles]

Of the more technical articles TIMBER (Vol. 26, p. 978) shows the comparative advantages of all the varieties of wood used for furniture; and, as the list at the end of this chapter shows, there is a separate article on each kind. TOOL (Vol. 27, p. 14), by J. G. Horner, is of great importance. It would fill 75 pages of this Guide, and contains 79 illustrations. The furniture maker will find in it complete information about all the hand tools and machine tools used in the industry. JOINERY (Vol. 15, p. 476), by James Bartlett, describes, with practical diagrams, every variety of joint and dovetail. Sound guidance for the workshop will be found in GLUE (Vol. 12, p. 143), PAINTER-WORK (Vol. 20, p. 457), LAC (Vol. 16, p. 35), LACQUER (Vol. 16, p. 53), in regard to which there is also information in the article JAPAN (Vol. 15, p. 188), FRENCH POLISH (Vol. 11, p. 154), WEAVING, _Industrial Technology_ (Vol. 28, p. 440), DYEING (Vol. 8, p. 744), by Profs. J. J. Hummel and Edmund Knecht; REP (Vol. 23, p. 105), TAPESTRY (Vol. 26, p. 403), with numerous illustrations, by A. S. Cole; SILK, _Manufacture_ (Vol. 25, p. 102); PLUSH (Vol. 21, p. 857), VELVET (Vol. 27, p. 979), MARBLE (Vol. 17, p. 676), by J. S. Flett; ONYX (Vol. 20, p. 118); and ALABASTER (Vol. 1, p. 466).

[Sidenote: Decoration and Ornament]

[Sidenote: Biographical Articles]

Although wood, ivory, precious stones, bronze, silver and gold have been used from antiquity for the decorations of furniture, the modern maker will be more concerned with WOOD-CARVING (Vol. 28, p. 791), illustrated,

by F. A. Crallan, author of *_Gothic Wood-carving_*. In this article materials and methods are described, and there is much information as to the domestic use of wood-carving. The article will be most valuable to manufacturers and dealers who have to do with church fittings. GILDING (Vol. 12, p. 13) and CARVING AND GILDING (Vol. 5, p. 438) impart knowledge of a practical nature as to these processes. The art of inlaying is described in MARQUETRY (Vol. 17, p. 751) and BOMBAY FURNITURE (Vol. 4, p. 185); see also VENEER (Vol. 27, p. 982). Materials other than wood used for inlaying are described, as, for example, PEARL (Vol. 21, p. 25) for pearl and mother of pearl; IVORY (Vol. 15, p. 92), LAPIS LAZULI (Vol. 16, p. 199), TORTOISESHELL (Vol. 27, p. 71), BRASS (Vol. 4, p. 433), etc. The mention of the last two materials naturally suggests the name of BOULLE and the Britannica's biography of that artist. Such biographies, as anyone interested in the subject knows, are most difficult to find, and they are included in much detail in the new Britannica. BOULLE (Vol. 4, p. 321) was the most distinguished of modern cabinet-makers before the middle of the 18th century; and, beginning with that date, both France and England produced a number of men whose renown is scarcely less than that of the great painters, sculptors, architects or musicians of the period. The Britannica's accounts of their lives, ideas and work will be of much value and interest to those who make or deal in furniture. For the French schools we get the essential facts about, for example, OEBEN (Vol. 20, p. 11), to whom Louis XV's famous desk owes its general plan; RIESENER (Vol. 23, p. 324), his more celebrated pupil, who completed the desk; RÖNTGEN, DAVID (Vol. 23, p. 693), the maker of "harlequin furniture," several of whose ingenious mechanical devices are described; and GOUTHIERÈRE (Vol. 12, p. 291), the metal-worker whose furniture mounts are among the most noted art products of the Louis XV and XVI periods. CHIPPENDALE (Vol. 6, p. 237), with whom arose the marvellously brilliant school of English cabinet-makers, is the subject of a biography describing fully the

characteristics of his designs; and the history of this school is continued under such headings as HEPPLEWHITE (Vol. 13, p. 305), whose taste at its best “was so fine and so full of distinction, so simple, modest and sufficient that it amounted to genius”; ADAM, ROBERT (Vol. 1, p. 172), who left so deep and enduring a mark upon English furniture, and SHERATON (Vol. 24, p. 841), “the most remarkable man in the history of English furniture,” whose extravagant creations marked the end of the great school. Many other biographies are included in the list appended.

ALPHABETICAL LIST OF ARTICLES, INCLUDING BIOGRAPHIES, IN THE
ENCYCLOPAEDIA BRITANNICA WHICH ARE OF SPECIAL INTEREST TO FURNITURE
MANUFACTURERS AND DEALERS

Acacia
Adam, Robert
Agate
Ailanthus
Alabaster
Alder
Algum
Arabesque
Arbor Vitae
Armoire
Arts and Crafts
Ash
Bahut
Bamboo
Baroque
Barry, Sir Charles
Basin-stand

Basket

Bed

Beech

Bérain, Jean

Birch

Bombay Furniture

Bonheur du Jour

Bookcase

Boulle, André Charles

Box

Boxwood

Brass

Brocade

Buffet

Carving and Gilding

Casket

Cassone

Casuarina

Cedar

Chair

Cheffonier

Chenille

Cherry

Chest

Chestnut

Chintz

Chippendale, Thomas

Coco-nut Palm

Coffer

Console

Copal

Copeland, Henry

Corduroy

Cradle

Crash

Cressent, Charles

Cretonne

Crunden, John

Cryptomeria

Cupboard

Curtain

Cushion

Cypress

Damask

Dammar

Date Palm

Design

Desk

Divan

Dresser

Dumb-Waiter

Duramen

Dyeing

Ebony

Electroplating

Elm

Embossing

Encoignure

Etagère

Fir

Footman

Frame

French Polish

Furniture

Gilding

Gillow, Robert

Glue

Gouthière, Pierre

Halfpenny, William

Hazel

Hepplewhite, George

Hickory

Holly

Huon Pine

Ince, William

Ingle-nook

Inlaying

Iron

Ivory

Japan, _Art_

Japanning

Jarraah Wood

Johnson, Thomas

Juniper

Kauri Pine

Lac

Lacquer

Lampstand

Lapis Lazuli

Larch

Leather

Leather, Artificial

Le Pautre, Jean

Lime, or Linden
Linen-press
Liquidambar
Lock
Lock, Matthias
Lowboy
Mahogany
Mammee Apple
Manwaring, Robert
Maple
Maple, Sir John B.
Marble
Marot, Daniel
Marquetry
Mastic, or Mastich
Mayhew, Thomas
Meissonier, J. A.
Mirror
Moreton Bay Chestnut
Morris, William
Nettle Tree
Oak
Oeben, J. F.
Olive
Onyx
Ormolu
Ornament
Osier
Ottoman
Overmantel
Painter-work

Pearl

Pergolesi, M. A.

Pigments

Pine

Plane

Plush

Prie-dieu

Rep

Resin

Riesener, J. H.

Rococo

Röntgen, David

Rosewood

Rousseau de la Rottière, J. S.

Sabicu Wood

Satin Wood

Screen

Sequoia

Settee

Settle

Shearer, Thomas

Sheraton, Thomas

Sideboard

Silk

Sofa

Spruce

Stall

Stool

Table

Tallboy

Tapestry

Tea-caddy
Teak
Tea-poy
Textile Printing
Throne
Ticking
Timber
Tortoiseshell
Tray
Triclinium
Tripod
Turpentine
Upholsterer
Varnish
Velvet
Velveteen
Vernis Martin
Walnut
Wardrobe
Washstand
Weaving
What-not
Willow
Window-cornice
Window-seat
Wine Table
Wood-carving

CHAPTER IX

FOR MERCHANTS AND MANUFACTURERS OF LEATHER AND LEATHER GOODS

The purpose of the department of the Guide in which this chapter appears, addressed to persons engaged in certain important occupations, is not only to show them how Britannica-reading will enlarge their knowledge of some aspects and relations of their business, but also to show how Britannica-reading will help them to realize the importance of _educating the general public_ in regard to that business. This education of the public is not necessarily confined to advertising, although the best form of advertising that can be used by anyone who sells a good article, or an article that is good at its price, is probably to tell the public what it really is and how it is really made. In the direct personal intercourse between salesman and purchaser there is opportunity for the imparting of information which, if it possesses genuine interest, will be gladly received and will stimulate trade. Mere praise of an article is uninteresting and unconvincing; while facts that explain _why_ that article is adapted to a particular use, and _why_ it is better than another article sold at a lower price will always receive attention.

[Sidenote: About Selling Leather Goods]

All this is especially true of leather goods, for the public ignorance on the subject of leather is abysmal. Nothing is more universally used, yet ninety-nine out of a hundred who use it not only do not know what lies beneath the surface of it, but do not know that there is any difference in value between a natural grain surface and a mechanically grained false surface, and it is quite certain that nearly all the men and women who walk out of a store after buying skiver would be

nonplussed if they were asked whether the upper or lower part of a split skin was the best.

Both the leather merchant and the public would be delighted to hear some of the curious things that the Britannica tells about leather, which is, from any point of view, one of the most interesting of all commodities; although few of those who use it, and perhaps as few of those who deal in it, ever stop to think how curious a relation there is between the original nature of the material and the qualities of the finished product. In cattle and sheep, the hide is a garment that covers every part of the body but the feet. Adapted to our own use, its most important service as a garment is to cover our feet. It is so far a natural product that no imitation of it possesses any of its chief merits, and yet so far an artificial product that when the hide has been removed from an animal, it requires treatment in order that it may not lose the flexibility which makes it, for a thousand purposes, more valuable than wood or metal, and in order that it may not decay.

[Sidenote: What Skin Is]

Skin is waterproof because its surface consists of scales, and although in most quadrupeds, as in man, these scales are so small as to be invisible, they will so resist the entrance of any tan liquor or other preservative fluid that they must be scraped away before the skin can be treated. Under these horny scales there is a layer of soft cells, and under this a membrane which makes the natural grain surface of leather. Under this, again, lies the "true" skin, in In the upper of these two, the white fibres lie parallel with the grain. In the lower, the white fibres, which are here coarser, lie in bundles, bound together by yellow fibres, so that this layer is really a woven fabric. The spaces in the weave are filled with a soft jelly, and the fibres do not multiply among

themselves, as cells do, but are developed, as they are needed, from this jelly. Tan liquor has the peculiar property of converting this jelly into a “leathery” substance, which although it does not then assume the shape of fibres, becomes nearly as tough as the fibres themselves, and thus makes leather more solid and stronger than the original skin; and the virtue of leather depends largely on the presence of this jelly. [Sidenote: Naturally Woven Fibres] two layers. The body of an old bull will have absorbed it, just as fat is absorbed in old age, so that the spaces in the weave of the fibre are left vacant, and (as the scaly outer surface of the skin has been scraped away to admit the tan liquor) any water with which the hide comes into contact will be soaked up. That is why old bull leather is not waterproof and is lacking in substance. Again, the weave of this innermost layer of skin, lying next to the flesh, varies in different animals. In sheepskin the fibres are very loosely woven, and for this reason great care is needed in preparing the leather, and when the skin is split, the under half is only fit for the light usage to which “chamois” leather is restricted. But however the quality, surface or thickness of the skin may differ, its true structure is the same in all animals used for leather, save the horse, which is exceptional in possessing, over the loins, a third skin, very closely woven and very greasy, which makes horsehide taken from this part of the body peculiarly waterproof, pliable and durable.

As you are in the leather business, you probably knew all these facts already, but perhaps they were not arranged in your mind in a form in which you could explain them to others as clearly as you will be able to do after reading the articles in the Britannica from which this general statement is summarized. And when you are reading about any other business, or about any other subject of any kind, you will find that the Britannica _goes to the root of the subject_ in the same thorough way in which it deals with the fibres and the jelly that make up the substance

of leather. Now for the articles in detail—or the principal ones; the others are sufficiently indicated by the list at the end of this chapter.

SKIN (Vol. 25, p. 188), by Dr. F. G. Parsons, vice-president of the Anatomical Society of Great Britain and Ireland, with illustrations from microscopic enlargements, covers the comparative anatomy of the skin in all groups of animals, and the process of skin development in the embryo. The articles mentioned in the chapter *_For Stock-Raisers_* tell you about the domestic animals whose hides are chiefly used for leather. The chapter on *_Zoology_* in this Guide gives a list of the articles on the other animals whose skins are tanned for fancy leathers. The main article LEATHER (Vol. 16, p. 330), equivalent to 50 pages of this Guide, is by Dr. James G. Parker, principal of the Leathersellers' Technical College, London, and author of *_Principles of Tanning_* and other standard trade text-books. After explaining the distinctions between tanned, tawed, and chamoised leathers, it takes up the subject of sources and qualities of hides and skins, and describes the structure of skin in relation to the finished product. The characteristics and peculiarities of hides and skins from different parts of the world are thoroughly explained. We learn why hides from animals bred in mountainous districts are the best, and where the finest sheep- and goat-skins come from.

[Sidenote: Processes of Tanning]

Tanning Materials is the subject of the next section. These are classified into pyrogallols, catechols, and subsidiary materials; and the article describes their composition and preparation by grinding, with explicit directions for their testing, including the latest official method of the International Association of Leather Trades

Chemists. The processes of making heavy leathers are next discussed. We learn the many ways of cleaning, softening, depilating, and fellmongering (or dewooling) by liming, rounding and scudding, and finally the process of actual tanning in its three steps of colouring, handling, and laying away. In connection with depilation, it is interesting to note that it has been discovered that it is not the lime but the action of bacteria in the lime which causes the hair to fall out. The finishing of sole leather, harness leather and other grades is explained, also the theory of the formation of the "bloom" and its removal, as well as the process of "scouring." The art of Currying has a section to itself, and the preparations for tanning or dressing hides for trunks and suit cases by bating, puering, scudding, plumping, drenching and splitting, receive detailed attention. The tanning of light leathers, and all the varieties of basils, skivers, Russia leather, seal, alligator, snake, frog and kangaroo leathers, Japan and enamel leathers are fully treated. Tawing, Wooling, Dressing, Chrome Tanning, Combination Tannages, Oil Tanning (Chamoising), Preller's Helvetia or Crown Leather, Transparent Leather, Parchment, Tar and Peat Tanning, Dyeing, Staining and Finishing, Glove Leathers, and Bookbinding Leathers are some of the other sections of this excellent treatise. LEATHER, ARTIFICIAL (Vol. 16, p. 345) is a separate article.

[Sidenote: Chemistry of Leather Manufacture]

TANNIN, or TANNIC ACID (Vol. 26, p. 399) is a general account of the vegetable products which have the property of converting raw hide into leather. Specific information about the materials from which the pyrogallol tannins are obtained will be found under MYROBALANS (Vol. 19, p. 114), CHESTNUT (Vol. 6, p. 112), DIVIDIVI (Vol. 8, p. 332), SUMACH (Vol. 26, p. 70), OAK (Vol. 19, p. 931), GALLS (Vol. 11, p. 422) a full

and interesting account of the insect produced vegetable excrescence which yields a high percentage of tannin, by Francis H. Butler, of the Royal School of Mines; and WILLOW (Vol. 28, p. 688). For the catechol tannins see HEMLOCK (Vol. 13, p. 262), CATECHU (Vol. 5, p. 507), MANGROVE (Vol. 17, p. 572), MIMOSA (Vol. 18, p. 500), LARCH (Vol. 16, p. 211), BIRCH (Vol. 3, p. 958), which yields the empyreumatic oil used in the preparation of Russia leather, to which the pleasant odor is due.

There are numerous articles in the Britannica on the chemicals used in the process of tawing, chrome tanning, etc., such as ALUM (Vol. 1, p. 766), ACETIC ACID (Vol. 1, p. 135), GLAUBER'S SALT (Vol. 12, p. 114), BICHROMATES AND CHROMATES (Vol. 3, p. 912).

[Sidenote: Dyeing]

The chief classes of dyes used for leather are the acid; basic, or tannic; direct, or cotton; and mordant dyes, and these are described at great length in a valuable article DYEING (Vol. 8, p. 744), equivalent to 20 pages of this Guide, by the late J. J. Hummel, professor of Dyeing, University of Leeds, and Dr. Edmund Knecht, professor of Technological Chemistry, University of Manchester. The section on the _Theory of Dyeing_ shows how the dyeing property of a substance depends upon its chemical composition. Separate articles go more deeply into the chemistry of dyeing materials used with leather, and some of the more important of these are SULPHONIC ACIDS (Vol. 26, p. 60), SULPHURIC ACID (Vol. 26, p. 65), FORMIC ACID (Vol. 10, p. 668), ANTIMONY (Vol. 2, p. 127), TITANIUM (Vol. 26, p. 1017), IRON (Vol. 14, p. 796), LOGWOOD (Vol. 16, p. 922), FUSTIC (Vol. 11, p. 375), BRAZIL WOOD (Vol. 4, p. 463), and TUMERIC (Vol. 27, p. 474). Comparatively few of the coal-tar colours have as yet been adapted to leather manufacture, but their characteristics are discussed in such articles as AZO-COMPOUNDS (Vol. 3,

p. 81), ANILINE (Vol. 2, p. 47), INDULINES (Vol. 14, p. 507), FUCHSINE (Vol. 11, p. 273), and SAFRANINE (Vol. 23, p. 1000).

[Sidenote: Special Leathers]

PARCHMENT (Vol. 20, p. 798), by Sir E. Maunde Thompson, Principal Librarian, British Museum, is an interesting historical account of the skins and their preparation. Their use as writing material was widespread at a very early period. "The Jews made use of them," says the article "for their sacred books, and it may be presumed for other literature also; and the old tradition has been maintained down to our own day, requiring the Synagogue rolls to be inscribed on this time-honoured material." The difference between parchment and vellum is explained. SHAGREEN (Vol. 24, p. 769) tells about a species of untanned leather used for ornamental purposes. It is a curious fact that the addition of the word "chagrin," for anxiety or annoyance, to the English language was due to the unpleasant sensation that came from touching the rasping surface of this leather. Stamped leather for wall hangings is described in the section _Stamped Leather_ of the article MURAL DECORATION (Vol. 19, p. 19), by William Morris and Walter Crane. SHOE (Vol. 24, p. 992) contains an illustrated section on the _Manufacture of Leather Shoes_. SADDLERY AND HARNESS (Vol. 23, p. 988), by Cecil Weatherly, and GLOVE (Vol. 12, p. 135) are treated both from an historical and a practical point of view. BOOKBINDING (Vol. 4, p. 216), illustrated, by C. J. H. Davenport, of the British Museum, has a great deal of interesting information about the leathers used in this art. The flexible binding, which has been applied for the first time on a large scale in the new Britannica, originated when vellum instead of paper was used for books, and it possesses the great advantage that a volume sewed in this way can be opened flat, and lies flat without being held.

ALPHABETICAL LIST OF ARTICLES AND OF SUBJECTS IN THE ENCYCLOPAEDIA
BRITANNICA OF SPECIAL INTEREST TO THOSE IN THE MANUFACTURE AND SALE OF
LEATHER AND LEATHER GOODS

Acetic Acid

Acid dyes

Aldehyde tanning

Algarobilla

Alligator Leather

Alum

Angols

Aniline

Antimony

Azo Compounds

Barkometer

Basic, or Tannin dyes

Basils

Bates

Bating

Bichromates and Chromates

Birch

Bleaching

Bloom

Bookbinding

Bookbinding Leathers

Bottle-tanning

Brazil Wood

Canaigre

Catechols

Catechu

Chamoising
Chestnut
Chestnut Oak
Chrome Box
Chrome Tanning
Colouring Pits, or Suspenders
Combination Tannages
Crust Stock
Currying Apparatus
Currying Processes
Dash-wheel
Depilation
Direct, or Cotton, Dyes
Dividivi
Dongola Leather
Drenching
Dressing
Drum Dyeing
Dusting Material
Dyeing
Enamel Leather
Erodiin
Fatliquoring
Fellmongering, or Dewooling
Finishing
Formic Acid
Frog Skin
Fuchsine
Fustic
Galls
Gambier

Glauber's Salt
Glazing (Glacé leather)
Glove
Glove Leathers
Grinding Machinery and Leaching
Handlers, or Floaters
Heavy Leathers
Hemlock
Hide Mill, or Double-Acting Stocks
Hide-powders
Hides and Skins
Indulines
Iron
Iron Tannage
Janus Colours
Japan Leather
Kangaroo Leather
Kaspine Leather
Kips
Larch
Leather
Leather, Artificial
Levant Morocco
Liming
Logwood
Mangrove
Mimosa, or Golden Wattle
Mordant dyes
Morocco Leather
Myrobalans
Oak bark

Oak wood

Oil Tanning

Parchment

Payne and Pullman Process

Peat Tanning

Pigskin

Portmanteau

Power Transmission, _Belts_

Preller's Helvetia or Crown Leather

Puering

Pyrogallols

Quebracho

Roans

Russia Leather

Saddlery and Harness

Safranine

Sammying

Scudding

Seal Leathers

Setting

Shagreen

Shoe

Skin

Skivers

Snakeskin

Splitting Machines

Staining

Sulphonic Acids

Sulphuric Acid

Sumach

Sweating

Tan Liquors
Tanner's Beam
Tanner's Hook
Tanner's Knives
Tannin, or Tannic Acid
Tawing
Tiffany Bate
Titanium
Transparent Leather
Tray Dyeing
Turmeric
Upper Leather
Valonia
Vellum
Vidal Colours
Waxing
Willow
Willow Calf
Tannin Precipitation
Tanning Materials
Tar Tanning
Wilson Scouring Machine
Wool-rug Dressing

CHAPTER X

FOR JEWELLERS, CLOCK AND WATCH MAKERS AND MERCHANTS

By long established custom, watches and the higher grade of clocks form part of the jeweller's stock, and he sells a few other articles of utility, such as purses and bags, but to all intents and purposes he shares with the artist and art-dealer the distinction of making a living by adding pleasure to the lives of others. The very word "jewelry" carries, in its root form, the idea of joy; and when a Senwosri princess, 43 centuries ago, smiled happily as she raised her brown arms to fasten the clasp of a new necklace, the jeweller of Memphis on the Nile no doubt took his little profit, as the jeweller of Memphis on the Mississippi takes his to-day, all the more gladly for being, in the oriental phrase, a "Distributor of delights." Sour philosophers have always sneered at women for loving jewels, and most of all for piercing their ears and noses to vary its display, but the nose-ring that overhangs a thick Nubian lip is an expression of the same charming instinct that makes a child diversify the arrangement of her daisychains. And jewelry plays its part in the higher emotions as well as in the pretty vanities; witness the engagement ring, the marriage ring and all the uses, described in the Britannica, of jewels as religious symbols.

[Sidenote: Specimens Reproduced]

The article JEWELRY (Vol. 15, p. 364), by A. H. Smith, the official in charge of the great jewel collection in the British Museum, contains nearly a hundred illustrations, half of them on plate paper, which include examples of every period and every variety of the jeweller's art, and these, with the illustrations in other articles mentioned in this chapter, are so full of interest to the jeweller's customers that he ought really to keep his Britannica at his place of business rather than at his house. It is, at any rate, amusing to recall that in a speech made by the Editor-in-chief of the Britannica, on the occasion of

a banquet given to celebrate the completion of the new edition, he remarked that when he had chanced to take home the proof sheets of this article, to read them at night, he carefully kept them out of his wife's sight lest they might suggest too tempting possibilities. The article divides modern jewelry into three classes:

(1) objects in which gems and stones form the principal portions, and in which the work in silver, platinum or gold is really only a means for carrying out the design by fixing the gems or stones in the position arranged by the designer, the metal employed being visible only as a setting;

(2) when gold work plays an important part in the development of the design, being itself ornamented by engraving (now rarely used) or enamelling or both, the stones and gems being arranged in subordination to the gold work in such positions as to give a decorative effect to the whole;

(3) when gold or other metal is alone used, the design being wrought out by hammering in repoussé, casting, engraving, chasing or by the addition of filigree work, or when the surfaces are left absolutely plain but polished and highly finished.

[Sidenote: The "Personal Art" Movement]

The second of these three classes includes the work which has completely revolutionized the theory of design, so far as the best class of trade is concerned, since the Paris International Exposition of 1900 first drew general attention to the exquisite creations of Lalique and his school. L. C. Tiffany, in the United States, and Philippe Wolfers, in Belgium, have done more than any designers other than the French to

extend this new movement; but in England, Germany, Austria, Russia and Switzerland there has been a notable increase of individual effort and purpose, and a recognition of the possibilities of personal art as at any rate an important factor in the business. Side by side with this development new standards have been established in mechanical work.

“Nearly every kind of gold chain now made is manufactured by machinery, and nothing like the beauty of design or perfection of workmanship could be obtained by hand at, probably, any cost.” The article, equivalent in length to about 35 pages of this Guide, contains a full review, amplified by the results of the most recent excavations (some of them undertaken expressly for the archaeological purposes of this edition of the Britannica) of the history of jewelry, Egyptian, Assyrian, Mycenaean, Greek, Etruscan, Roman, Merovingian, Oriental and Renaissance.

[Sidenote: Rings for Love and Murder]

RING (Vol. 23, p. 349), of which Prof. Middleton, long art director of the South Kensington Museum, is the chief contributor, is another copiously illustrated article. Among the curious items of information it contains, there is the unromantic origin of the engagement ring (which may be cited by the jeweller to prove that it should always be a costly one), the ancient Romans regarding it as a pledge to assure the donor's fulfilment of his promise; the fact that the modern rheumatism ring had its medieval forerunner in the rings, blessed by the sovereign, which were worn as a preservative against cramp; and the description of the old poison rings, which were of two kinds: those merely affording, in the bezel, a secret receptacle so that the poison might be always at hand for suicide, and those provided with a hollow point to which, on touching a spring, the venom ran as in a snake's fang, so that the murderer could give a fatal scratch while shaking hands with his victim.

BROOCH (Vol. 4, p. 641) traces, with many illustrations of typical

specimens, the “fibula” or safety pin from its origin in Central Europe during the Bronze Age, through the modifications which introduced the bow shape, providing space for thicker folds of cloth, to the modern ornament. The long brooch is not a new fashion, for silver brooches no less than 15 inches in length have been found in Viking hoards of the 7th, 8th and 9th centuries. EAR-RING (Vol. 8, p. 798) describes ear “ornaments” of the most grotesque size. In Borneo the hole in the ear lobe is stretched to a calibre of 3¾ inches, but the Masai tribes in equatorial Africa far outdo this, stretching the lobes, year after year, until they can wear stone ear-plugs weighing 2 lbs. 14 ozs. each, with a diameter of 4½ inches; and they thus achieve the supreme elegance of making the two long flaps of flesh meet above their heads. It is also curious to note the custom of some oriental tribes of wearing one ear-ring only. BRACELET (Vol. 4, p. 359) describes the three distinct models worn by the Israelites, all of which the Authorized Version calls “bracelet,” although the original Hebrew has separate names for them. Armlets have always been conspicuous in the regalia of Eastern kings, and the pair captured at Delhi and taken to Persia by Nadir Shah in 1739 contain jewels valued at more than \$5,000,000, including the famous “Sea of Light” diamond, which, although it weighs only 186 carats as against the 516½ of the largest fraction into which the Cullinan stone was cut, is unique as possessing the finest lustre of any known specimen. The 24 plate illustrations in the article SCANDINAVIAN CIVILIZATION (Vol. 24, p. 287), by Miss B. S. Phillpotts, show some exquisite designs of clasps, collars and pins exhumed in Denmark, Norway and Sweden, and supposed by some authorities to antedate the oldest Egyptian jewelry.

[Sidenote: Precious Metals]

The article GOLD (Vol. 12, p. 192) is a thorough workshop treatise, as well as a detailed study of existing mines and of the influence their

production exerts upon the “price,” if it can be so called, of a metal which is its own standard of value. SILVER (Vol. 25, p. 112) and PLATINUM (Vol. 21, p. 805) are treated with similar comprehensiveness. The articles ALLOYS (Vol. 1, p. 704), ASSAYING (Vol. 2, p. 776), METAL (Vol. 18, p. 198), METALLOGRAPHY (Vol. 18, p. 202), and METALLURGY (Vol. 18, p. 203), all by noted authorities, are full of information useful to the jeweller. METAL-WORK (Vol. 18, p. 205), fully illustrated, incidentally touches upon the art of the silver- and gold-smith; and this branch of the subject is also treated in such articles as PLATE (Vol. 21, p. 789), with over 30 typical illustrations—a most interesting historical account, by several well-known experts, of works in gold and silver which belong to any class other than those of personal ornament and coins; and DRINKING VESSELS (Vol. 8, p. 580), illustrated, by Dr. Charles H. Read of the British Museum, which discusses gold and silver cups. Mention must also be made of the description of American work in precious metals before the time of Christopher Columbus, in the section _Archaeology_ of the article AMERICA (Vol. 1, p. 812), by the late Dr. O. T. Mason, of the National Museum, Washington; also of MEXICO, _Ancient Civilization_ (Vol. 18, p. 335), by the famous ethnologists, Dr. E. B. Tylor of Oxford and Dr. Walter Lehmann, of the Royal Ethnographical Museum, Munich; EGYPT, _Ancient Art_ (Vol. 9, p. 73), by W. M. Flinders Petrie; GREEK ART (Vol. 12, p. 470), illustrated, by Dr. Percy Gardner, of Oxford; ROMAN ART, _Work in Precious Metals_ (Vol. 23, p. 483), illustrated, by H. Stuart Jones, director of the British School at Rome; JAPAN, _Art, Sculpture and Carving_ (Vol. 15, p. 176), by Capt. Frank Brinkley, author of _A History of the Japanese People_; and CHINA, _Bronzes_ (Vol. 6, p. 215), by C. J. Holmes, formerly Slade professor of fine art at Oxford.

FILIGREE (Vol. 10, p. 343) describes the delicate jewel work of twisted gold and silver threads, and also the “granulated” work which consists

of minute globules of gold soldered to form patterns on a metal surface.

In India the filigree worker has retained the patterns used by the ancient Greeks and works in the same way they did. Wandering workmen are given so much gold, coined or rough. This is weighed, heated and beaten into wire, and worked in the courtyard or on the verandah of the customer's house. The worker reweighs the complete work when finished and is paid at a specified rate for his labor. Repoussé (Vol. 23, p. 108), by M. H. Spielmann, editor _Magazine of Art_; CHASING (Vol. 5, p. 956) and INLAYING (Vol. 14, p. 574) are other articles dealing with certain processes in jewel work. The jeweller also must not overlook two superb articles, MEDAL (Vol. 18, p. 1), illustrated, by M. H. Spielmann, and NUMISMATICS (Vol. 19, p. 869), which is by three specialists, and is most fully illustrated by designs inviting practical use. ENAMEL (Vol. 9, p. 362), illustrated, by Alexander Fisher, author of _The Art of Enamelling on Metals_, goes very fully and practically into this interesting subject, which is further discussed in JAPAN, _Cloisonné Enamel_ (Vol. 15, p. 189). MOSAIC (Vol. 18, p. 883), illustrated, by Professor Middleton and H. Stuart Jones, deals in part with the ornamentation of jewelry by this method. In BRAZING and SOLDERING (Vol. 4, p. 463) the composition of silver solder used for jewelry is described, and in CEMENT there is an account of _Jeweller's_ or _Armenian Cement_ (Vol. 5, p. 659).

[Sidenote: Precious Stones]

The article GEM treats the subject in two sections, of which the first (Vol. 11, p. 560), by F. W. Rudler, of the Museum of Practical Geology, London, deals with _Mineralogy_ and _General Properties_. Here are discussed hardness, specific gravity, crystalline forms and cleavage, colour, refraction, chemical composition, etc., and there is an interesting section on superstitions in regard to gems, the medical and

magical powers with which they were reputed to be endowed. These beliefs are very remarkable, and it has even been suggested by archaeologists that jewelry did not have its origin so much in a love for personal decoration, as in the belief that the objects used possessed magical virtue. The article MINERALOGY (Vol. 18, p. 509), by L. J. Spencer, of the British Museum, and editor of the *Mineralogical Magazine*, will be found especially valuable for reference in the workshop. It gives, among other things, the scale of hardness, and nomenclature and classification of minerals. The crystal formation of gems as well as their optical properties—characteristics by which the genuineness of precious stones may be tested—are discussed and explained in the article CRYSTALLOGRAPHY (Vol. 7, p. 569), with over 100 illustrations, also by L. J. Spencer.

The cutting of gem stones is treated under LAPIDARY AND GEM CUTTING (Vol. 16, p. 195), by Dr. George F. Kunz, the famous gem expert to Tiffany & Co., New York,—an article of uncommon historical interest and practical value, in which diamond cutting is considered at much length.

The second section of the article GEM, *Gems in Art* (Vol. 11, p. 562), by Dr. A. S. Murray, the famous British archaeologist, and A. H. Smith, gives an account of precious stones engraved with designs. The illustrations show more than 90 examples, including Cretan and Mycenaean intaglios, Greek, Phœnician and Etruscan scarabs and scarabæoids, cameos, seals, Oriental, Christian, and modern gems. This subject is further discussed in separate articles, such as SCARAB (Vol. 24, p. 301), by Dr. F. Ll. Griffith, the Egyptologist, an account of the designs which, originating in Egypt during the Fourth Dynasty, have exercised a lasting influence on the design and shape of gems; CAMEO (Vol. 5, p. 104), INTAGLIO (Vol. 14, p. 680), SEALS (Vol. 24, p. 539), illustrated, by Sir E. Maunde Thompson, formerly principal librarian, British Museum, as well as in the articles on ancient and Oriental civilizations, already mentioned.

[Sidenote: Synthetic Stones]

The artificial duplication of certain gems by chemical processes which yield products identical in composition and physical properties with the natural stones is a subject of growing importance to the jeweller, and the latest developments are described in GEM, ARTIFICIAL (Vol. 11, p. 569), by Sir William Crookes. This famous chemist and authority on precious stones does not hesitate to declare that although the artificial diamonds so far produced have been microscopic in size, scientists have now found the right method and that "there is no reason to doubt that, working on a larger scale, larger diamonds will result."

The artificial production of rubies, sapphires, Oriental emeralds, amethysts, topazes and zircons is also discussed. Descriptions of the several gem stones are found under their respective headings, for example DIAMOND (Vol. 8, p. 158), illustrated, by H. A. Miers, principal of the University of London, and former editor of the Mineralogical Magazine. Here are given its scientific characters, its uses (especially for faceting softer precious stones), distribution, and mining, and the wonderful history of the most famous diamonds of the world. RUBY (Vol. 23, p. 812), the most valued of gem stones, is often called "Oriental ruby" to distinguish it from SPINEL (Vol. 25, p. 684), an aluminate stone of inferior hardness, density and value. It is interesting to note that many historic stones described as monster rubies were really spinels. The great ruby set in the Maltese Cross in front of the Imperial State Crown of England is a spinel. SAPPHIRE (Vol. 24, p. 201) was known to the Greeks as "hyacinthus," and the present name was formerly applied to lapis lazuli. ASTERIA or STAR STONE (Vol. 2, p. 792) tells how the luminous star comes to be seen in sapphires, rubies and topazes. The name EMERALD (Vol. 9, p. 332) is used for a number of stones, of which the most valued is not a true emerald at all;

see CORUNDUM (Vol. 7, p. 207). The same is true of the TOPAZ (Vol. 27, p. 48), the more prized Oriental topaz being a yellow corundum, harder and denser than the stone from which it takes its name. "Scotch" or "Spanish" topazes are yellow or smoke-tinted quartz, or cairngorm. The AMETHYST (Vol. 1, p. 852) is violet or purple quartz, and the sapphire of a purple colour is often called an Oriental amethyst. The many varieties of the beautiful ZIRCON (Vol. 28, p. 989), such as HYACINTH (Vol. 14, p. 25) and JARGOON (Vol. 15, p. 276) are carefully described and distinguished. These valuable articles on the precious stones have been contributed by F. W. Rudler, of the Museum of Practical Geology, London. PEARL (Vol. 21, p. 24) discusses the results of the latest researches on the cause of pearl formation, and gives a graphic account of pearl-fishing.

[Sidenote: Semi-Precious Stones]

The material in the Britannica on the semi-precious stones is as complete. There are many articles, specified in the list at the end of this chapter. ALEXANDRITE (Vol. 1, p. 576) is remarkable for its property of appearing dark green by daylight, and red by candle-light, which makes it especially popular in Russia where green and red are the military colors; CHRYSOBERYL (Vol. 6, p. 320), of which alexandrite is a variety; CHRYSOLITE (Vol. 6, p. 320), often mistaken for chrysoberyl; PERIDOT (Vol. 21, p. 147), like chrysolite, a variety of olivine; BERYL (Vol. 3, p. 817), much prized by the ancients as a gem stone, and of which the EMERALD (see above) and the AQUAMARINE (Vol. 2, p. 237) are the chief "precious" varieties; TOURMALINE (Vol. 27, p. 103), the remarkable stone of as much interest to the physicist as to the jeweller, on account of its optical and electrical properties; and RUBELLITE (Vol. 23, p. 804), its much prized red variety GARNET (Vol. 11, p. 470), together with ALMANDINE (Vol. 1, p. 712), which, when cut

with a convex face is known as carbuncle; CINNAMON-STONE (Vol. 6, p. 376), the light red garnet, so easily mistaken for a variety of zircon (the article tells how to distinguish them); DEMANTOID (Vol. 7, p. 979), the green garnet from the Urals, and PYROPE (Vol. 22, p. 695), usually known as Bohemian garnet; JADE (Vol. 15, p. 122), which occupies in China the highest place as a jewel, and whose many varieties are here clearly distinguished; JET (Vol. 15, p. 358); HAEMATITE (Vol. 12, p. 804); MOONSTONE (Vol. 18, p. 807); CAT'S-EYE (Vol. 5, p. 537), a term applied to several distinct minerals of which CROCIDOLITE (Vol. 7, p. 477) has recently become very popular; OPAL (Vol. 20, p. 120), an article in which the brilliant flashes of colour in this stone are explained; QUARTZ (Vol. 22, p. 715), with its many ornamental varieties such as AGATE (Vol. 1, p. 368), AMETHYST (Vol. 1, p. 852), AVENTURINE (Vol. 3, p. 54), BLOODSTONE (Vol. 4, p. 85), CAIRNGORM (Vol. 4, p. 952), CARNELIAN (Vol. 5, p. 365), CHALCEDONY (Vol. 5, p. 803), CHRYSOPRASE (Vol. 6, p. 320), HELIOTROPE (Vol. 13, p. 232), MOCHA STONE (Vol. 18, p. 637), ONYX (Vol. 20, p. 118), ROCK-CRYSTAL (Vol. 23, p. 433), SARD (Vol. 24, p. 209), and SARDONYX (Vol. 24, p. 218).

[Sidenote: Watches and Clocks]

The article WATCH (Vol. 28, p. 362), illustrated, by Lord Grimthorpe, the great authority on watches and clocks, and Sir H. H. Cunynghame, vice-president of the British Institute of Electrical Engineers, is full of interest. There is a very valuable historical account beginning with the invention of portable time pieces in the 15th century. The parts of a modern watch are described, with details as to the mainspring, different types of escapement, the balance-wheel and hair-spring, compensation adjustments and secondary compensation. Methods of correcting temperature errors are discussed, and a simple means for demagnetizing a watch which has been near a dynamo is given. The proper

materials used for jewelled bearings are described in the articles DIAMOND, CORUNDUM, etc. LUBRICANTS (Vol. 17, p. 88) contains a valuable paragraph on the properties and preparation of the fluid oils used on the spindles of watches and clocks.

The article CLOCK (Vol. 6, p. 536) is by the same distinguished authorities as WATCH, with an additional section on _Decorative Aspects_ (p. 552), by James Penderel-Brodhurst. It is equivalent to 55 pages of this Guide and is fully illustrated. Among the topics considered are the earliest clocks and their gradual improvement; the essential components of a clock; the mechanics of the pendulum; methods of compensation, including the use of the new nickel-steel alloy—described in the article INVAR (Vol. 14, p. 717)—the barometrical error, and methods of counteraction; suspension of pendulums; balance, anchor, dead, pinwheel, detached or free, and gravity escapements; the remontoire systems for abolishing errors in the force driving the escapement; testing of clocks; clock wheels; striking mechanism; the watchman's clock, church and turret clocks, electrical clocks, miscellaneous clocks, including magical clocks and other curious designs. The section on _Decorative Aspects_ tells about styles of cases and mountings, the origin and development of the “grandfather” clock, etc. In connection with long-period clocks, attention should be given to the new and ingenious, if not commercially practical, device invented by the Hon. R. J. Strutt. Electrified particles emitted by a radioactive substance separate two strips of gold leaf, and these, falling together after the charge has been conducted away upon contact with metal, are extended again, the process being constantly repeated. If some way could be found to utilize this motion to work an escapement, we should have a clock that would go on indefinitely, since 1000 years must elapse before even half the small amount of radium used has disappeared. A description of this so-called “radium” clock will be found in PERPETUAL MOTION (Vol. 21, p. 181).

ALPHABETICAL LIST OF ARTICLES IN THE ENCYCLOPAEDIA BRITANNICA OF SPECIAL
INTEREST TO MANUFACTURERS OF AND DEALERS IN JEWELRY, CLOCKS, AND WATCHES

Agate

Aigrette

Aiguillette

Albite

Alexandrite

Alloys

Almandine

Amazon-stone

Amber

America, _Archaeology_

Amethyst

Andalusite

Anhydrite

Apatite

Apostle Spoons

Aquamarine

Arabesque

Arts and Crafts

Assaying

Asteria, or Star-stone

Aventurine

Axinite

Azurite

Bead

Benitoite

Beryl

Beryllonite
Biddery
Bloodstone
Bort
Bracelet
Brazing and Soldering
Britain, _Ornaments_
Bronzite
Brooch
Cairngorm
Cameo
Campani-Alimenis, M.
Carbonado
Carnelian
Cat's-eye
Cellini, Benvenuto
Cement
Chain
Chalcedony
Chasing
Chessylite
China, _Art_
Chrysoberyl
Chrysolite
Chrysoprase
Cinnamon-stone
Clock
Collar
Congreve, Sir William
Coral
Corundum

Costume

Cressent, Charles

Crocidolite

Cross

Crown

Crystallography

Cyanite

Demantoid

Diallage

Diamond

Dioptase

Drinking Vessels

Dumortierite

Ear-ring

Egypt, _Ancient Art_

Electroplating

Emerald

Emery

Enamel

Epidote

Euclase

Felspar

Filigree

Finiguerra, Maso

Fluorescence

Fluor-spar

Franklin, Benjamin

Galileo Galilei

Garnet

Gem

Gem, Artificial

Gold

Gold beating

Göthite

Gouthière, Pierre

Greek Art

Grimthorpe, 1st Baron

Haematite

Hiddenite

Hyacinth, or Jacinth

Hypersthene

Inlaying

Intaglio

Invar

Iolite

Ivory

Jade

Japan, _Art_

Jargoon

Jasper

Jet

Jewelry

Knighthood and Chivalry

Kunzite

Labradorite

Lapidary and Gem Cutting

Lapis Lazuli

Leucite

Line-engraving

Lubricants

Malachite

Marot, Daniel

Meissonier, J. A.

Medal

Metal

Metallography

Metallurgy

Metal-Work

Mexico, _Ancient Civilization_

Microcline

Mineral Deposits

Mineralogy

Miniature

Mint

Mocha-stone

Monogram

Monteith

Moonstone

Morel-Ladeuil, Léonard

Mosaic

Nepheline

Niello

Numismatics

Oligoclase

Olivine

Onyx

Opal

Orthoclase

Palladium

Paste

Pearl

Peridot

Perpetual Motion

Phenacite
Phosphorescence
Plagioclase
Plate
Plated Ware
Platinum
Pollaiuolo
Prehnite
Pyrope
Pyroxene
Quartz
Regalia
Repoussé
Ring
Rock-crystal
Roman Art
Rubellite
Ruby
Sapphire
Sard
Sardonyx
Scandinavian Civilization
Scarab
Seals
Sheffield Plate
Silver
Sphene
Spinel
Spodumene
Staurolite
Sunstone

Tassie, James

Tiffany, C. L.

Time, Measurement of

Time, Standard

Topaz

Tourmaline

Turquoise

Variscite

Vesuvianite

Watch

Weighing Machines

Weights and Measures

Wyon, Thomas

Zircon

Zoisite

CHAPTER XI

FOR MERCHANTS AND MANUFACTURERS OF ELECTRICAL MACHINERY AND SUPPLIES

[Sidenote: Construction and Operation]

Electrical machinery and supplies include three main groups of appliances: The apparatus by which electricity is originally generated; the apparatus by which current is transmitted and, if necessary, modified before it is used; and the infinitely various appliances for its final employment. In connection with any one of the latter, information may be needed as to its structure and its mechanical or

electrochemical method of operation, or as to its uses, and in the treatment of these two aspects of a vast number of subjects the advantages of the encyclopaedic plan of the Britannica are obvious. One article will explain the method by which the same principles are applied to a number of different machines. Another article will deal with a group of appliances all used for similar purposes; and a reference to the Index of 500,000 entries (Vol. 29) will at once guide the reader who turns to the name of any electrical appliance to either kind of information he desires at the moment, whether he wants to know how the machine is made and operated, or what kind of work it does and how efficiently it does it.

The reader to whom this chapter is addressed is already familiar with the general subject of electricity, but he may at any moment desire to review or to supplement his general knowledge in connection with some new appliance which, for the first time, applies to commercial use one of the many and intricate laws of electrical vibration. The whole subject of the nature and action of electricity is outlined in the article ELECTRICITY (Vol. 9, p. 179), by Prof. J. A. Fleming, of the University of London, one of the world's foremost authorities. In a space equivalent to hardly more than 30 pages of this Guide, the field covered in detail by many other articles is so concisely and clearly surveyed that you get a complete view of the theoretical and practical developments by which electrical science and industry have reached their present position. The same contributor then considers ELECTROSTATICS (Vol. 9, p. 240) and ELECTROKINETICS (Vol. 9, p. 210); and, in CONDUCTION, ELECTRIC (Vol. 6, p. 855), deals with metallic, non-metallic, dielectric and gaseous conductors. One section of this article is by Sir J. J. Thomson, winner, in 1906, of the Nobel Prize for Physics. The form in which metal is chiefly employed for the conduction of electricity is the subject of a separate article, WIRE (Vol. 28, p.

738); and the articles on the individual metals deal with their electrical properties.

[Sidenote: Batteries and Dynamos]

The whole subject of the chemical production of electricity is discussed in ELECTROLYSIS (Vol. 9, p. 217), by W. C. D. Whetham, of the technical staff of Cambridge University. BATTERY (Vol. 3, p. 531), fully illustrated, deals with all the forms of primary battery, and ACCUMULATOR (Vol. 1, p. 126), also illustrated, by Walter Hibbert, of the London Polytechnic, with all the secondary types. The alkaline accumulators, of which the Edison apparatus is a well known type, are the subject of a special section. Turning to mechanically produced electricity, the first article to read is ELECTROMAGNETISM (Vol. 9, p. 226). This brings you naturally to the article DYNAMO (Vol. 8, p. 764), by C. C. Hawkins, author of one of the best practical text-books on the subject. This copiously illustrated article, in length equivalent to 50 pages of this Guide, discusses _continuous current dynamos_, _lap-winding_, _commutators_, _field-magnets_, _forgings and castings for magnets_, _air-gaps_, _armature cores_, _carbon brushes_, _cooling surfaces_ and _alternators_.

Having thus covered the subject of obtaining current, the group of articles next to be considered is that dealing with its measurement and the examination of resistances. The general article UNITS, PHYSICAL (Vol. 27, p. 740), contains a section on _electrical units_. Then come POTENTIOMETER (Vol. 22, p. 205); METER, ELECTRIC (Vol. 18, p. 291); VOLTMETER (Vol. 28, p. 206), illustrated; AMPEREMETER (Vol. 1, p. 879), illustrated; OHMMETER (Vol. 20, p. 34); WATTMETER (Vol. 28, p. 419); GALVANOMETER (Vol. 11, p. 428), illustrated; ELECTROMETER (Vol. 9, p. 234), illustrated; ELECTROSCOPE (Vol. 9, p. 239), illustrated;

WHEATSTONE'S BRIDGE (Vol. 28, p. 584), illustrated; and OSCILLOGRAPH (Vol. 20, p. 347), illustrated.

[Sidenote: Lighting Appliances]

The commercial supply of current is covered by a series of articles of which the first to be read is ELECTRICITY SUPPLY (Vol. 9, p. 193), to which Emile Garcke, the famous electrical engineer, contributes a section. POWER TRANSMISSION, _Electrical_ (Vol. 22, p. 233), is by Louis Bell, chief engineer of the General Electric Co., Boston; and contains full details as to the use of both two-phase and three-phase generators in transmission. INDUCTION COIL (Vol. 14, p. 502) and TRANSFORMERS (Vol. 27, p. 173) are both fully illustrated. LIGHTING, _Electric_ (Vol. 16, p. 659) deals with arc, incandescent and vapour lamps, and with wiring. The section on household work gives excellent practical information about the best arrangements of lights. A special class of electric light supplies is discussed in LIGHTHOUSE (Vol. 16, p. 627), by W. T. Douglass, who erected the new Eddystone and the Bishop's Rock lights, and by N. G. Gedye, another practical expert.

The appliances used to convert current back again into the mechanical energy from which it had been derived are described in the article MOTORS, ELECTRIC (Vol. 18, p. 910). This article divides continuous current motors into five classes: _Separately excited_; _series-wound constant current_; _series-wound constant potential_; _series-wound interdependent current and potential_; and _shunt-wound constant potential_. Alternating current motors are similarly classified as _Synchronous constant potential_; _induction-polyphase constant potential_; _induction monophasic constant potential_; _repulsion commutating_, and _series-commutating_.

[Sidenote: Trolley Cars and Railroads]

Machinery for applying electric power to transportation, both for trolley cars and heavy railroad traffic, is described in the article TRACTION (Vol. 27, p. 118), by Prof. Louis Duncan, who designed the first electric locomotives employed with large loads—those introduced in 1895 by the Baltimore & Ohio R.R. for its track in the tunnel under Baltimore. The article gives, with many mechanical diagrams, accounts of the appliances by which the current is taken from trolley wires, conduits and third rails, and of the types of motors and controllers employed. CRANE (Vol. 7, p. 368), by Walter Pitt, describes the peculiar type of “crane-rated” motor, by the aid of which steam and hydraulic cranes can be displaced. The electric furnaces used for the reduction of ores and for manufacturing processes in which exceptionally high temperatures are required, are treated in ELECTROMETALLURGY (Vol. 9, p. 232), by W. G. M’Millan, lecturer on metallurgy at Mason College, Birmingham. Electric machinery for the refining of metals is dealt with in the article ELECTROCHEMISTRY (Vol. 9, p. 208). Under SURGICAL INSTRUMENTS (Vol. 26, p. 133) there is a description of the apparatus used for cautery and for illuminating parts of the interior of the body. The appliances used in ELECTROTHERAPEUTICS are dealt with under that heading (Vol. 9, p. 249). Information as to other medical and surgical apparatus will be found under RÖNTGEN RAYS (Vol. 23, p. 694), X-RAY TREATMENT (Vol. 28, p. 887), by Dr. H. L. Jones, of St. Bartholomew’s Hospital, London; and FLUORESCENCE (Vol. 10, p. 575), by Prof. J. R. Cotter, of Trinity College, Dublin.

[Sidenote: Telegraph and Telephone]

TELEGRAPH (Vol. 26, p. 510), equivalent in length to 70 pages of this Guide, and fully illustrated, is by a number of contributors, and

discusses both land lines and submarine cables. The section on instruments, by H. R. Kempe, electrician to the General Post Office, London, includes a full description of the transmitters and receivers employed in the various systems of wireless telegraphy. TELEPHONE (Vol. 26, p. 547) deals with the fixed and portable instruments, the batteries and switchboards, the new automatic exchange "selectors," and with special applications of the microphone.

A number of other electric appliances are discussed in separate articles, such as BELL (Vol. 3, p. 692), by H. M. Ross, in which burglar alarm devices are described; and VENTILATION, _Fan_ (Vol. 27, p. 1011), by James Bartlett; while sparking plugs and other ignition appliances are treated under OIL ENGINE (Vol. 20, p. 35).

There are also a number of appliances used mostly in experimental and educational work. Such, for instance, are ELECTRICAL OR ELECTROSTATIC MACHINE (Vol. 9, p. 176), with many illustrations; ELECTROPHORUS (Vol. 9, p. 237), and LEYDEN JAR (Vol. 16, p. 528).

The metals, chemicals and other materials sold by dealers in electrical supplies, and their properties and uses, are described in COPPER (Vol. 7, p. 102), ZINC (Vol. 28, p. 981), LEAD (Vol. 16, p. 314), SULPHURIC ACID (Vol. 26, p. 65), SODIUM, _Compounds_ (Vol. 25, p. 341); CHROMIUM (Vol. 6, p. 296); NITROGEN, _Compounds_ (Vol. 19, p. 715); SAL AMMONIAC (Vol. 24, p. 59), BICHROMATES AND CHROMATES (Vol. 3, p. 912), CARBON (Vol. 5, p. 305), RUBBER (Vol. 23, p. 795), and GUTTA PERCHA (Vol. 12, p. 743).

The following is a partial list, in alphabetical order, of articles of peculiar interest to dealers in electrical supplies.

Accumulator
Amperemeter, or Ammeter
Armature
Battery
Bell
Bichromates and Chromates
Carbon
Chromium
Condenser
Conductor, Electric
Copper
Dielectric
Dynamo
Electricity
Electrical, or Electrostatic, Machine
Electricity Supply
Electrokinetics
Electrolysis
Electromagnetism
Electrometer
Electrophorus
Electroscope
Electrotherapeutics
Fluorescence
Fuze, or Fuse
Galvanometer
Gutta Percha
Induction Coil
Lead
Leyden Jar
Lighting

Meter, Electric

Motors, Electric

Nitrogen, _Compounds_

Ohmmeter

Oil Engine

Oscillograph

Potentiometer

Power Transmission

Röntgen Rays, _Apparatus_

Rubber

Sal Ammoniac

Sodium, _Compounds_

Sulphuric Acid

Surgical Instruments

Telegraph

Telephone

Thermometry, _Electrical_

Traction, _Electric_

Tramway

Transformers

Units, Physical

Vacuum Tube

Ventilation

Voltmeter

Wattmeter

Wheatstone's Bridge

Wire

Zinc

CHAPTER XII

FOR MERCHANTS AND MANUFACTURERS OF CHEMICALS AND DRUGS

[Sidenote: A Factor in All Industries]

The chemical and drug industry is not only in itself an enormous business, but it supplies essential materials for almost every branch of manufacturing. Chemical products are employed in our buildings, our clothing, our food; we come into the world and go out of the world with the odour of chemicals about us. The manufacturer or dealer cannot analyze all the influences that affect his market, and when he tries, as he must, to consider the future of the trade, to reckon with the channels of demand that will arise in the course of new applications of chemical products, he is facing all the problems of all the industries.

The variety of raw materials from which chemical products are derived, and the activity with which new sources are discovered and developed, are almost as bewildering. Only a century has passed since coal-tar was first distilled, and to-day no chemist would venture to fix the limits of its industrial possibilities. Electrolysis has been in use since 1804, and yet the future of the world's wheat supply probably depends upon processes, as yet hardly beyond the experimental stage, of utilizing atmospheric nitrogen.

In connection with so comprehensive an industry, the uses of the Britannica are so manifold that this whole Guide might be devoted to them. Articles on every manufacturing process touch upon the use of chemicals. The articles on countries, states and cities are full of relevant information; and there is hardly a scientific article that

would not be helpful. But the 40 general articles on chemistry, the 350 on chemical compounds, and the 75 on manufactured products call most immediately for attention; and, with the aid of other chapters in the Guide, the reader who desires to go further will easily find his way.

[Sidenote: Articles on Chemicals]

The article CHEMISTRY (Vol. 6, p. 33), equivalent to 135 pages of this Guide, is divided into 6 sections. The first, *_History_*, traces the general trend of the science from its infancy to the foundations of the modern theory. The second section, *_Principles_*, treats of nomenclature, formulæ, chemical equations and chemical changes. It provides a brief but complete introduction to the terminology and methods of the chemist, and there is not a line in it which will not prove of value in some way or other to the chemical manufacturer. Sections 3 and 4 are devoted to *_Inorganic and Organic Chemistry_*, giving a history of the subjects and the principles underlying the structure of compounds, with cross references to all articles dealing with their preparation and properties. Sections 5 and 6 deal, respectively, with *_Analytical_* and *_Physical Chemistry_*.

Dr. Walter Nernst, professor of physical chemistry, University of Berlin, is the author of CHEMICAL ACTION (Vol. 6, p. 26), which deals specifically with the nature of chemical forces and deduces the laws of chemical statics and kinetics. Of interest and importance in connection with the manufacture of chemicals is SOLUTION (Vol. 25, p. 368), by W. C. D. Whetham, of Cambridge University, author of *_Theory of Solution_*, etc. Another theoretical article which will be found widely useful is THERMOCHEMISTRY (Vol. 26, p. 804), by Prof. James Walker, of Edinburgh University. For further details see the chapter on CHEMISTRY in this Guide.

[Sidenote: Manufacture of Chemicals]

It is possible here to mention only a small amount of the material dealing with the manufacture of chemicals. At the end of this chapter there is a fuller alphabetical list. It may be noted, however, that the articles on the elements, metallic and non-metallic, give much consideration to their compounds, how these are made and how used in the arts and in medicine. But in addition to this there are many noteworthy contributions dealing with chemical manufacture. For instance, ALKALI MANUFACTURE (Vol. 1, p. 674), by Dr. Georg Lunge, professor of technical chemistry, Zurich Polytechnic, 11 pages in length and with 10 illustrations. The chief processes described are the Leblanc, ammonia-soda, and electrolytic, together with others dependent upon them. The facts about the manufacture of the carbonate, hydrate, and sulphate of soda, chlorine, hydrochloric acid, etc., are fully given. POTASSIUM (Vol. 22, p. 197) treats of the commercial compounds of this metal in the same manner. NITROGEN (Vol. 19, p. 714) explains the new process for the commercial manufacture of nitric acid from atmospheric air—a matter of enormous industrial importance—and also the conversion of nitrogen into ammonia, which has been done successfully only within the past few years.

The manufacture of chemical products by the use of electricity is the subject of ELECTROCHEMISTRY (Vol. 9, p. 208), and a still larger field is covered by ELECTROMETALLURGY (Vol. 9, p. 232). Both of these valuable articles are by W. G. M'Millan, formerly secretary of the Institute of Electrical Engineers of Great Britain. SULPHURIC ACID (Vol. 26, p. 65), illustrated, by Dr. Lunge, describes the properties, reactions and manufacture of the most important of all chemicals, including the more modern contact processes.

[Sidenote: Drugs, Origin and Manufacture]

As a key to the subject of the origin and manufacture of drugs, the article PHARMACOLOGY (Vol. 21, p. 347), by Dr. Ralph Stockman, professor of materia medica and therapeutics in the University of Glasgow, presents a great amount of interesting and valuable information on the action of chemical substances (apart from foods) on all kinds of animals, from bacteria up to man. A short history of pharmacology is given and a large part of the article concerns the action of drugs. There is also a classification of drugs according to the latest and most scientific methods into twenty-eight groups, describing the effects of each group. An appendix to the article, by Dr. H. L. Hennessy, is entitled Terminology in Therapeutics, and is a general explanation of the common names used in the therapeutic classification of drugs.

Since therapeutics is concerned with the remedial power of drugs and the conditions under which they are to be used, the article THERAPEUTICS (Vol. 26, p. 793), by Dr. Sir Lauder Brunton, of St. Bartholomew's Hospital, London, and author of the well-known treatise, Modern Therapeutics, should not be overlooked, nor POISON (Vol. 21, p. 893), by Dr. Sir Thomas Stevenson, lecturer on chemistry and forensic medicine at Guy's Hospital, London, wherein all poisons are classified and their antidotes are indicated.

PHARMACY (Vol. 21, p. 355), by E. M. Holmes, of the Pharmaceutical Museum, London, is largely historical in its nature, and yields much interesting and valuable information about the pharmacist. We learn that an Egyptian papyrus of the date 2300 B.C. gives direction as to the preparation of prescriptions, and that diachylon plaster, invented by Menecrates in A.D. 1, is used for the same purposes to-day. A great deal

of curious knowledge about ancient remedies, such as the thigh bone of a hanged man, moss grown on a human skull, the ashes of the head of a coal-black cat, etc., renders this article especially entertaining.

PHARMACOPŒIA (Vol. 21, p. 353), also by Mr. Holmes, tells about the pharmacopœias in use in different countries, the standardization of drugs, etc.

In the list at the end of this chapter are noted the numerous separate articles on drugs, their preparation and use that appear in the Britannica. Mention should be made of the articles on the elements, such as IRON (Vol. 14, p. 799), ARSENIC (Vol. 2, p. 653), MERCURY (Vol. 18, p. 158), IODINE (Vol. 14, p. 725), BROMINE (Vol. 4, p. 633), SODIUM (Vol. 25, p. 343), POTASSIUM (Vol. 22, p. 200), MAGNESIUM (Vol. 17, p. 321), BISMUTH (Vol. 4, p. 11). Separate sections dealing with pharmacology are found in the articles on very many plants, such as ALOE (Vol. 1, p. 720), ANISE (Vol. 2, p. 55), ARROWROOT (Vol. 2, p. 649), _Iceland Moss_ (Vol. 14, p. 241), CINCHONA (Vol. 6, p. 369), COCA (Vol. 6, p. 614), COLCHICUM (Vol. 6, p. 661), DANDELION (Vol. 7, p. 801), HOP (Vol. 13, p. 678), HOREHOUND (Vol. 13, p. 692), LOBELIA (Vol. 16, p. 837), MINT (Vol. 18, p. 557), MUSTARD (Vol. 19, p. 97), PEPPERMINT (Vol. 21, p. 128), etc.

[Sidenote: Biographies of Eminent Scientists]

The scientific biographies include not a few subjects which will be of interest, owing to familiarity with the names, to those engaged in the chemical and drug business. Among these are LISTER, BARON JOSEPH L. (Vol. 16, p. 777), to whose work and teaching the present importance of the manufacture of antiseptics is largely due; PASTEUR, LOUIS (Vol. 20, p. 892); CURIE, PIERRE, and MME. MARIE CURIE (Vol. 7, p. 644), the physicists who first announced the existence of radium; LIEBIG, BARON J.

VON (Vol. 16, p. 590), the great physiological chemist; LUNGE, GEORG (Vol. 17, p. 126), the noted expert in technical chemistry, already mentioned as a contributor to the Britannica, and GLAUBER, J. R. (Vol. 12, p. 114), the German chemist who made a living chiefly by the sale of secret chemical and medicinal preparations.

ALPHABETICAL LIST OF ARTICLES IN THE ENCYCLOPAEDIA BRITANNICA OF SPECIAL
INTEREST TO THOSE ENGAGED IN THE MANUFACTURE AND SALE OF CHEMICALS AND
DRUGS

Abel, Sir Frederick A.

Acacia

Acenaphthene

Acetic Acid

Aceto-Acetic Ester

Acetone

Acetophenone

Acetylene

Achard, F. C.

Acid

Acid Amides

Aconite

Acorus Calamus

Acridine

Adenine

Adipocere

Affinity, Chemical

Albumin, or Albumen

Alcohol

Alcohols

Aldehydes

Alembic

Algaroth, Powder of

Alizarin

Alkahest

Alkali

Alkali Manufacture

Alkaline Earths

Alkaloid

Alkanet

Allantoin

Alloxan

Alloxantin

Allyl Alcohol

Almond

Aloe

Alum

Aluminium

Amidines

Amines

Ammonia

Ammoniacum

Amygdalin

Amyl Alcohols

Amyl Nitrite

Anaesthesia and Anaesthetics

Analysis

Anatto

Andrews, Thomas

Angelica

Aniline

Animé
Anise
Anthracene
Anthraquinone
Antimony
Antipyrine
Antiseptics
Apothecary
Araroba Powder
Archil
Argol
Aristolochia
Arnica
Arrowroot
Arsenic
Asafetida
Asparagine
Aspen
Asphodel
Azo Compounds
Azoximes
Baeyer, Adolf von
Balard, Antoine J.
Balsam
Barium
Base
Baumé, Antoine
Bdellium
Becher, J. J.
Bell, Jacob
Belladonna

Benzaldehyde

Benzine

Benzidine

Benzoic Acid

Benzoin

Benzophenone

Benzyl Alcohol

Berberine

Bergman, Torbern Olof

Berthelot, M. P. E.

Berthollet, C. L.

Beryllium, or Glucinum

Berzelius, J. J.

Betaine

Betel Nut

Bhang

Bibirine or Bebeerine

Bichromates and Chromates

Bismuth

Bittern

Black, Joseph

Borage

Borax

Boric Acid or Boracic Acid

Boron

Boussingault, J. B. J. D.

Brande, William Thomas

Bromine

Brown, S. M.

Brucine

Buchu

Bunsen, P. W. von
Butyl Alcohols
Butyric Acid
Cadmium
Caesium
Caffeine
Cajuput Oil
Calabar Bean
Calcium
Calomel
Calvert, F. Crace
Camphors
Cannizzaro, Stanislao
Cantharides
Capsicum
Capsule
Caraway
Carbazol
Carbide
Carbohydrate
Carbolic Acid
Carbon
Carbonates
Carbon Bisulphide
Carbonic Acid
Cardamon
Carvacrol
Cassia
Castor Oil
Catalysis
Catechu

Caustic

Cavendish, Henry

Cayenne Pepper

Cellulose

Cerium

Chamomile, or Camomile Flowers

Charcoal

Chemical Action

Chemistry

Chevreul, M. F.

Chloral

Chlorates

Chlorine

Chloroform

Chlorpicrin

Chromium

Chrysene

Cimicifuga

Cinchona

Cinnamic Acid

Cinnamon

Cinnolin

Citric Acid

Clark, Thomas

Cloves

Coal-tar

Cobalt

Coca, or Cuca

Cocaine

Coco-nut Palm

Cod-liver Oil

Colchicum
Colcothar
Collodion
Colocynth
Colt's-Foot
Columbium
Combustion
Condenser
Conine
Copaiba
Copal
Copper
Copperas
Coriander
Corrosive Sublimate
Coumarin
Coumarones
Creosote
Cresols
Crookes, Sir William
Crotonic Acid
Croton Oil
Crystallization
Cubebs
Cumin
Curie, Pierre
Cyanamide
Cyanic Acid and Cyanates
Cyanide
Cyanogen
Cytisine

Dalton, John

Dammar

Dandelion

Daniell, John F.

Davy, Sir Humphry

Decolourizing

Depilatory

Dessication

Dewar, Sir James

Dextrine

Diazo Compounds

Didymium

Digitalis

Dill

Diphenyl

Disinfectants

Distillation

Dividivi

Döbereiner, J. W.

Dragon's Blood

Drug

Dulong, Pierre Louis

Dumas, J. B. A.

Durene

Earth

Ecgonine

Elaterium

Elecampine

Electrochemistry

Electrolysis

Electrometallurgy

Element

Elixir

Elm

Epsom Salts

Equivalent

Erbium

Erdmann, Otto Linné

Ergot, or Spurred Rye

Erythrite

Esters

Ether

Ethers

Ethyl

Ethyl Chloride

Ethylene

Eucalyptus

Eugenol

Euphorbium

Eupion

Europium

Fehling, Hermann von

Fennel

Fenugreek

Fig

Filter

Fir

Fischer, Emil

Fittig, Rudolf

Flamel, Nicolas

Flavin

Fluoranthene

Fluorene
Fluorescein
Fluorine
Formalin, or Formaldehyde
Formic Acid
Formula
Fourcroy, A. F., Comte de
Foxglove
Frankland, Sir Edward
Frémy, Edmond
Fresenius, Karl R.
Friedel, Charles
Fructose, or Fruit-sugar
Fuchs, Johann N. von
Fulminic Acid
Fumaric and Maleic Acids
Fumitory
Furazanes
Furfurane
Fusel Oil
Gadolinium
Galangal
Galbanum
Gallic Acid
Gallium
Gamboge
Gannal, J. N.
Garlic
Gay-Lussac, J. L.
Geber
Gelatin

Gelsemium

Gentian

Geoffroy, E. F.

Gerhardt, Charles F.

Germanium

Gibbs, Oliver Wolcott

Gilbert, Sir Joseph H.

Ginger

Ginseng

Gladstone, John Hall

Glaser, Christopher

Glauber, Johann R.

Glauber's Salt

Glucinum

Glucose

Glucoside

Glutaric Acid

Glycerin, or Glycerol

Glycols

Gmelin (family)

Gold

Graham, Thomas

Grains of Paradise

Greenheart

Guaco, Huaco or Guao

Guaiacum

Guanidine

Guarana

Guelder Rose

Guimet, Jean B.

Gum

Guyton de Morveau, Baron

Harcourt, W. Vernon

Hartshorn, Spirits of

Hashish

Hellebore

Helmont, Jean B. van

Hemp

Henbane

Henna

Henry, William

Herb

Hippuric Acid

Hofmann, A. W. von

Homberg, William

Homoeopathy

Hop

Horehound

Houseleek

Hydantoin

Hydracrylic Acid

Hydrastine

Hydrate

Hydrazine

Hydrazone

Hydrocarbon

Hydrochloric Acid

Hydrogen

Hydroxylamine

Hyposulphite of Soda

Hyssop

Iatrochemistry

Iceland Moss

Imidazoles, or Glyoxalines

Indazoles

Indene

Indicator

Indigo

Indium

Indole

Indulines

Inulin

Iodine

Iodoform

Ipecacuanha

Iron

Isatin

Isomerism

Isoxazoles

Jaborandi

Jalap

Juniper

Kámalá

Kekulé, F. August

Kelp

Kermes

Ketenes

Ketones

Kino

Klaproth, M. H.

Kolbe, A. W. Hermann

Kopp, Hermann F. M.

Kousso

Kunkel, or Kunckel von Lowenstjern, J.

Lactic Acid

Lactones

Laevulinic Acid

Lanolin

Lanthanum

Laudanum

Lavender

Lavoisier, A. L.

Lead

Le Blanc, Nicolas

Lemery, Nicolas

Lemon

Liebig, Baron J. von

Lime

Linseed

Liquorice

Lister, Baron

Lithium

Litmus

Lobelia

Lunge, Georg

Madder

Magnesium

Magnus, H. G.

Malic Acid

Mallow

Malonic Acid

Malt

Mammee Apple

Mandelic Acid

Mandrake

Manganese

Mangosteen

Manna

Marggraf, Andreas S.

Marignac, Jean C. G. de

Mastic

Mayow, John

Medical Jurisprudence

Medicine

Mellitic Acid

Mandeléeff, Dmitri I.

Mercaptans

Mercury

Mesoxalic Acid

Methyl Alcohol

Meyer, J. Lothar

Meyer, Victor

Microcosmic Salt

Mineral Waters

Mint

Mitscherlich, E.

Mohr, K. Friedrich

Moissan, Henri

Molybdenum

Mond, Ludwig

Morphine

Mucic Acid

Murexide

Murray, John

Musk

Muspratt, J. and J. S.

Mustard

Mustard Oils

Myrrh

Myrtle

Naphtha

Naphthalene

Naphthols

Naphthylamines

Nepenthes

Newlands, John A. R.

Nickel

Nightshade

Niobium

Nitre

Nitric Acid

Nitrobenzene

Nitro Compounds

Nitrogen

Nitroglycerin

Nobel, Alfred B.

Nux Vomica

Officinal

Oils

Olefine

Oleic Acid

Opium

Orcin

Orpiment

Orris-root

Oxalic Acid
Oxazoles
Oxide
Oximes
Oxygen
Oxyhydrogen Flame
Palladium
Palmitic Acid
Paraffin
Paraldehyde
Pasteur, Louis
Pelouze, T. Jules
Pennyroyal
Peppermint
Pepsin
Perfumery
Perkin, Sir W. H.
Pettenkofer, Max J. von
Pharmacology
Pharmacopoeia
Pharmacy
Phenacetin
Phenanthrene
Phenazine
Phenol
Phenolphthalein
Phosphates
Phosphorus
Phthalazines
Phthalic Acids
Picene

Picric Acid

Picrotoxin

Pilocarpine

Pimento

Pine

Piperazin

Piperine

Piperonal

Platinum

Plattner, K. F.

Podophyllin

Poison

Polymethylenes

Pomade

Potashes

Potassium

Priestley, Joseph

Primuline

Propiolic Acid

Propyl Alcohols

Proust, Joseph Louis

Prout, William

Prussic Acid

Pumice

Purin

Purslane

Pyrazines

Pyrazoles

Pyrene

Pyrethrum

Pyridine

Pyrimidines
Pyrocatechin
Pyrogallol
Pyrones
Pyrophorous
Pyrrol
Pyruvic Acid
Quassia
Quercitron
Quinazolines
Quinine
Quinoline
Quinones
Quinoxalines
Radium
Ramsay, Sir William
Raoult, François M.
Rare Earths
Regnault, H. V.
Resorcin
Retene
Rhamnus Purshiana
Rhatany, or Krameria Root
Rhodium
Rhubarb
Richter, J. B.
Roebuck, John
Roscoe, Sir H. E.
Rose
Rouelle, G. F.
Rouge

Rubidium
Ruthenium
Saccharic Acid
Saccharin
Safflower
Saffron
Safranine
Sainte-Claire Deville, E. H.
Sal Ammoniac
Salep
Salicin, Salicinum
Saliscylic
Salt
Saltpetre
Samarium
Sandalwood
Sandarach
Santonin
Sarsaparilla
Scammony
Scandium
Scheele, K. W.
Schlippe's Salt
Schönbein, C. F.
Schützenberger, P.
Senega
Senna
Sesame
Silica
Silicon
Silliman, Benjamin

Silver

Snake-root

Soap

Soap-bark

Sodium

Solution

Spectroscopy

Spikenard, or Nard

Spirits

Sponges

Squill

Stahl, G. E.

Stas, J. S.

Stearic Acid

Stereochemistry

Stero-isomerism

Stoichometry

Stramonium

Strontium

Strophanthus

Strychnine

Styrolene

Succinic Acid

Sugar

Sulphonal

Sulphonic Acids

Sulphur

Sulphuric Acid

Sumbul, or Sumbal

Supra-renal extract

Talc

Tamarisk
Tannin or Tannic Acid
Tantalum
Tar
Taraxacum
Tartar
Tartaric Acid
Tellurium
Tennant, Charles
Tennant, Smithson
Terbium
Terpenes
Tetrazines
Tetrazoles
Thénard, L. J.
Therapeutics
Thermochemistry
Thiazines
Thiazoles
Thiophen
Thomsen, Julius
Thomson, Thomas
Thorium
Thymol
Thyroid
Tin
Tincture
Titanium
Toilet Powders
Toluene
Tonqua Bean

Tooth Powders and Pastes

Triazines

Triazoles

Triphenylmethane

Trophine

Tungsten

Turmeric

Upas

Uranium

Urea, or Carbamide

Urethane

Urotropin

Valency

Valerian

Valeric Acid

Vanadium

Vanilla

Van't Hoff, J. H.

Vaseline

Vauquelin, L. N.

Veratrum

Veronal

Viburnum

Vitriol

Weighing Machines

Weights and Measures

Weldon, Walter

Wenzel, K. F.

Williamson, A. W.

Wine

Wintergreen

Winter's Bark
Wislicenus, J.
Witch-hazel
Wöhler, Friedrich
Wollaston, W. H.
Wormwood
Wurtz, C. A.
Xanthic Acid
Xanthone
Xylene
Yew
Young, James
Ytterbium
Yttrium
Zinc
Zirconium

CHAPTER XIII

FOR MERCHANTS AND MANUFACTURERS OF FOOD PRODUCTS

The manufacturer of or dealer in food products must of necessity be interested in questions of transportation by land and sea, of taxation, of agriculture, stock-raising and fishing, for example. For all such subjects as these he is referred to other chapters of this Guide. Here he will find only the chief articles on the subjects most closely related to the study of food products. But on these he may glean a wealth of information that will be of greatest value to him, and from

them he can turn readily and with profit to a survey of the larger area covered by other chapters.

As a general introduction to the subject the student should read DIETETICS (Vol. 8, p. 214), by the late Dr. W. O. Atwater, who was in charge of the Nutrition Investigation of the U. S. Department of Agriculture, and R. D. Milner, also of that Department. This article deals with the composition and nutritive values of foods, their fuel value, quantities of nutriments needed, hygienic and pecuniary economy of foods (with tables showing the percentage composition of common food materials), conditions of digestibility, and other matters of equal importance. NUTRITION (Vol. 19, p. 920), by Prof. D. N. Paton and Dr. E. P. Cathcart, both of the University of Glasgow, discusses more particularly digestion and the utilization of the different food constituents.

[Sidenote: Food Preservation]

After establishing the value and relative importance of the various substances used as food, it is of great interest to everyone in the business to consider the subject of FOOD PRESERVATION (Vol. 10, p. 612), an article by Otto Hehner, formerly president of the Society of Public Analysts, in which there are separate sections on _Preservation by Heat_ (which includes all canning processes); _by Chemicals_; _by Drying_; _by Refrigeration_; _by Pickling_. The sterilization of milk, condensed milk and milk powder all fall within the scope of this article. The preservation of food by cold is described in fuller detail in the article REFRIGERATING AND ICE MAKING (Vol. 23, p. 30), by T. B. Lightfoot, author of the standard technical book on that subject. Among the separate articles on preservative materials are VINEGAR (Vol. 28, p. 96), ACETIC ACID (Vol. 1, p. 135), CITRIC ACID (Vol. 6, p. 397), OILS

(Vol. 20, p. 43), SALT (Vol. 24, p. 87), SALTPETRE (Vol. 24, p. 93), SUGAR (Vol. 26, p. 32), BORAX (Vol. 4, p. 243), FORMALIN OR FORMALDEHYDE (Vol. 10, p. 667), BENZOIC ACID (Vol. 3, p. 756), SALICYLIC ACID (Vol. 24, p. 69), SULPHUR, _Compounds_ (Vol. 26, p. 63), ALCOHOL (Vol. 1, p. 525).

[Sidenote: Adulteration]

The objections to the use of some of these chemicals are discussed in ADULTERATION (Vol. 1, p. 218), by Otto Hehner. This article is about as long as 50 pages of this Guide. There is an interesting historical introduction, from which we learn that the first legal statute in which the adulteration of food is noticed dates from the reign of King John in England (1203). There is an elaborate account of all the subsequent legislation in Great Britain, the United States, and Germany. The effects upon digestion of the chemical preservatives mentioned above are shown in the light of the very latest investigations. There is a section on colouring matter in food, with information about harmless and harmful dyes; and the last part of the article considers adulteration as recently applied to the more important articles of food, such as milk (with tests for borax and formaldehyde), cream, butter, cheese, lard, oils, flour and bread, sugar, marmalade and jams, tea, coffee, cocoa and chocolate, wine, beer, spirits, non-alcoholic drinks, and vinegar.

The properties of adulterants and colouring matters are described in separate articles, such as GLUCOSE (Vol. 12, p.